ACE MANUFACTURING SYSTEMS LTD

AMS® In Pursuit of Excellence

INDEX - MACHINE HISTORY

MACHINE NUMBER : M11-930

SL NO	DESCRIPTION	FORMAT NUMBER	ENCLOSED
1	MACHINE DISPATCH CLEARANCE	F-QA-FA -16	✓
2	PACKING LIST / DISPATCH DETAILS	F-TSG-18 & F-PAC-01	✓
3	MINUTES OF MEETING	F-TSG-17	✓
4	WORK ORDER	F-TSG-15	✓
5	MACHINE SEQUENCE RUNNING REPORT	F-QA-FA-12	✓
6	SPINDLE STABILISATION REPORT	F-QA-FA-11	✓
7	MACHINE HEALTH CHECK POINTS	F-QA-FA-09	✓
8	CHECK LIST - MECHANICAL	F-QA-FA-07	✓
9	MACHINE HISTORY - MECHANICAL	F-QA-FA-10	✓
10	CHECK LIST - ELECTRICAL	F-QA-FA-08	✓
11	MACHINE HISTORY - ELECTRICAL	F-QA-FA-14	✓
12	FINAL ASSEMBLY REPORT - MECHANICAL	F-QA-FA-05/15	✓
13	GEOMETRICAL TEST REPORT	DS-DSN-PROD-TC 01~ 06	✓
14	PRACTICAL TEST REPORT	F-TSG -34	✓
15	BALL BAR REPORT	F-QA-FA-02	✓
16	AXIS CALIBRATION REPORT	F-QA-FA-01	✓
17	SPINDLE BALANCING REPORT	F-QA-FA-04	✓
18	SPINDLE ASSEMBLY	F-QA-IPR-SPL- 01	✓
19	HYDRAULIC ASSEMBLY REPORT (OPTIONAL)	F-QA-IPR-HYD-01	✓
20	GEAR BOX / ATC ASSEMBLY	F-QA-IPR-GBA- 01 / SUPPLIER REPORT	✓
21	BALL SCREW ASSEMBLY	F-QA-IPR-BSA- 01	✓
22	BASE BUILDING	F-QA-IPR-BBA- 01/ 02	✓

Date : 22/11/13

CHECKED BY

APPROVED BY

ACE MANUFACTURING SYSTEMS LIMITED.

400-NB

DISPATCH ADVICE

AMS® In Pursuit of Excellence

Sl No. 30/11	Description Of Dispatch Details	D.A No. : DNo#012489
1	Customer Name	**WABCO INDIA LIMITED., - PLANT-I [WABIN002]**
2	Telephone No.	044 - 30902600 / 42245000
3	Fax No.	044 - 26241287 / 30902609
4	E-Mail	
5	Customer's ECC NO.	AAFCA6421PXM001
6	Customer's Sales Tax No. & Date	CST : 871408 25/03/2008 TIN : 33030461698
7	Sales Order No.	MO10003718
8	Machine Model	MCV-400
9	Machine No.	M11-932
10	Invoice Address	WABCO INDIA LIMITED., - PLANT-I [WABIN002] PLOT NO. 3 (SP), III MAIN ROAD, , AMBATTUR INDUSTRIAL ESTATE, , , CHENNAI , 600058
11	Place Of Delivery	Default Address
12	Purchase Order No. & Date	4800004849, Date : 03-SEP-13
13	P.O Value(Excluding Taxes & Duties)	Rs. 33,75,000.000
14	Transportation	Door delivery through any reliable carriers and freight on paid basis
15	Insurance	To be arranged & borne by AMS. Dispatch details to be faxed to the customer.
16	Transportation Budget	17,000.00
17	Hypothecation	
18	Special Scheme	NONE
	Scheduled Dispatch	22/11/2013
18	Payment	Refer payment collection detail sheet
19	Special Instructions	Keep the machine ready for dispatch. Contact HOD before dispatch. Shortage list to be made while preparing machine Invoice All pending points to be attended before machine dispatch.
20	**Special Remarks**	

For ACE MANUFACTURING SYSTEMS LTD.

Authorised Signatory

ACE MANUFACTURING SYSTEMS LIMITED

PACKING LIST

AMS® In Pursuit of Excellence

Packing List No.	: DNo#012489	Date :	22/11/2013 1
Sales Order No.	: MO10003718	Date :	22/10/2013 1
Customer	: WABCO INDIA LIMITED., - PLANT-I CHENNAI		
P.O. No.	: 4800004849 Date : 03-SEP-13		

Sl No.	Item Id.	Model	Qty.
1	M11-932	MCV-400	1

Sl No.	Description	Short Supply?
1	CNC Vertical Machining Center Model "MCV-400"	
	Spindle nose taper 7/24 No.40	
	AC Servo axes drive (Fanuc)	
	Twin Arm (Random) type High Speed Auto tool changer	
	Tool shank BT-40	
	LM Guideways for all the three axes	
	CNC System Fanuc 0IMATE-MD with 8.4" Colour LCD	
	Ball screw stretching for X, Y & Z Axes	
	Panel cooler for electrical cabinet	
	Fail safe brake for Z-Axis	
	Standard work light	
	Coolant tank and chip tray	
	Three tier indication lamp	
	Flexible coupling for X, Y & Z Axes	
	Manual pulse generator	
	Rigid tapping	
	Total machine enclosure	
	Chip disposal to the rear side of the machine through two chutes on the sides	
	One set of manuals	
	Rapid traverse of 30 / 30 / 30 m/min. for X, Y & Z Axes	
	Tool life management	
	Air drier unit (Small)	
2	Spindle speed of 80-8000 rpm with Indirect drive	
3	Higher spindle power 11/7.5 kw in lieu of standard spindle 7.5/5.5 kw For spindle speed of 80-8000 rpm with Indirect drive	
4	Flood coolant system around the spindle (In lieu of flexinox hose arrangement)	
5	Chip flushing system on the base	
6	Coolant gun	
7	Front auto door	
8	Increased day light area (NO increase in Z-stroke) - 160mm	
9	30 station tool magazine in lieu of 20 station tool magazine	
10	High pressure coolant through spindle system (16 lpm at 21 bar) with Grundfoss centrifugal pump and bag type filtration system	

For ACE MANUFACTURING SYSTEMS LTD.

Authorised Signatory

11	SPECIAL COOLANT TANK FOR CTS APPLICATION	
12	Fanuc Beta4 Motor & Drive along with 4th Axis feature + interfacing suitable for Nikken Rotary table 202 LFA	
13	Hydraulic Rotary Joint (4 Ports) fitted on rotary table (Customer will supply NIKKEN 202 LFA Rotary Table)	
14	Wider guarding	
15	Aluminium tank Hydraulic power pack for fixture with 4 station @ Maximum of 70 Bar & minimum of 5 bar from REXROTH	
16	Hydraulic interfacing (only electrical wiring) with looping assembly for 4 lines on the machine	
17	Fixture interfacing with necessary M-codes	
18	TPM friendly machine (see through panel, markings etc.)	
19	Proximity Switches - Single (plug & socket type) & Multi Switches - BALLUF Make in lieu of Standard	
20	Telemechanique MCCB in lieu of Standard	
21	Push Button, Selector Switches, Illuminated Push button Switch (LED type) only, Illuminated Selector Switch (LED type)	
	Illuminated pilot lamp (LED type) only in lieu of our standard	
	Data backup in flash ROM including part programs, Macro programs, Parameters and ladder backup must to be supplied along the machine	
23	ELCB in main power circuit	
24	Fanuc CD (Soft copy of Fanuc manual)	
25	Pull Studs - 30 Nos	
26	SERVO SPIN - 12 lubricating oil (1 liter)	
27	1 set of glass fuses(inside the cabinet)	
28	Electrical documentation manual & Fanuc system Warranty Card	
29	Electrical Back up CD (Inside the cabinet pouch)	
30	Geometrical test report	
31	NAS test report	
32	Laser Calibration test report	
32	Heavy cut & Rigid tapping test report	

For ACE MANUFACTURING SYSTEMS LTD.

Authorised Signatory

ACE MANUFACTURING SYSTEMS LIMITED				AMS In Pursuit of Excellence
PACKING LIST ANNEXTURE-II				
Doc No:	PAN#000906	**M/c No:**	**M11-932 / MCV-400**	
Date:	22/11/2013	**So No:**	MO10003718	
M/s	**WABCO INDIA LIMITED., - PLANT-I., CHENNAI**			

Sl. No	Description	Qty	Remarks	Part No
1	O'Ring ø8.1 x 1.6	6		S15-005-00
2	O'Ring ø24.5 x 3.0	6		S15-005-17
3	O'Ring ø57.6 x 2.4	6		S15-005-18
4	O'Ring ø19.1 x 1.6	2		S15-003-33
5	O'Ring ø11.1 x 1.6	4		S15-005-01
6	O'Ring ø51.6 x 2.4	2		S15-003-70
7	Cord of O'Ring ø3 X 2 Mtrs	1		S15-015-20
8	Spanner Double Ended 10 X 11	1		S31-001-01
9	Spanner Double Ended 12 X 13	1		S31-001-02
10	Spanner Double Ended 18 X 19	1		S31-001-03
11	Spanner Double Ended 30 X 32	1		S31-001-04
12	Spanner Double Ended 36 X 41	1		S31-001-11
13	Allen Key Set	1		S31-002-101
14	Extended Allen Key (A/F 8)	1		C99-5106-8904-00
15	Split Collet ,Stud Push With Spring	2		C15-005,C74-427&C85-3101
16	Fuse Set (Fanuc Make)	1		S22-010-11
17	Glass Fuse Set (Kept in Electrical Cabinet)	1		
18	Lubrication Oil (Spin-12 / Process Oil-12)	1		S04-005-07-CAN
19	Pull Stud BT-40 (DIN)	30		S13-010-16
20	FANUC Series Oi Model D / Oi Mate-Model D USER`S MANUAL	1		D00-003-00-27
21	FANUC Series Oi Model D / Oi Mate-Model D MAINTENANCE MANUAL	1		D00-003-00-28
22	Fanuc AC SERVO & SPINDLE MOTOR Bis series /alpha series MAINTENANCE MANUAL	1		D00-003-00-03
23	Operating Instructions And Maintainance Manual	1		D00-003-00-05
24	Spare Parts Manual	1		D00-003-00-04
25	Air Cooler Manual	1		D00-003-00-06
26	Pin Locator For CTS	2		C57-3129-8114

Packed by: .....

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ACE MANUFACTURING SYSTEMS LIMITED				AMS® In Pursuit of Excellence
PACKING LIST ANNEXTURE-III				
Doc No:	PAN#000904	**M/c No:**	**M11-932 / MCV-400**	
Date:	22/11/2013	**So No:**	MO10003718	
M/s	**WABCO INDIA LIMITED., - PLANT-ICHENNAI**			

Sl. No	Description	Qty	Remarks	Part No
1	Electrical Documentation Manual	1		D00-003-00-00-000-E0
2	Maintenance Service Note (Fanuc Warranty Card)	1		D00-003-00-23
3	MACHINE BACK UP CD (Kept Machine Electrical Cabinet Pouch)	1		D00-003-00-00-000-E3

Packed by: .....

Printed Date & Time: 22/11/2013 2:27:41 PM

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ACE MANUFACTURING SYSTEMS LIMITED

AMS In Pursuit of Excellence

PACKING LIST ANNEXTURE-III

Doc No:	PAN#000937	**M/c No:**	**M11-932 / MCV-400**
Date:	29/11/2013	**So No:**	MO10003718
M/s	**WABCO INDIA LIMITED., - PLANT-ICHENNAI**		

Sl. No	Description	Qty	Remarks	Part No
1	SERVO MOTOR ß4IS/4K(A06B-0063-B003)	1		S21-011-29
2	FEEDBACK CONNECTOR PLUG (SIGNAL)(A06B-6114-K204#S)	1		S21-021-04
3	CONNECTOR(POWER & BRAKE)(A06B-6114-K220#E)	1		S21-021-38
4	1GB FLASH CARD (transcend make) 80X	1		S26-001-20
5	ADOPTER FOR FLASH ROM CARD SSD-PCMCIA-CF	1		S26-001-12
6	Fanuc CD (Soft copy of Fanuc manual)	1		

Packed by: GIRISH.S.V

Printed Date & Time: 29/11/2013 3:35:51 PM

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ACE MANUFACTURING STSTEMS LTD

PACKING CASE IDENTIFICATION LIST

AMS

Machine Number : MH-932 Date : 30/11/013

Sl. No.	Case - 1 Contents	Quantity
1	Machine Assembly	01NO
2	Standard Spares & Machine Manuals	01BOX
3	Servo Spin - 12 Oil	01Liter
4	Levelling Screws & Lock Nuts	04+4NOS
5	Levelling Blocks - (100) / 180 / 220 / 300	08NOS
6	Cylinder - Hydraulic / Pneumatic	[illegible]
7	Machine Guards	02NOS
8	Side Door Handles	—
9	Power Pack	—
10	Power Pack Stand	01NO
11	A P C Levelling Pads	01NO
12	Machine Lifting Ropes	—
13	Fixture parts	—
14	Customer Components	02NOS
15	Rotary Table	—
16		—
17		

	Case - 2 Contents	
1	Coolant Tank / Conveyor Assembly	01NO
2	Coolant Pumps	05NOS
3	Mesh Top Plates	02NOS
4	Chip Bin	01NO
5	Chip Bin Mesh	02NOS
6	Chucker Plates	01NO
7	Oil Bin	01NO
8	Chip Trolley Electrical Spare.	01BOX
9	Conveyor Top Guard	—
10	Bag Filters - (50) / 100 Bar	02NOS
11	Grund Fos Pump	01NO
12	Base Guard	—
13		
14		
15		
16		
17		

	Case - 3 Contents	
1		/
2		
3		
4		
5		
6		
7		

Checked By : Jayanna.

ACE MANUFACTURING SYSTEMS LTD.

MACHINE DISPATCH CHECK POINTS

AMS — For Pursuit of Excellence

CUSTOMER'S NAME: WABCO

M/C No.: MH-932 MCV-400 DATE: 30/11/2013

S.No	DESCRIPTION	YES/ NO/NA	S.No.	DESCRIPTION	YES/ NO/NA
1	Packing list available	Yes	21	Chip Trolley cleaned	NA
2	Hydraulic / Pneumatic Counter Balance Cylinder removed	NA	22	Standard Spares & m/c manual available	yes
3	Z- Motor Cover Removed & Packed	NO	23	Coolant tank assly. packed	Yes
4	Side door handles removed & Packed	NO	24	Index Table Cover packed properly	NA
5	Base guard removed & Packed	NA	25	Oil skimmer bin cleaned	NA
6	Coolant Gun tied properly	yes	26	Voltage Stabilizer packed	NA
7	Air gun tied properly	NA	27	Machine Platform cleaned	NA
8	Leveling Screws, Nuts removed & Packed	Yes	28	Wanner Pump & cover cleaned & put back	NA
9	Leveling blocks (100)/180/220/300 Ht. Qty: - 4/6/7/8/11/20	Yes	29	Customer components packed properly	NA
10	Automatic Tool Changer	Yes	30	Index Table packed properly	NA
11	APC Assembly packed Properly	NA	31	Chiller unit cleaned	NA
12	Pallet removed & Packed	NA	32	Rotary Table packed properly	NO
13	APC leveling screws - 1set	NA	33	Toolings packed properly	NA
14	Power pack stand	Yes	34	Tail Stock packed properly	NA
15	Lub Tank oil drained - retain 2 ltrs.	Yes	35	Practical Test piece packed	NA
16	Machine oil Bin cleaned	Yes	36	Fixture/s packed properly	NA
17	Coolant tank mesh cleaned & put back	Yes	37	Air removed from cover with Vaccum cleaner	NA
18	Chip Bin cleaned & put back	Yes	38	Desiccant Bags Hung	Yes
19	Chip Bin mesh cleaned & put back	Yes	39	Fumigation Certificate available	NA
20	TPM points stickers fixed properly	Yes	40		

REMARKS / SHORTAGES IF ANY:

NIL

ASSEMBLY	STORES	FIXTURE	ELECTRICAL	DISPATCH	AUTHORISED SIGNATORY
				Jayanna	

ACE MANUFACTURING SYSTEMS LTD.

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DISPATCH CHECK POINTS

CUSTOMER'S NAME : NABCO

MACHINE NUMBER : M11-932 DATE : 30/4/2013

S.No	CHECK POINTS	STATUS - YES / NO	REMARKS
1	PACKING CASE IS IN GOOD CONDITION.	Yes	
2	CHIP CONVEYOR IS FREE FROM CHIPS & UNWANTED MATERIALS.	Yes	
3	PAINT SHOULD NOT BE PEELED OFF IN ANY PART OF THE MACHINE.	Yes	
4	ALL POCKETS IN ATC ARE EMPTY & CLEANED	Yes	
5	NO TOOL IN SPINDLE & CLEANED	Yes	
6	NO LOOSE PARTS KEPT ANYWHERE IN THE MACHINE.	Yes	
7	ENSURE RUST PREVENTIVE OIL IS APPLIED TO ALL MACHINED SURFACE.	Yes	
8	BOLT & NUT ARE FITTED AT ALL RELAVENT PLACES ON PACKING CASE.	Yes	
9	SPARE BOX KEPT INSIDE THE PACKING CASE.	Yes	
10	GEOMETRICAL, AXIS CALIBRATION, CUTTING TEST REPORTS ARE SENT WITH THE MACHINE (in side the cabinet)	Yes	
11	ELECTRICAL CABINET CLEANED.	Yes	
12	ENSURE MACHINE IS CLEANED & NO DUST ON ANY WHERE ON M/c	Yes	
13	SUPRABHA COVER IS SEALED PROPERLY.	Yes	
14	ENSURE PACKING BOX SHOULD HAVE MINIMUM 3" GAP ALL OVER	Yes	
15	COOLANT WATER FROM ALL PIPES REMOVED BY AIR JET.	Yes	
16	NO RUST OBSERVED ANYWHERE IN THE MACHINE & RUST PREVENTIVE SPRAYED TO ALL FASTENERS & BLACKEDNED PARTS	Yes	
17	THERE IS NO TOUCHUP PAINT DEPOSITED ON MATCHING PARTS.	— NO —	
18	MACHINE LIFTING PROCEDURE CHART FIXED ON THE PACKING CASE	Yes	
19	CABINET KEY SENT WITH MACHINE TIED TO THE AC UNIT EYE BOLT	yes	
20	PANEL COOLER FILTER IS CLEANED	Yes	
21	CHECK ITEMS AS PER PACKING LIST	Yes	
22	WOODEN PACKING PROVIDED BELOW THE APC STATION.	— NO —	
23	WOODEN Support given bdow the Grundfoss pump Mounting bracket	———	Thus proposal can Lwr A 23/4/2013
24.	grundfoss pump Removed and packed seperately	Yes / NA / NO	
25	MACHINE LIFTING PROCEDURE SKTCH PASTED ON THE GOURD	— Yes —	

Jayarma — V. Gopal — [signature]

ATTENDED BY | QUALITY ASSURANCE | DISPATCH - INCHARGE

F-PAC-02-RO

16	Hydraulic power pack to be tested before dispatch	OK	Before dispatch	Mech.Assy.
17	MCCB to be shifted outside below the panel cooler of electrical cabinet	OK	Before dispatch	Elec.Assy.

With the above, the machine stands clear for dispatch and the machine will be dispatched after attending all the above points.

AMS requests M/s. Wabco India Limited to arrange for the release of balance 60% payment along with taxes & duties before machine dispatch.

AMS thanks M/s. Wabco India Limited representatives for their kind co-operation extended during machine inspection.

AMS

WABCO INDIA LIMITED

S. Palaniappan → PMD

V. Parthasarathy - V.Dir. - PED

ACCEPTANCE FOR DISPATCH : [INTERNAL USE]

TSG DEPT. **ASSEMBLY DEPT.** **QUALITY DEPT.** **DESIGN DEPT.**

F.A.

ACE MANUFACTURING SYSTEMS LIMITED

WORK ORDER

AMS® In Pursuit of Excellence

Customer :
M/s. WABCO INDIA LIMITED., - PLANT-I
PLOT NO. 3 (SP), III MAIN ROAD,, AMBATTUR INDUSTRIAL ESTATE,, , CHENNAI, 600058, TAMILNADU

SO.No MO10003718
PO.No 4800004849

WO.Date : 26/10/2013 **PO.Date :** 03/09/2013

S.O.TYPE : Standard

Sl No.	Item Id.	Item Description	Unit	Qty.
1	VMC	CNC VERTICAL MACHINING CENTER - MCV-400	Nos	1

MACHINE COLOUR :

Machine Colour : ACE Blue & Light Grey combination.

WARRANTY :

a) One year (i.e. 12 months) from the date of commissioning of the machine or 15 months from the date of supply of the machine whichever is earlier against faulty material and workmanship. Guarantee is not applicable for part, which has normal wear & tear (like springs, balls etc.) and certain other items like Bulbs, Lamps, Indicator Lamps, Revolving Lamps, Seals and Glass Fuses.
b) For CNC System, motors and drives, it is Two Years from the date of dispatch. This warranty is applicable only for the first installation of the machine.

Scope of Supply, Technical Specs & CNC Features As Per Annexure I

Machine Inspection	Geometrical accuracies and machine acceptable as per test chart and NASA test
CustomerInspection Type	DIRECT
Customer Inspection Date	31/10/2013
Component Trials Required?	NOT APPLICABLE

For ACE MANUFACTURING SYSTEMS LTD.,

P.Ramadas
Managing Director

L.S.Umesh
Chief Executive Officer

ACE MANUFACTURING SYSTEMS LTD.

PRODUCT SPECIFICATION SHEET

MODEL: MCV-400

Sl. No.	Description	Unit	MCV-400
1	**TABLE**		
1.1	TABLE SIZE	mm	900 x 450
1.2	T-SLOT DIM.	N x W x P	4 x 18 x 100
1.3	MAX. LOAD ON TABLE	Kgf	300
1.4	DIST. FROM FLOOR TO TABLE TOP	mm	925
2	**MACHINE CAPACITY**		
2.1	X - AXIS TRAVEL	mm	600
2.2	Y - AXIS TRAVEL	mm	450
2.3	Z - AXIS TRAVEL	mm	500
2.4	DIST. FROM SPINDLE FACE TO TABLE TOP WITH INCREASED DA LIGHT AREA (160mm)	mm	335 - 835
2.5	DIST. FROM SPINDLE CENTER TO COLUMN GUIDEWAYS	mm	510
2.6	COOLANT SUPPLY-FLOOD COOLANT AROUND THE SPINDLE	lpm	50
3	**SPINDLE**		
3.1	SPINDLE NOSE TAPER		7 / 24 No.40
3.2	SPINDLE SPEED - INDIRECT DRIVE	rpm	80 - 8000
3.3	SPINDLE BEARING DIAMETER	mm	70
4	**FEED**		
4.1	RAPID TRAVERSE X, Y & Z AXES	m / min.	30 / 30 / 30
4.2	CUTTING FEED	mm / min.	1 - 10000
4.3	GUIDEWAYS FOR X, Y & Z AXES		Linear Motion
5	**AUTOMATIC TOOL CHANGER**		
5.1	NUMBER OF TOOLS	Nos.	30
5.2	MAX. TOOL DIA	mm	80
5.3	MAX. TOOL DIA. WITH ADJACENT POCKETS EMPTY	mm	155
5.4	TOOL SELECTION		Bi - Directional
5.5	TOOL LENGTH MAX.	mm	250
5.6	TOOL WEIGHT MAX.	Kg.	8
5.7	CHIP TO CHIP TIME (WITH Z ONLY / WITH X & Y ALSO)	Sec.	3.9 / 4.8
5.8	TOOL CHANGE SYSTEM		Twin Arm
5.9	TOOL SHANK TYPE		BT - 40
5.10	PULL STUD TYPE		DIN 69872 Form-B
6	**ACCURACY WITH ROTARY ENCODER**		
6.1	POSITIONING FOR X, Y & Z AXES AS PER VDI DGQ 3441	mm	0.015
6.2	POSITIONING FOR X, Y & Z AXES AS PER ISO 230-2	mm	0.010
6.3	REPEATABILITY FOR X,Y&Z AXES AS PER VDI DGQ 3441	mm	± 0.005
6.4	REPEATABILITY FOR X, Y & Z AXES AS PER ISO 230-2	mm	± 0.003
7	**MOTORS**		
7.1	SPINDLE POWER 15 MIN./CONT. (FANUC)	kW	11 / 7.5
7.2	FEED MOTOR X, Y & Z AXES (FANUC)	Nm	11 / 11 / 20

ACE MANUFACTURING SYSTEMS LTD.

PRODUCT SPECIFICATION SHEET

MODEL: MCV-400

Sl. No.	Description	Unit	MCV-400
8	**INSTALLATION DATA - STANDARD MACHINE**		
8.1	MACHINE WIDTH	mm	1900
8.2	MACHINE DEPTH	mm	3300
8.3	MACHINE HEIGHT	mm	2750
8.4	MACHINE WEIGHT - NET	Kgf	4100
8.5	POWER SUPPLY FOR BASIC MACHINE	Kva	14
8.6	CNC CONTROL (WITH 8.4" COLOUR LCD)		FANUC 0IMATE-MD
8.7	PNEUMATIC PRESSURE REQUIRED	Kgf / Cm2	5
8.8	PNEUMATIC SUPPLY REQUIRED AT ATMOSPHERIC PR.	Lpm	300
9	**TANK CAPACITY - STANDARD MACHINE**		
9.1	COOLANT TANK	litr	300
9.2	LUBRICATION TANK (USEFUL QUANTITY)	litr	5 (4)
10	**MACHINE FEATURES / ACCESSORIES - STANDARD**		
10.1	PANEL COOLER FOR ELECTRICAL CABINET		
10.2	FAIL SAFE BRAKE FOR Z - AXIS		
10.3	STANDARD WORK LIGHT		
10.4	COOLANT TANK & CHIP TRAY		
10.5	THREE TIER INDICATION LAMP TO INDICATE CYCLE RUNNING READY FOR NEXT CYCLE & MACHINE IN ALARM STATUS		
10.6	FLEXIBLE COUPLING FOR X, Y & Z AXES		
10.7	MANUAL PULSE GENERATOR		
10.8	RIGID TAPPING		
10.9	TOTAL MACHINE ENCLOSURE		
10.10	CHIP DISPOSAL TO THE REAR SIDE OF THE MACHINE THROUGH TWO CHUTES ON THE SIDES		
10.11	MANUALS - ONE SET		
10.12	BALL SCREW STRETCHING FOR X, Y & Z AXES		
10.13	TOOL LIFE MANAGEMENT		
10.14	AIR DRIER UNIT (SMALL)		
11	**STANDARD CNC FEATURES OF FANUC-0IMATE-MD**		
11.1	THREE CONTROLLED AXES		
11.2	SIMULTANEOUS CONTROLLABLE AXES - 3 AXES		
11.3	FEED RATE / RAPID OVERRIDE FACILITY		
11.4	CIRCULAR INTERPOLATION - ANY TWO OUT OF THREE AXES		
11.5	JOG MODE		
11.6	BACKLASH COMPENSATION		
11.7	PITCH ERROR COMPENSATION		
11.8	MANUAL DATA INPUT		
11.9	DRY RUN		
11.10	SINGLE BLOCK		
11.11	PROGRAM PROTECT KEY		
11.12	SELF DIAGNOSTIC FUNCTION WITH SYSTEM & MACHINE ALARM STATUS DISPLAY		
11.13	EMERGENCY STOP		
11.14	MIRROR IMAGE (X & Y CO-ORDINATES)		

Sl. No.	Description	Unit	MCV-400
11.15	TOOL OFFSET MEMORY - 400 NOS.		
11.16	TOOL LENGTH COMPENSATION		
11.17	PART PROGRAM MEMORY 1280 MTS (512K) BATTERY BACKUP		
11.18	CUTTER DIAMETER COMPENSATION		
11.19	CANNED CYCLES FOR DRILLING, TAPPING AND BORING		
11.20	PROGRAMMABLE DATA INPUT (G10)		
11.21	HELICAL INTERPOLATION		
11.22	RIGID TAPPING		
11.23	CLOCK FUNCTION		
11.24	LOOK AHEAD CONTROL		
11.25	READER PUNCHER INTERFACE (RS 232)		
11.26	WORK CO-ORDINATE SYSTEM (G53 TO G59)		
11.27	ANGLE CHAMFERING / CORNER RADIUS		
11.28	CUSTOM MACRO-B		
11.29	CO-ORDINATE SYSTEM ROTATION		
11.30	NUMBER OF REGISTERED PROGRAMS TOTAL 200		
11.31	BACK GROUND EDITING		
11.32	EXTENDED PART PROGRAM EDITING		
11.33	RUN HOUR PARTS COUNT DISPLAY		
11.34	MANUAL PULSE GENERATOR		
11.35	ALARM HISTORY & OPERATION HISTORY		
11.36	SPINDLE LOAD DISPLAY		
12	**MACHINE FEATURES / ACCESSORIES - ADDITIONALLY ORDERED BY M/S WABCO INDIA LIMITED - PLANT-1, CHENNAI.**		
12.1	SPINDLE SPEED OF 80 8000 RPM WITH INDIRECT DRIVE		
12.2	HIGHER SPINDLE POWER 11/7.5 KW IN LIEU OF STANDARD SPINDLE 7.5/5.5 KW		
12.3	FLOOD COOLANT SYSTEM AROUND THE SPINDLE		
12.4	CHIP FLUSHING SYSTEM ON THE BASE		
12.5	COOLANT GUN		
12.6	FRONT AUTO DOOR		
12.7	INCREASED DAY LIGHT AREA (NO INCREASE IN Z STROKE)160MM		
12.8	30 STATION TOOL MAGAZINE		
12.9	HIGH PRESSURE COOLANT THROUGH SPINDLE SYSTEM (16 LPM AT 21 BAR) WITH GRUNDFOSS CENTRIFUGAL PUMP AND BAG TYPE FILTRATION SYSTEM		
12.10	SPECIAL COOLANT TANK FOR CTS APPLICATION		
12.11	FANUC BETA4 MOTOR & DRIVE ALONG WITH 4TH AXIS FEATURE + INTERFACING SUITABLE FOR NIKKEN ROTARY TABLE 202 LFA		
12.12	HYDRAULIC ROTARY JOINT (4 PORTS) FITTED ON ROTARY TABLE (CUSTOMER WILL SUPPLY NIKKEN 202 LFA ROTARY TABLE)		
12.13	WIDER GUARDING		
12.14	ALUMINIUM TANK HYDRAULIC POWER PACK FOR FIXTURE WITH 4 STATION @ MAXIMUM OF 70 BAR & MINIMUM OF 5 BAR FROM REXROTH		
12.15	HYDRAULIC INTERFACING (ONLY ELECTRICAL WIRING) WITH LOOPING ASSEMBLY FOR 4 LINES ON THE MACHINE		
12.16	FIXTURE INTERFACING WITH NECESSARY M-CODES		

ACE MANUFACTURING SYSTEMS LTD.			
PRODUCT SPECIFICATION SHEET			
MODEL: MCV-400			
Sl. No.	Description	Unit	MCV-400
12.17	TPM FRIENDLY MACHINE (SEE THROUGH PANEL, MARKINGS ETC.)		
12.18	PROXIMITY SWITCHES SINGLE (PLUG & SOCKET TYPE) & MULTI SWITCHES BALLUF MAKE IN LIEU OF STANDARD (E)	(E) Except Orientation Rig	
12.19	TELEMECHANIQUE MCCB IN LIEU OF STANDARD (C)	(C) Telmechanique is Now Schneider	
12.20	PUSH BUTTON, SELECTOR SWITCHES, ILLUMINATED PUSH BUTTON SWITCH (LED TYPE) ONLY, ILLUMINATED SELECTOR SWITCH (LED TYPE) ILLUMINATED PILOT LAMP (LED TYPE) ONLY IN LIEU OF OUR STANDARD (P)	(P) illuminated PB (LED Type	
12.21	DATA BACKUP IN FLASH ROM INCLUDING PART PROGRAMS, MACRO PROGRAMS, PARAMETERS AND LADDER BACKUP		
12.22	ELCB IN MAIN POWER CIRCUIT		
	SPECIAL NOTE: EQUIPMENT NOTE ENCLOSED		

V.G. dd 24/11/13

MINUTES OF MEETING REGARDING THE PENDING POINT IN THE EQUIPMENT NOTE

	WABCO REQUIREMENT	wabco remarks	AMS REMARKS	WABCO COMMENTS	Rev AMS Confirmation
1	Temperature detection of electrical/electronics power components(Panel cooler given)	why panel cooler for control panel is required	If difference in temp is more then 5 to 8 deg. Then condensation will take place inside the cabinet which will spoil the electronic components, to avoid this in AMS recommends to use panel cooler		AMS recommends to use panel cooler
2	Tool breakage sensor need to be provided-Omron Make	OMRON MAKE IS REQD	Metrol make contact type tool breakage sensor can be provided with additional price of Rs 70000	We explained OMRON HAS got reliability issue , so we are offering Metrol make, But WABCO will CONFIRM the OMRON model, AMS has to provide the same model and make.	AMS do not recommend OMRON as it is having reliability issue, so we are offering Metrol make, M/s. Wabco to give model number of OMRON if same to be used
3	Location of all pressure gauges at front side of machine(place the all gauge left or Right side middle portion)	Fix all pressure gauges at front side facing to operator	No	Pressure gauge should be kept at eye contact level.AMS HAS TO CLARIFY THE NUMBER OF PRERSSURE GAUGES TO BE KEPT IN THE FRONT	Filter & lubrication units mounted either side of the machine at bottom front side of the machine. Hydraulic related hydraulic valves can be given at front side at the top of the machine hence visible to operator.
4	Routing of cables & wires (All cables must in proper routing ensure the safety) Cable should not enter the top of electrical cabinet	M.S Cable duct / PVC Cable duct	Only transformer cable is taken from top with gland and metal cover	Wabco want transformer cable also to be routed in the side or bottom side of the machine.	Only transformer cable is provided in addition other cable are provided on the bottom and side of the machine. Transformer cable also covered with metal gland and cover.
5	Colour identification for DC and AC Circuits	Use Blue colour wire for DC and red colour for AC CircuitsAnd Black for 3Phase	Neutral of 220 vac is used black wire	OK	OK
6	Hydraulic power pack not with intergral system, individual pressure switch and valves with flow control.		Direction control valve, Flow control valve, individual pressure switch will be provided	Our old power pack design should be supplied (Not Intermittent power pack as applied recently)	OK
7	hydraulic unit/coolant motor/hydraulic motor should be in required color (Refer attached photo) Tank should be in aluminium and temp control and oil cooler		Air cooler for power pack & MS Tank	ALUMINIUM TANK Should be PROVIDED Tpm feautures to be confirmed by AMS,TEMPERATURE CONTROL BY SENSOR AND IT WILL CUT OFF IF THE SET TEMP EXCEEDS, AIR COOLER should BE PROVIDED FOR OIL COOLING	OK Rexroth make 70 bar power pack
8	Coolant tank top cover must be in aluminum checker plate	[illegible]	not possible	DISCUSSED WITH Mr.Umesh and confirmed can be given. intermediate enforcement should be taken care	OK
9	Extra plug and socket with switch should be provided In Electrical Cabinet for any other purpose		That should be protected with cover	we confirmed it CAN BE GIVEN AS PER WABCO REQUEST. BUT SOCKET WILL NOT BE PROVIDED IN THE ELECTRICAL PENDENT. Wabco also agreed	OK
10	All the manuals must be in soft copy in CD And Hard copy: ie, partprogramme backup, machine logic backup,electrical manual, assy manual, hydraulic circuit, pneumatic circuit, lubrication circuit,geomentrical parameter, Machine calibration certificates.		only electrical connection diagram will be given in hard copy.	OK DISCUSSED WITH Mr.Umesh and confirmed canbe given	OK

ACE MANUFACTURING SYSTEMS LTD

MACHINE SEQUENCE RUNNING TEST REPORT

AMS® In Pursuit of Excellence

MACHINE NUMBER :

Preconditions for Machine before starting Machine Sequence Running

1. Machine should be fitted with all accessories as per Work Order
2. Machine should be cleared from all Electrical & Mechanical Remarks
3. Machine should be run with Sufficient Coolant.
4. ATC should be full with tool holders
5. No Bypass of any Safety switches and cabinet locked.

Date	Time	No of Hours Running	Date	Time	No of Hours Running	Remarks / Alarm Observed
21/11/13	6PM	1 ✓			17 ✓	
		2 ✓			18 ✓	
		3 ✓			19 ✓	
		4 ✓			20 ✓	
		5 ✓			21 ✓	
		6 ✓			22 ✓	
		7 ✓			23 ✓	
		8 ✓			24 ✓	
		9 ✓				
		10 ✓				
		11 ✓				
		12 ✓				
		13 ✓				
		14 ✓				
		15 ✓				
		16 ✓				

Nature of Remarks Related to	1.Hydraulic leakage	2. Coolant leakage	3.Electrical alarm	4. ATC	5. Spindle	6. Axes	7. Others

Note : In case M/c is stopped due to any reason, write the Sl. No. of the above remarks against corresponding number of hours, set right the M/c & continue to run for a total period of 24 hrs.

In charge - Mechanical

V. Gadde

In charge - QA

ACE MANUFACTURING SYSTEMS LTD

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SPINDLE TEMPERATURE STABILIZATION REPORT

M/c. NUMBER : MH-932	MAX. RPM : 8000	DRIVE: ID/DD/IDCTS/DDCTS
SPINDLE SL No. : 13966	PULLEY RATIO : 1:1 / 1:1.25 / 1:1.5 /1:1.75 / 1:2	

Note : Refer TM-QA-07 for test procedure. **Recommended Temp. Rise : 20°C max. above ambient**

% of current drawn at max RPM of the spindle : (Recommended max : DD : β3 = ≤ 15%, β6 / α6 & above = ≤10%. ID : β3 = ≤ 20%, β6 / α6 & above = ≤15%.	Measured = 11 %
Spindle noise at max RPM [Recommended : 80 dB(A) upto 10 kW, 85 dB(A) for > 10 kW]	Measured = 79 dB (A)

SPINDLE RPM	TIME	AMBIENT TEMPERATURE	SPINDLE TEMPERATURE
500	12:00	26°C	
8000	12:30	26°C	30°C
	01:00	27°C	33°C
	01:30	27°C	35°C
	02:00	27°C	36°C
	02:30	27°C	37°C
	03:00	28°C	38°C
	03:00	28°C	38°C
	03:30	28°C	39°C
	04:00	28°C	39°C

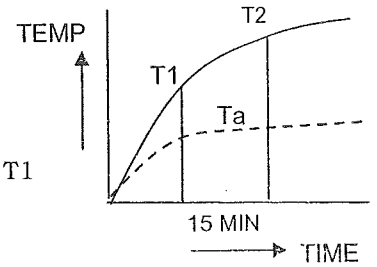
1. The Stabilization point, Calculated as follows.

Is the temperature at which the rate of rise over 15 minutes is less than 5%

= [(T2 – T1) / (T1 –Ta)] X 100

= < 5%

T1 = Previous Temp., T2 = Temp. after 15 Min & Ta = Ambient Temp. at T1



2. RUN SPINDLE AFTER STABILIZATION FOR 3 HOURS.

11/11/13 Date	Checked By

ACE MANUFACTURING SYSTEMS LTD	AMS® In Pursuit of Excellence
MACHINE HEALTH CHECK POINTS	

MACHINE NUMBER : M11-932 **DATE :** 20-11-13

I	AXIS	
1	Ensure plugs are fitted for L M Guide ways.	✓
2	Ensure the lube timer is set 12 min for HG and 20 min for LM Guide ways.	✓
	In Grease Lubrication lube timer is set to 6 Hours.	— N.A —
3	Ensure leveling bolts are fixed with locknut.	✓
4	Check the Table tenon slot 14H8 / 18H8 (Tolerance 0 to +0.027 mm)	✓
5	Ensure butt reference function.	✓
6	Check all the axes reference arrows are matched.	✓
7	Check all Emergency push button function of Machine / APC / Double Hand	✓
8	Ensure marking of all screws with Yellow paint marker	✓
9	Ensure cover fixed to Index table, rotary joint & tool breakage sensor	— N.A —
10	Ensure smooth movement of sliding door without play and unhealthy noise	✓
11	Ensure effective function of Ball catch assembly on sliding doors.	✓
II	**SPINDLE**	
1	Ensure air coming through all air jet (4/8 holes) at oriented spindle position	✓
2	Ensure supply of air to spindle Air Purge system	— N.A —
3	Check the tightness of push stud.	✓
4	Ensure grub screws are fitted to spindle flange.	✓
5	Check spindle orientation position and note down the value. (Parameter No. : 4031 for Fanuc controller, ... for Siemens controller)	2861
6	Ensure spindle balancing is completed	✓
7	Ensure presence of gap between draw bar bush & ejection screw / roller (visually or by feel)	✓
8	Check the "Tool not in spindle" proximity message.	✓
III	**ATC**	
1	Ensure ATC / APC / Counter balance bracket pinning.	✓
2	Ensure oil level is max. in ATC gear box.	✓
3	Check the Tool Number display on panel board is same as in spindle	✓
4	Check tool change position limit switch for proper functioning & band width (Max. 2.5 mm on each side).	✓
5	Ensure Proper working of Tool Ejector in HSK Magazine	— N.A —
6	Ensure ATC Auto door smooth function	— N.A —
7	Ensure Magazine rotates clockwise direction in manual mode	✓
8	Ensure recovery of ATC function during power failure / Emergency	✓
9	Ensure all Proximity switch & Pinch clamp mounting screws tightness to be checked and paint with Yellow marker	✓
	N. Gr-d CHECKED BY	1...4 PAGE NO

ACE MANUFACTURING SYSTEMS LTD MACHINE HEALTH CHECK POINTS		AMS® In Pursuit of Excellence
IV		COOLANT
1	✓	Ensure no coolant leakage around the machine & No presence of coolant in lubrication oil drain pipe
2	✓	Ensure proper working of float switch (by shifting the sensor upwards).
3	✓	Check function of Flood coolant (M7), Chip push (M182), Base flush (M181), CTS (M52) & Coolant gun (through selector switch).
4	NA	Ensure drum wash in conveyors (Optional)
5	NA	Check conveyor overload message & Direction (Optional)
6	NA	Check oil in conveyor gear box. (Optional)
7	NA	Ensure functioning of screw conveyor and chip conveyor in forward & reverse directions (through selector switch).
8	✓	Ensure CTS flow switch setting by opening the bag filter air vent ball valve (Optional)
9	✓	Ensure suction strainer & Check Valve in CTS tank & pr. gauge functioning (Optional)
10	NA	Ensure CTS pin locator hole is plugged (Optional)
11	✓	Ensure that twin arm plunger & CTS pin locator screw do not foul (Optional)
12	✓	Ensure bag filter clog message by closing bag filter outlet ball valve (Optional)
13	NA	Ensure High pressure air blast after CTS switched off (Optional)
V		APC
1	NA	Check pallet initialization
2	NA	Ensure supply of air jet to pallet rest pad cleaning.
3	NA	Ensure Pallet clamp seat check and arm seat check signal
4	NA	Ensure pallet change takes place smoothly with specified load (as per work order) without abnormal / unhealthy noise
5	NA	Ensure Auto door smooth function
6	NA	Ensure Clamp line pressure relief valve
VI		HYDRAULIC, PNEUMATIC & LUBRICATION
1	✓	Ensure air pressure switch is working properly.
2	✓	Ensure proper setting of oil (1 drop / Two tool change) in FRL unit & Oil Level low message
3	✓	Check the FRL drain hose is fixed.
4	✓	Ensure identification name plates are fixed on power pack hoses.
5	✓	Check the Hyd.oil level low message (by adjusting sensor position) & Pr. gauge function
6	✓	Ensure hydraulic flow control valve grub locking in APC machine.
7	✓	Ensure no oil leakage in hydraulic valves & fittings
8	✓	Ensure lubrication pressure switch is set to 15 bar. \ 10 Bar for Grease lubrication
9	NA	Ensure pneumatic quick change coupling fixed to 4th axis provision
10	✓	Check Hydraulic specification is as per circuit diagram
VII		GENERAL
1		Check all the points of MOM are attended.
2	✓	Ensure coolant tank capacity engraved or marked in name plate
3		
4		

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MACHINE HEALTH CHECK POINTS

MACHINE No. : M11-932 ASSEMBLY TECHNICIAN : Mr Thejas

S.No	REMARKS OBSERVED	STATUS	SIGN
1	Aluminium power pack 4 station to be	✓	✓
2	Hydraulic looping 4 lines	✓	✓
3	Cartridge dummy plate mtg	✓	✓
5	Sadle dummy holes to be closed	✓	✓
6	All Pressure gauges and FRL unit to be mtd at front side of the mc	✓	✓
7	Mc base guard mtg.	✓	✓
8	FRL drain hose to be fixid		
9	Axies butt refereennce valves are not matching with internal check list valve.	✓	✓
10	All Screws tightness to be checked and mark with yellow marker.	✓	✓
11	Customer TPM check list points to be attended	✓	✓
12	Coolant float switch setting to be done	✓	✓
13	Base build, Ball screw, Power pack Spindle assy reports to be filed.		✓

Date	ASSEMBLY IN CHARGE	CHECKED BY	PAGE No.
			3...4

F-QA-FA-09-R2

ACE MANUFACTURING SYSTEMS LTD.

MACHINE HEALTH CHECK POINTS

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MACHINE No. :

ASSEMBLY TECHNICIAN :

S.No	REMARKS OBSERVED	STATUS	SIGN
1)	PCB 107 to be fixed on DIN Rail	✓	
2)	Sticker for X26 Board to be put	✓	
3)	Tag For R.T Drive thermal Cable	✓	
4)	Cable Routing - inside Cable Carrier bracket	✓	
5)	Hyd thermal Sw wiring	✓	

Date

ASSEMBLY IN CHARGE

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PAGE No. 4...4

F-QA-FA-09-R2

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - MECHANICAL

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MACHINE NUMBER : M11-932 **DATE :** 20/11/13

1	ATC Details : Model	ISO	4030V
		HSK	A50 20V/A50 24V/A50 30V/A63 20V/A63 24V/A63 30V/A63 40V/A100 20V/A100 24V/A100 30V
1.1	ATC : Turret Type/Twin Arm	Sl.No	19458
1.2	ATC Torque Motor	TM 50 / 63 / 80 / 100 / 125 / Servo ATC	
2	**COOLANT SYSTEM DETAILS :**		
2.1	Make : Rajmane Industries	Model	Type
	Flood Coolant Pump - Crown/Flexinox/Disc Nozzle	RKM 306 - 206 / RV 160 - 270	Centrifugal Pump
2.2	Chip Pushing Coolant Pump	RKM 206 / RV 160 - 270	
2.3	Base Flushing Coolant Pump	RKM 206 / RV 160 - 270	
2.4	Coolant Gun & Drum Wash Pump	RKM 206 / RV 160 - 270	
2.5	Arm Cleaning N.A		
2.6	Shower N.A		
2.7	Oil Skimmer - Make / Model N.A		
2.8	High Pressure Coolant Pump	Make	GF/ WP / BP
		Model A96514892P	11312
2.9	Pressure Relief Valve	Make	Polyhydron
		Model	DPRH-10T-50-11
2,10	Rotary Joint	Make	Deublin / RIX
		Model	
2.11	Chiller unit Details	Make :	Sl. No:
3	**FLOAT SWITCH :**		
	Make	Filpro Electronics	
4	**CHIP CONVEYOR DETAILS :**		
4.1	Type of Conveyor	Scraper / Slat cum Scraper / Magnetic along X / Y / Z - axis	
4.2	Model	N.A	
4.3	Make	Universal Conveyor / SRV / Filcon	
4.4	Sl.No		
5	**CONVEYOR GEAR BOX DETAILS :**		
5.1	Model	N.A	
5.2	Make	Universal Conveyor / SRV / Rotomotive	
6	**FILTER SPECIFICATIONS :**		
6.1	Model / Working pressure	8/15 / 50 bar / 100 bar	
6.2	Make	Universal Conveyor	
6.3	Sl.No	4538 & 4539	
7	**PAPER BAND FILTER SPECIFICATIONS :**		
	Make	Sl No. N.A	
	Model		
8	**COIL / SCREW CONVEYOR DETAILS :** Qty : 1 / 2		
8.1	Make N.A	Universal Conveyor	
9	PRESSURE SWITCH FOR PNEUMATIC SYSTEM		
9.1	Operating Range	2 to 15 Kg /cm2	
9.2	Make	SUCO / SMC Digital	
9.3	Quantity	1 +1	

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MACHINE HISTORY - MECHANICAL

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10	**HYDRAULIC SYSTEM :**			
10.1	Make		Pascal / Hyd Tech / Rexorth / Yuken /Jai Chamundi /Hymech	
10.2	Product No		QC. No : AP 9280	
10.3	Tank Capacity (Liters)		32 / 40 / 63 / 70 / 100	
10.4	Maximum Working Pressure		50 / 70 / 150 Bar	
10.5	HYDRAULIC PUMP	Make Rexroth / Yuken Corotech / Vickers	Model	
	MOTOR	Rotomotive / Siemens	90L-4-1400 / 90S-4-1390 / 100LA-4	
11	**PRESSURE SWITCHES FOR HYDRAULIC SYSTEM :**			
11.1	Make		Polyhydron / Yuken	
11.2	Model		1PS 30-50 / 30-100 / 10 - 30 / 30-350	
11.3	Quantity		04	
12	**HYDRAULIC ACCUMULATOR :**			
12.1	Make	FOX / BOSCH / HYDAC	FOX / BOSCH / HYDAC	
12.2	Set Oil Pressure	Tool de clamp :	Counter balance:	
12.3	Gas Pressure	Tool de clamp :	Counter balance:	
12.4	Volume (Litres)	Tool de clamp : 1.4	Counter balance : 4 or 10	
13	**HYDRAULIC CYLINDERS :**			
	Make	Model	Function	Qty
13.1	Island Hydraulics / Preac	Ø 40 x 22 L - 125/152/180/1150	Pallet Forward / Reverse	
14	**FRL UNIT**			
	Make	Model	Operating Range	Grade of Oil
	Shavo / Festo / Norgan / Rexroth	MINI / MIDI	0-16 Kg/cm^2	Servo Way Spin 12
15	DRYER	Make		Specification
		Trident / Norgan		VXD - 2
16	**TOOL DECLAMP SYSTEM :**		Mechanical / Pneumatic / Hydraulic / Hydro-Pneumatic	
17	**TURRET / INDEXING TABLE (OPTIONAL)**			
	Make	Model	Sl.No	No of Station
	Pragathi / AMS			8 /12 /
18	**ROTARY TABLE**	Make NIKKEN / UCAM	Model	Sl.No
19	**TAIL STOCK / TRUNION SUPPORT**	Make UCAM	Model	Sl.No
20	**COUNTER BALANCING**			
	Pneumatic		**Hydraulic**	
20.1	Make	SMC / Rotex / Festo	Island Hydraulic	
20.2	Cylinder Diameter :	Ø 80 / 100 / 125	Ø 50 X 32 X 670	
20.3	Air Reservoir used	Yes / No		
	ADDITIONAL IF ANY			

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CHECK LIST - MECHANICAL

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MACHINE NUMBER : M11-932 **DATE :** 18-11-13

I Vertical axis drop during power off : —

' - ve' direction < 0.005 mm	'+ ve' <0.015mm direction
System Reading : 0.001	System Reading. : 0.007
Dial Reading. : 0.002	Dial Reading. : 0.007

II % of Axis current drawn (Recommended as per Data sheet DS -QA -02)

A	Axis	25 % Rapid Rate	50% Rapid Rate	100% Rapid Rate	600 mm/m Feed rate
	X	14%	17%	50%	10%
	Y	18%	22%	48%	16%
	Y / Z Up	57%	59%	68%	55%
	Y / Z Down	36%	34%	32%	38%
	B - Forward				
	B - Reverse				

B % of Axis holding current (Recommended as per Data sheet DS -QA -02)

Axis	100% Rapid Rate		2000 mm/min. Feed Rate	
	Motor end	Brake end	Motor end	Brake end
X	07%	08%	00%	00%
Y	21%	23%	01%	03%
Z	54%	39%	45%	46%

C Over travel against Axis dead stopper in mm (Recommended as per Drawing)

Axis	Brake end	Motor End
X	18.376	28.833
Y	15.411	16.610
Z	12.902	19.718

III **SPINDLE**

	Push stud stroke length for Tool ejection (mm)	Recommended Values	Measured Value
A	BT – 30/40 /50	Refer DS-DSN-PROD-01	0.75 MM
	HSK A50/63/100		—
B	Tool clamping force-Kgf	Refer DS-DSN-PROD-01	840 Kgf

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C	Axial play in the spindle (Ref. TM-QA-10 for procedure) (Recommended max.except BT50 : 0.005 mm , for BT 50 : 0.008 mm)		
	1. Dial Reading. 'R1' (Ref. Step D)	0 ⟶ +0.042 ⟶ 0	Difference = 0.002 mm (R1~R2)
	2. Dial Reading. 'R2' (Ref. Step G)	0 ⟶ -0.032 ⟶ 0.030	
	1. Dial Reading. 'R1' (Ref. Step D)	0 ⟶ + N.A ⟶	Difference = mm (R1~R2)
	2. Dial Reading. 'R2' (Ref. Step G)	0 ⟶ N.A ⟶	

IV	Turret type ATC	
A	Check the spring lever compression at :	
	1. After the magazine butts against the dead stopper (mm)	— N.A —
	2. Z – axis position below zero position (mm)	— N.A —
	3. Recommended : BT 40 - 2.5 to 3 mm , BT 30 - 1.5 to 2 mm (i.e. 1-2)	— N.A —
B	Tool picking position for Turret Type ATC Recommended : BT 40 - 60 to 80 mm , BT 30 - 50 to 65 mm	— N.A —

V	A.P.C	
A	Pressure setting for the pallet transfer cylinder in kg/cm^2 (As per circuit diagram)	— N.A —
B	Pressure setting for the pallet clamp-de clamp in kg/cm^2 (As per circuit diagram)	— N.A —

VI	Cycle Time in seconds (Refer Work order for recommended value)	
	1. Check the tool to tool time	2.00 Sec
	2. Check the chip to chip time with Z axis only	3.70 Sec
	3. Check the chip to chip time with all axes	4.55 Sec
	4. Check the pallet changing time	— N.A —

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CHECK LIST - MECHANICAL

M/c. NUMBER :

DATE:

1. MACHINE LEVEL READING :

Along X-Axis

Along Y-Axis

2. TABLE TOP READINGS:

Along X-Axis

Along Y-Axis

PALLET - II

Along X-Axis

Along Y-Axis

3. TENON SLOT READING :

PALLET - II

4. MANDREL READINGS:

Front

Side

RUNOUT:

a. TOP : (0.010mm)

1.

b. BOTTOM :(0.020mm)

1.

2.

3.

4.

5. TURN AROUND READING:

6. MANDREL READINGS:

Front

Side

RUNOUT:

a. TOP : (0.010mm)

1.

b. BOTTOM :(0.020mm)

1.

2.

3.

4.

7. TURN AROUND READING:

NOTE: 1. For recommended value refer Design Data sheet DS-DSN-PROD-TC 01-06.

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CHECK LIST - MECHANICAL

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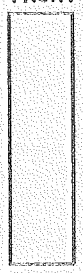
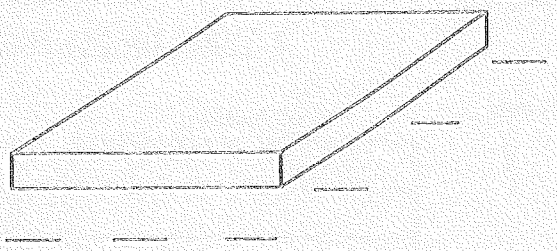
8.COLUMN READINGS :

FRONT



SIDE

9. SQUARENESS READING (X & Y):



10. PALLET HEIGHT VARIATION:

Pallet-1: _____

Pallet-2 : _____

11. VARIATION IN HEIGHT OF GAUGE PLANE OF SPINDLES:
w.r.t TABLE/PALLET SURFACE (MEASURED ON MANDREL FACE)

12. PARALLELITY OF SPINDLES PLANE w.r.t X-AXIS SIDE MOVEMENT:

13.VARIATION IN CENTER DISTANCE BETWEEN SPINDLES:

NOTE: 1. For recommended value refer Design Data sheet DS-DSN-PROD-TC 01~06.

CHECKED BY

nagaraj bhat.

From: Balasubrahmanya I G.
Sent: Monday, November 11, 2013 1:24 PM
To: lakshmikanth nr.; Guruprasad.A.R; shashidhara k. aithal; Suresh V.; nagaraj bhat.
Cc: Ganesh S.; Ramprakash H S.; maruthesh pt.
Subject: RE: BALLUFF PXS SENSORS

Sir,
Since BALLUFF proximity sensors are coming with 52mm thread length, it is not possible to use Tool Orientation proximity in Wabco machine.(MCV-400).Tool declamping bracket to be machined and then used for Future machines.

-----Original Message-----
From: lakshmikanth nr.
Sent: 17 October 2013 15:28
To: Guruprasad.A.R; Balasubrahmanya I G.
Subject: BALLUFF PXS SENSORS

Dear sir,

Please check and confirm the same.

With Regards,
LAKSHMIKANTH.NR
DESIGN-ELECTRICAL
ACE MANUFACTURING SYSTEMS LTD
BANGALORE.560058
Mobile no- 9008424155,9481618545
E-mail-lakshmikanthnr@amslindia.co.in

-----Original Message-----
From: yathishhr
Sent: Thursday, October 17, 2013 3:11 PM
To: lakshmikanth nr.
Subject: FW: REQUIREMENT OF PXS SENSORS

Dear Lakshmi,

Pls. go through the attachment & confirm the same.

REGARDS,
Yatheesh H.R
Executive –Ele.purchase
Ph.No. 9880189809
ACE MANUFACTURING SYSTEMS LIMITED.
Web :- www.amsl.in

FONT LISTS

PCL Internal Fonts

Font Name	Symbol	spacing	Style	StrokeW.	Pitch/Point Typeface	Sample
Courier					scalable	
	0		0	0	4099	ABCDEFGHIJKLMNOPQRSTUVWXYZ
CG Times					scalable	
	1		0	0	4101	ABCDEFGHIJKLMNOPQRSTUVWXYZ
CG Times Bold					scalable	
	1		0	3	4101	ABCDEFGHIJKLMNOPQRSTUVWXYZ
CG Times Italic					scalable	
	1		1	0	4101	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
CG Times Bold Italic					scalable	
	1		1	3	4101	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
CG Omega					scalable	
	1		0	0	4113	ABCDEFGHIJKLMNOPQRSTUVWXYZ
CG Omega Bold					scalable	
	1		0	3	4113	ABCDEFGHIJKLMNOPQRSTUVWXYZ
CG Omega Italic					scalable	
	1		1	0	4113	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
CG Omega Bold Italic					scalable	
	1		1	3	4113	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
Coronet					scalable	
	1		1	0	4116	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
Clarendon Condensed					scalable	
	1		4	3	4140	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Univers Medium					scalable	
	1		0	0	4148	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Univers Bold					scalable	
	1		0	3	4148	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Univers Medium Italic					scalable	
	1		1	0	4148	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
Univers Bold Italic					scalable	
	1		1	3	4148	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
Univers Medium Condensed					scalable	
	1		4	0	4148	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Univers Bold Condensed					scalable	
	1		4	3	4148	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Univers Medium Condensed Italic					scalable	
	1		5	0	4148	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
Univers Bold Condensed Italic					scalable	
	1		5	3	4148	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
Antique Olive					scalable	
	1		0	0	4168	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Antique Olive Bold					scalable	
	1		0	3	4168	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Antique Olive Italic					scalable	
	1		1	0	4168	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
Garamond Antiqua					scalable	
	1		0	0	4197	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Garamond Halbfett					scalable	
	1		0	3	4197	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Garamond Kursiv					scalable	
	1		1	0	4197	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*
Garamond Kursiv Halbfett					scalable	
	1		1	3	4197	*ABCDEFGHIJKLMNOPQRSTUVWXYZ*

51506

FANUC Series 0i Mate-M Model D　　DATA SHEET　　1/2

得意先名:			
CUSTOMER:	FANUC INDIA PRIVATE LIMITED	DATE OF MANUFACTURE:	2013-07
CONTRACT NO.:	YD03818- 000	DATE OF TEST	:2013/07/31
MODEL NO.:	A02B-0321-B500	CHECKED BY	:A. Koike
SERIAL NO.:	E137G2690	APPROVED BY	: K.Miura
SETTING OF INPUT:			

CNC IDENTIFICATION NUMBER

3C7B5D01
0C91D6A0
5E34C389
7C92C421

EDITION OF PRINTED BOARD & ROM

NAME	EDIT	NAME	EDIT	NAME	EDIT
A20B-8201-0088	01A	A02B-0319-H500#60W3	05		
		A02B-0319-H580#40B0	09		
		A02B-0319-J541#60VM	07		
		A02B-0319-J571#658E	06		
		A02B-0319-J572#658F	05		
		A02B-0319-J575#659B	03		
		A02B-0319-J579#BW50	03		
		A02B-0320-J578#DX50	03		
		A02B-0320-J578#DX51	03		
		A02B-0322-H501#D5F1	29		
		A06B-6057-H507#90C5	06		
		A06B-6057-H507#90C8	04		

LIST OF CARD , DIMM & BACK BOARD

A20B-3300-0768	1		
A20B-3900-0286	1		
A20B-8002-0703	1		
A20B-8200-0560	1		

本データシートはファナック工場出荷時のCNC装置の装置情報です
Information contained in this Data Sheet is system configuration at the time of FANUC factory shipment.

NAME	NAME	SPECIFICATION	ORDR
BASIC CONTROL UNIT	ﾍﾞｰｼｯｸ ﾕﾆｯﾄ	A02B-0321-B500	1
AXIS CONTROL CARD A2 FROM/SRAM MEMORY A1 POWER SUPPLAY UNIT MAIN BOARD DISPLAY UNIT NAME PLATE	ｼﾞｸ ｾｲｷﾞｮ ｶｰﾄﾞA2 FROM/SRAMﾒﾓﾘ A1 ﾃﾞﾝｹﾞﾝ ﾕﾆｯﾄ ﾒｲﾝﾎﾞｰﾄﾞ ﾋｮｳｼﾞ ﾕﾆｯﾄ ﾒｲﾊﾞﾝ	A02B-0319-H032 A02B-0319-H051 A02B-0319-H111 A02B-0319-H125 A02B-0319-H164#M A02B-0322-H190	1 1 1 1 1 1
PACKAGE 3 CUSTOM SOFTWARE SIZE 512K BYTE MACRO EXECUTOR CONTROLLABLE AXES EXPANSION	ﾊﾟｯｹｰｼﾞ 3 ｶｽﾀﾑｿﾌﾄ ﾖｳﾘｮｳ 512Kbyte ﾏｸﾛ ｴｸﾞｾﾞｷｭｰﾀ ｾｲｷﾞｮｼﾞｸｽｳ ｶｸﾁｮｳ	A02B-0322-H532 A02B-0322-J738#512K A02B-0322-J888 A02B-0322-R689	1 1 1 1
I/O UNIT FOR 0i *MANUAL PULSE GENERATOR	I/O UNIT *ｼｭﾄﾞｳ ﾊﾟﾙｽ ﾊｯｾｲ	A02B-0309-C001 A860-0203-T001	1 1
DIGITAL SERVO UNIT DISCHARGE RESISTOR AC LINE FILTER	ﾃﾞｼﾞﾀﾙｻｰﾎﾞﾕﾆｯﾄ ｶｲｾｲﾃｲｺｳﾕﾆｯﾄ AC ﾗｲﾝ ﾌｨﾈﾀｰ	A06B-6130-H002 A06B-6130-H402 A81L-0001-0083#3C	1 1 1
	SPINDLE AMP.	A06B-6164-H343#H580	1
	BIS4/4000 BIS12/3000 BIS12/3000 BIS22B/3000 SPB6/100001	A06B-0063-B003 A06B-0078-B103 A06B-0078-B103#0100 A06B-0082-B403 A06B-1445-B103	1 1 1 1 1

ACE MANUFACTURING SYSTEMS LTD

AMS — For Pursuit of Excellence

CABINET ENERGIZING REPORT

Route card No.: 392	Date: 11/10/13
Sales order No.: 3624	M/C No.: MM-932

WIRING OPTIONS

DESCRIPTION		CHECKPOINTS
STANDARD MACHINE	☑	STD CLNT,
CHIP CONVEYORS	☐	SCRAPER, SLAT, SCREW, HC CONNECTOR,
APC	☐	I/O CARD, RELAY BOARD, EI/O3CABLE, TB1 TERMINATION,
ROTARY TABLE	☐	SYSTEM OPTION, DRIVE INTERFACE, MS CONNECTORS,
INDEX TABLE	☑	MPCB, CONTACTOR, SSR, ACR3,
FIXTURE	☑	I/O CARD, RELAYBOARD, EI/O2 CABLE, TB2 TERMINATION,
SPECIAL COOLANT	☑	F9, F7, F4
	☐	
	☐	
	☐	

SHORT CIRCUIT TEST

Test to be done with the help of continuity tester/ multimeter/ megger .

1)BETWEEN PHASE & EARTH: No	2)BETWEEN PHASE & NEUTRAL: No
3)BETWEEN TWO PHASES: NO	4)BETWEEN 24V & EARTH: NO

VOLTAGE TEST

Max. allowed tolerance For 415V ac - ±15V, For 220/200V ac - ±10V, For 24V dc- ±1V,

1)INCOMING MAINS: 412.8V	2)SINGLE PHASE SUPPLY:
3) TFR 1 SECONDARY: TAPPING1: 222.7V	TAPPING2: 221.4V
4) TFR 4 SECONDARY: TAPPING1:	TAPPING2: 223.7V
5)REG.POWER SUPPLY(PS1)O/P: 24.09V DC	5)REG.POWER SUPPLY(PS2)O/P: 24.11V DC

AXIS CURRENT TEST

Check with MPG feed, Check for noise& current less than 10%

1)X -AXIS -	2)Y -AXIS -
3)Z -AXIS -	

SYSTEM LOADING DATA

Please refer data sheef for running version Nos.

1)LOGIC VER No.: TMA01-02-08-M00	2)MACRO VER No.: 2,T1.01
3)O9001 VER No.: A2.2.03	4)O9010 / O9013 VER No.:
5)O9003 VER No.: C1.A.02	6)O9011 / O9012 VER No.:
7)O9002 VER No.: B2.2.02	8)O8020 VER No.:

ACE MANUFACTURING SYSTEMS LTD

CABINET ENERGIZING REPORT

In Pursuit of Excellence

CHECKPOINTS

SL No.	Description	status
1	Cabinet colour - As per work order	YES
2	Panel layout - As per layout diagram	YES
3	Ferruling- As per circuit diagram	YES
4	Identification stickers /name plates/ tags - As per circuit diagram	YES
5	Lug Crimping - Physical check	YES
6	Terminal tightness - Physical check for loose connections	YES
7	Ratings of main transformer& related contactor,mcb - As per circuit diagram	YES
8	Ratings of other mcb's like TFR4 control & 230v 1ph- As per circuit diagram	—
9	Ratings of mpcb's& contactors - As per circuit diagram	YES
10	Cabinet door setting- check for lsw setting, Beading, locking,allignment	YES
11	Health check of elements- For physical damage, energization, mounting/ fixing	YES
12	Cabinet mounting - Check for Screw tightness, rubber beading if required	YES

ADDITIONAL CONTENTS WITH CABINET

1)MACHINE HISTORY	☑	2)PRE WORK ORDER	☑
3)M/C No. PLATE	☐	4)PVC CHANNEL COVER	☑
5)PENDENT BEADINGS	☐	6)SPLIT PLATE	☑
7)SPINDLE POWER,FAN,F/B CABLE	☑	8)	☐
	☐		☐

SHORTAGE/ PENDING WORK DECLARATION

SL No.	DESCRIPTION	ACTION PLAN
1	22. Resister Yet clarify	
2	main S/W (Spical) telemechanique	☑
3		
4		
5		
6		
7		

CHECKED BY:- KS/WS

VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS® In Pursuit of Excellence

M/C NO.: M11-732	DATE: 15/10/13
MODEL: MCV-1000	CUSTOMER: WABCO INDIA LIMITED.

M/C BACKUP (PATH) : \\arya3\E-Backup-II\ELECTRICAL MACHINE DATA\AMS DISPATCH MACHINE BACKUPS (20013-14)\FANUC 11.09

CNC SYSTEM

0i-MD ☐ 0i-MATE-MD ☑ ☐

PART No A02B-0321-B500

Sl.No.: ~~E0B[illegible]~~ E137G2690 ✓

SYSTEM OPTIONS:

β SERVO DRIVE SVPM

7.5i (20/20/40) ☐ 15i (40/40/80) ☑

PART No A06B-6164-H343

Sl.No.: V13744053 #H580

α SERVO DRIVE PSM

11i ☐ 15i ☐ 26i ☐ ☐

PART No A06B-[ ][ ][ ][ ]-[ ][ ][ ][ ]

Sl.No.:

α SERVO DRIVE SPM

11i ☐ 15i ☐ 22i ☐ ☐

PART No A06B-[ ][ ][ ][ ]-[ ][ ][ ][ ]

Sl.No.:

α SERVO DRIVE SVM2

40/40i ☐ 80/80i ☐ 160/160i ☐

N/A

PART No A06B-[ ][ ][ ][ ]-[ ][ ][ ][ ]

Sl.No.:

α SERVO DRIVE SVM1(1'st)

40i ☐ 80i ☐ 160i ☐

PART No A06B-[ ][ ][ ][ ]-[ ][ ][ ][ ]

Sl.No.:

α SERVO DRIVE SVM1(2'nd)

40i ☐ 80i ☐ 160i ☐

PART No A06B-[ ][ ][ ][ ]-[ ][ ][ ][ ]

Sl.No.:

FANUC I/O MODULE / I/O CARD

I/O MODULE WITH MPG ☐	I/O CARD WITH MPG ☐	I/O CARD WITHOUT MPG ☑	2 Nos
PARTNo.: A02B-0309-C001	PART No.: A20B-2002-0520	PART No.: A20B-2002-0521	
Sl No. E137D2370	Sl No.	Sl No.	QTY;

AC REACTOR1

0.11mH/62A ☑ 0.07mH/110A ☐ ☐

PART No A81L-0001-0156

Sl.No.: RT134239

CHECKED BY:- [illegible]

VERIFIED BY:- [illegible]

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS® In Pursuit of Excellence

SERVO MOTOR - X AXIS

β8/3K ☐ β12/3K ☑ β22/3K ☐ ☐

PART No.: A06B-0078-B103 #100

Sl.No.: ~~C137W2434~~ C137W1205

SERVO MOTOR - Y AXIS

β8/3K ☐ β12/3K ☑ β22/3K ☐ ☐

PART No.: A06B-0078-B103

Sl.No.: ~~C137W1205~~ C137B2434

SERVO MOTOR - Z AXIS

β12/3K ☐ β22/3K ☑ β22/2K ☐ ☐

PART No.: A06B-0082-B403

Sl.No.: C137A6BC7

B AXIS DETAILS

ROTARY TABLE ☑ INDEX TABLE ☐ ☐

DRIVE: βiSV-20 ☑ βiSV-40 ☐

PART No.: A06B-6130-H002

Sl.No.: 91380146 3

MOTOR: β4/4K ☑ β8/3K ☐

PART No.: A06B-0063-B003

Sl.No.:

REACTOR: A81L-0001-0083#3C (0.15mH/24A) ☐

Sl No.: T13060330

DISCHARGE RESISTOR: 16Ohms ☐ 30Ohms ☑

N/A

PMM AXIS DETAILS

SERVO ARM ☐ SERVO MAG ☐ TILTING TABLE ☐

DRIVE: βiSV-20 ☐ βiSV-40 ☐

PART No.: A06B-____-H002

Sl.No.:

MOTOR: β4/4K ☐ β8/3K ☐

PART No.: A06B-0____-____

Sl.No.:

REACTOR: A81L-0001-0083#3C (0.15mH/24A) ☐

Sl No.:

DISCHARGE RESISTOR: 16Ohms ☐ 30Ohms ☐

SPECIAL INFORMATIONS:

CHECKED BY:- KS/MZ

VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS® In Pursuit of Excellence

ISOLATTION SWITCH (SW1) — AC 3 Ph 415V ☑

40A ☐ 63A ☑ 100A ☐

MAKE: SALZER ☐ ABB ☐ ☐

TYPE: MCCB → ☑ Schneider Electric

ISOLATTION TRANSFORMER (TFR1) PRIMARY- 415V/3Ph SECONDARY - 220V/3Ph, 220V/1Ph, 230V/1Ph

11.3KVA ☐ 14.3KVA ☐ 18 KVA ☑ 22.5KVA ☐

MAKE: SALZER ☐ T.S INTERNATIONAL ☑

SL.No: 1328-07-13

PRIMARY MCB (F15) — AC 3 Ph 415V / 3POLE ☑

25A ☐ 32A ☐ 40A ☑ 50A ☐

MAKE: SCHNEIDER ☑ MERLIN GERIN ☐

TYPE:

SECONDARY MCB (F19) — AC 3 Ph 220V / 3POLE ☑

40A ☐ 50A ☐ 63A ☑ 80A ☐

MAKE: SCHNEIDER ☑ ABB ☐

TYPE:

DRIVE CONTROL MCB (F20) N/A — AC 3 Ph 220V / 2POLE ☐

1A ☐ 3A ☐ ☐ ☐

MAKE: SCHNEIDER ☐ MERLIN GERIN ☐

TYPE:

SPINDLE MOTOR FAN MPCB (F56) — AC 3 Ph 220V / 3POLE ☑

0.1-0.16A ☑ 0.25-0.4A ☐ ☐

MAKE: ABB ☐ SPRECHER &SCHU ☐

MPCB SETTING: 0.1A / 0.13A ☑ / 0.25A

PSM CONTACTOR (K19) — 2 30V AC ☑

CA3-30-10 ☐ A110-30-10 ☑ A75 - 30 ☐ ☐

MAKE: ABB ☑ SPRECHER &SCHU ☐

TYPE:

REGULATED POWER SUPPLY DETAILS

PS1 ☑ PS2 ☑ PS3 ☐

OUTPUT: 24V DC 10A ☑ 24V DC 20A ☐

MAKE: ABB (CP-E 24/10.0) ☐ OMRON(S8VE 24024) ☑ KB ELECTORNICS ☐

Sl. No.: PS1: PS2: PS3:

ISOLATTION TRANSFORMER (TFR4) PRIMARY- 415V/1Ph SECONDARY - 200V/1Ph, 230V/1Ph

N/A 900VA ☐ 450VA ☐ ☐

MAKE: SALZER ☐ T.S INTERNATIONAL ☐

SL.No:

N/A **TFR4 PRIMARY MCB (F16) :** 3A/2P ☐ ☐

MAKE:SCHNEIDER ☐ MERLIN GERIN ☐

TFR4 SEC. MCB (F58) : 6A/1P ☑ ☐

MAKE:SCHNEIDER ☑ MERLIN GERIN ☐

TFR4 SEC. MCB (F59) : 2A/1P ☐ 10A/1P ☑

MAKE:SCHNEIDER ☑ MERLIN GERIN ☐

SPECIAL INFORMATIONS:

CHECKED BY:- KS/MS

VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS® In Pursuit of Excellence

CONTROL CIRCUIT

SEALED KEY BOARD/MCP: Sl.No.: 2972

TYPE: (YSL-AMS-SKB-2-REV1-10/09) ☑ YSL-SPARK-MCP-5-REV0-03/10 ☑

ATC BRAKE RELAY (PCB 1) :

TYPE: 1 SSR RELAY -YSL-SSRP-BR-3-REV3-08/11 ☐ 2 SSR RELAY - YSL-S26-009-77-REV6-08/11 ☑

I/O DISTRIBUTON PCB(PCB2) (PCB104 /105 /106 /107 /156 /157 /256 /257) : QTY: 06 NOS

TYPE: YSL-S26-009-56-REV0-04/08 ☐ YSL IFB-F50-WT-P3-02-0313 ☑

CNC ON/OFF RELAY (PCB 3) :

TYPE: YSL -SSR-CNC ON / OFF-REV8-08/10 ☑

Z AXIS BRAKE RELAY (PCB 5) :

TYPE: YSL-SSRP-BR3-REV3-08/11 ☑

Z AXIS BRAKE CURRENT SENSINGRELAY (CSR1) :

TYPE: YSL ELECTRONICS (CSR1-2) ☑

AC RELAYS: Z BRAKE/EMG (ACR1) ☑ ATC/CAB. THERMAL(ACR2) ☑ I. TABLE THERMAL(ACR3) ☑

MAKE: OMRON ☑ ELESTA ☐

RELAY BOARD(STD-X20) N/A 1RELAY(ELECTRO MAGNETIC-DC) ☐

TYPE: YSL-UMK-1RM-24V- DC-CN-FI-2-REV1-12/09 ☐

RELAY BOARD(STD-X31) 1RELAY(ELECTRO MAGNETIC-DC/AC) ☑

TYPE: YSL-S26-009-082-REV1-06/10 ☑

RELAY BOARD(APC-X25) N/A 8 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-064-REV4-09/11 ☐

RELAY BOARD(FIXTURE-X26) 2 / 4 / 8 RELAY (SSR TYPE-DC) ☑

TYPE: YSL-S26-009-062-REV4-09/11 ☐ S26-009-063-REV4-09/11 ☑ S26-009-064-REV4-09/11 ☑

RELAY BOARD(TOOL DECLAMP - X33) 1 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-061-REV4-07/12 ☐

RELAY BOARD(CTS Robot Vessel - X 37) 1 RELAY (SSR TYPE-DC) ☑

TYPE: YSL-S26-009-061-REV4-07/12 ☑

RELAY BOARD(- X29) 1/2 RELAY (SSR TYPE-DC) ☑

TYPE: YSL-S26-009-062-REV4-07/12 ☑ N/A

CHECKED BY:- VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS® In Pursuit of Excellence

EMERGENCY CONTACTOR (K22)			24V DC ☑
CS8C-40E ☐	KC6-40E ☑	MAKE: ABB ☑	SPRECHER &SCHU ☐
		TYPE:	
CAPACITOR BANK (DP1) : N/A			24V DC ☐
		MAKE:	YSL-PF-CHU-1 ☐

SINGLE PHASE POWER SUPPLY CIRCUIT

MCB (F18) N/A			AC 1 Ph 230V / 1POLE ☐
10A ☐	☐ ☐	MAKE: SCHNEIDER ☐	MERLIN GERIN ☐
		TYPE:	
PANEL COOLER:		POWER: 500W ☐	1000W ☑
MAKE: ADVANCE (TURBO1000R33) ☑	ADVANCE (TURBO500R50) ☐		ACE (IGLOO - 750) ☐
Sl.No.: 51849 - 09 - 13			

UTILITIES (CABINET):			
CABINET LAMP:	PEARL(CFL-11W) ☑	PHILIPS(CFL-11W) ☐	☐
GANG BOX:	2 SWITCHES + 1SOCKET + I INDICATOR ☑ → Cabinet		
DOOR INTERLOCK:	LIMIT SW (HONEYWELL-ZLD A01D) ☑		
MCB FOR CAB. LAMP, 5 PIN SOCKET	(F66-3A/1PMAKE: SCHNEIDER) ☑		

UTILITIES (SWITCHGEARS):			
ADD ON BLOCK(PCB5/PSM(K19) INTERLOCK):	SPC&SCH(CS7-PV) ☐	ABB(CA5-10) ☐	
ADD ON BLOCK(STAR DELTA INTERLOCK):	SPC&SCH(CS7-PV) ☐	SPC&SCH(CA3-P) ☐	
ADD ON BLOCK(HYDRAULIC FAN POWER): (K9)	SPC&SCH(CS8-P22Z) ☐	ABB(CAF6-20-M) ☑	
ADD ON BLOCK(AUTO DRAIN SOL): (K4)	SPC&SCH(CS8-P22Z) ☐	ABB(CAF6-20-M) ☑	

SPINDLE STAR / DELTA CIRCUIT

CONTACTORS (K20 / K21) N/A			230V AC ☐
CA3-72N-L11 ☐	CA7 - 85 ☐ ☐	MAKE: ABB ☐	SPRECHER &SCHU ☐
		TYPE:	
RELAY BOARD(X32)			2 RELAYS (SSR TYPE) ☐
TYPE:	S26-009-80-CN--F-24DC-230AC ☐		

SPECIAL INFORMATIONS: F17 RCCB → 63 amps ☑ Schneider make. N/E

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MAGAZINE MOTOR POWER

MPCB (F1)	1-1.6A	✓	0.63-1A	☐	CONTACTOR (K1/K2)	CA8-9C ☐	VBC6-30-01	✓
MPCB SETTING:	1A	☐	1.2A	✓	MOTOR ID:	SA9715 ☐	SA4067 ✓	SA4060 ☐ TM80 ☐
SOLID STATE RELAY (SSR1):	✓ MAKE: YSL-ACSSC-2P-4 / ERI ELECTRONICS- SJDA - 482524(DSPST) ☐							

ARM MOTOR POWER

MPCB (F2)	1-1.6A	✓	MCB (F81-3A / 3P)	☐	CONTACTOR (K3/K76)	CA8-9C ☐	VBC6-30-01	✓
MPCB SETTING:	1.2A	✓	1.4A	☐	MOTOR ID:	SAG909 ☐	SA5907 ✓	SA4412 ☐ ☐
VFD(HITACHI-WJ200-1.5 KW)	☐	SPEED(50/ HZ)	☐		REACTOR/CHOKE:	2HP/ 6.5mH /3.4A(TSI) ☐		
SOLID STATE RELAY (SSR3):	✓ MAKE: YSL-ACSSC-2P-4 / ERI ELECTRONICS- SJDA - 482524(DSPST) ☐							

LUBRICATION MOTOR POWER

MCB (F3)	1A/3P	✓	MPCB(0.63-1A)	☐	CONTACTOR (K4)	CA8-9C ☐	BC6-30-01	✓
GREASE LUBRICAION SYSTEM (LUBE)	☐				MOTOR POWER:	0.09KW/0.5A ☐	0.12KW/0.6A	✓

FLOOD COOLANT MOTOR POWER

MPCB (F4)	0.63-1A	✓	2.5-4A	☐	CONTACTOR (K5)	CA8-9C ☐	BC6-30-01	✓
MPCB SETTING:	0.9A	✓	3.4A	☐	MOTOR POWER:	0.3KW/0.9A ✓	1.5KW/3.4A	☐
SURGE SUPRESSOR(RC4):	YSL-NIR-RC3-1/022-575VAC ✓							

CHIP PUSH COOLANT MOTOR POWER

MPCB (F5)	0.63-1A	☐	2.5-4A	✓	CONTACTOR (K6)	CA8-9C ☐	BC6-30-01	✓
MPCB SETTING:	0.9A	☐	3.4A	✓	MOTOR POWER:	0.3KW/0.9A ☐	1.5KW/3.4A	✓
SURGE SUPRESSOR(RC5): YSL-NIR-RC3-1/022-575VAC ✓					COMMON CONTACOR KIT - PUSH/FLUSH(ABB-VBC6 ☐			

CHIP FLUSH COOLANT MOTOR POWER

MPCB (F6)	0.63-1A	☐	2.5-4A	✓	CONTACTOR (K7)	CA8-9C ☐	BC6-30-01	✓
MPCB SETTING:	0.9A	☐	3.4A	✓	MOTOR POWER:	0.3KW/0.9A ☐	1.5KW/3.4A	✓
SURGE SUPRESSOR(RC6): YSL-NIR-RC3-1/022-575VAC ✓					COMMON MPCB WITH CHIP PUSH ☐			

COOLANT GUN / CONVEYOR DRUM WASH MOTOR POWER

MPCB (F7)	0.63-1A	✓	2.5-4A	☐	CONTACTOR (K8)	CA8-9C ☐	BC6-30-01	✓
MPCB SETTING:	0.9A	✓	3.4A	☐	MOTOR POWER:	0.3KW/0.9A ✓	1.5KW/3.4A	☐
SURGE SUPRESSOR(RC7):	YSL-NIR-RC3-1/022-575VAC ✓							

HIGH PRESSURE COOLANT MOTOR POWER

MPCB (F9)	4-6.3A	✓	2.5-4A	☐	CONTACTOR (K10)	CA8-12C ☐	BC7-30-01	✓
MPCB SETTING:	4.9A	☐	5.2A	☐	MOTOR POWER:	2.2KW/4.9A ☐	2.2KW/5.2A	☐
SURGE SUPRESSOR(RC9):	YSL-NIR-RC3-1/022-575VAC ✓							

HYDRAULIC 1 MOTOR POWER

MPCB (F8)	4-6.3A	☐	2.5-4A	✓	CONTACTOR (K9)	CA8-9C ☐	BC6-30-01	✓
MPCB SETTING:	2.7A	☐	3.5A	✓	MOTOR POWER:	1.1KW/2.7A ☐	1.5KW/3.5A	✓
SURGE SUPRESSOR(RC8):	YSL-NIR-RC3-1/022-575VAC ✓							

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INDEX TABLE MOTOR POWER

N/A

MPCB (F12)	1-1.6A	☑		CONTACTOR(K15/K16)	CA8-9C ☐	VBC6-30-01 ☑
MPCB SETTING:	1.2A ☐		1.4A ☐	MOTOR POWER:	1.1KVA/1.5A ☐	2.3KVA/3.2A ☐
SOLID STATE RELAY (SSR5):	MAKE: YSL-ACSSC-2P-4 / ERI ELECTRONICS- SJDA - 482524(DSPST) ☐					

CHIP CONVEYOR 1 MOTOR POWER

N/A

MPCB (F10)	0.63-1A ☐		1-1.6A ☐	CONTACTOR(K11/K12)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.65A ☐		☐	MOTORPOWER:	0.18KW/0.65A ☐	☐
SURGE SUPRESSOR(RC10):				YSL-NIR-RC3-1/022-575VAC ☐		

CHIP CONVEYOR 2 MOTOR POWER

N/A

MPCB (F11)	0.63-1A ☐		1-1.6A ☐	CONTACTOR(K13/K14)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.65A ☐		☐	MOTORPOWER:	0.18KW/0.65A ☐	☐
SURGE SUPRESSOR(RC11):				YSL-NIR-RC3-1/022-575VAC ☐		

COIL / SCREW CONVEYOR 1 MOTOR POWER

N/A

MPCB (F13)	0.63-1A ☐		1-1.6A ☐	CONTACTOR(K17/K18)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.65A ☐		1.1A ☐	MOTORPOWER:	0.18KW/0.65A ☐	0.37KW/1.1A ☐
SURGE SUPRESSOR(RC13):				YSL-NIR-RC3-1/022-575VAC ☐		

COIL / SCREW CONVEYOR 2 MOTOR POWER

N/A

MPCB (F30)	0.63-1A ☐		1-1.6A ☐	CONTACTOR(K28/K29)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.65A ☐		1.1A ☐	MOTORPOWER:	0.18KW/0.65A ☐	0.37KW/1.1A ☐
SURGE SUPRESSOR(RC30):				YSL-NIR-RC3-1/022-575VAC ☐		

OIL SKIMMER MOTOR POWER

N/A

MPCB (F21)	0.25-0.4A ☐		☐	CONTACTOR (K23)	CA8-9C ☐	BC6-30-01 ☐
MPCB SETTING:	0.25A ☐		☐	MOTOR POWER:	40W/0.22A ☐	☐
SURGE SUPRESSOR(RC21):				YSL-NIR-RC3-1/022-575VAC ☐		

Bagfilter Dumping MOTOR POWER (OPTIONAL-1)

MPCB (F49)	2.5-4A ☑		☐	CONTACTOR (K53)	CA8-9C ☐	BC6-30-01 ☑
MPCB SETTING:	☐		3.4A ☑	MOTOR POWER:	1.5KW/3.4A ☑	☐
SURGE SUPRESSOR(RC49):				YSL-NIR-RC3-1/022-575VAC ☑		

MOTOR POWER (OPTIONAL - 2)

N/A

MPCB (F)	☐		☐	CONTACTOR (K)	CA8-9C ☐	BC6-30-01 ☐
MPCB SETTING:	☐		☐	MOTOR POWER:	☐	☐
SURGE SUPRESSOR(RC):				YSL-NIR-RC3-1/022-575VAC ☐		

SPECIAL INFORMATIONS: MPCB'S- SPRECHER &SCHU(KTA7-25S) ☐ ABB (MS 116) ☑

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ATC & SPINDLE DETAILS

SPINDLE MOTOR:	MODEL:	β3 ☐	β6 ☐	β12 ☑	☐
	MOTOR RPM:	6000 ☐	8000 ☑	10000 ☐	☐
	MOTOR POWER:	15/11 KW			
	PART No:	A06B-1447-B103			
	Sl. No:	C137J133B #0102			
	GEAR RATIO:	1:1 ☑	1:1.25 ☐	1:1.5 ☐	DIRECT DRIVE ☐
	POWER LIMIT:	IN KW: 11/7.5 KW			
	SPEED LIMIT:	IN RPM: 80-8000			

PROXIMITY SENSORS-SPINDLE

MAKE; TEKNIC ☐ BALLUFF ☑ TYPE: (24V DC - PNP) ☐

		M12X40	M12X52	M12X68	M12X73
SPINDLE ORIENTATION (PXS1) ☑	TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐ TEKNIC	☑
TOOL CLAMP(PXS3) ☑	TYPE: CONNECTOR ☑ NON CONN. ☐	M12X40 ☐	M12X52 ☐	M12X68 M12X63 ☐ BALLUF	M12X73 ☑
TOOL DECLAMP(PXS4) ☑	TYPE: CONNECTOR ☑ NON CONN. ☐	M12X40 ☐	M12X52 ☐	M12X68 M12X63 ☐ BALLUF	M12X73 ☑
NO TOOL IN SPINDLE(PXS32) ☑	TYPE: CONNECTOR ☐ NON CONN. ☐	M08X50 ☐	M08X68 ☑	M12X73 M8X63 ☐ BALLUF	☐
ONE ROTATION(PXS39) ☑	TYPE: CONNECTOR ☑ NON CONN. ☐	M12X40 ☐	M12X52 ☐	M12X68 M12X63 ☐ BALLUF	M12X73 ☑

N/A

PROXIMITY SENSORS-ATC(DISC TYPE)

MAKE; TEKNIC ☐ TYPE: (24V DC - PNP) ☐

		M12X40	M12X52	M12X68	M12X73
MAGAZINE IN POSITION (PXS2) ☐	TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐
MAGAZINE REFERENCE (PXS46) ☐	TYPE: CONNECTOR ☐ NON CONN. ☐	M12X73 ☐	FG2X05APO24-5000LFLAT ☐		☐

PROXIMITY SENSORS-ATC(TWIN ARM TYPE)

MAKE; TEKNIC ☐ Balluff ☑ TYPE:EGT 12 X 02 AP - 024 SEM4L(24V DC - PNP) ☑

MAGAZINE IN POSITION PROXIMITY (PXS2) ☑	MAG REFERENCE PROXIMITY (PXS46) ☑
POCKET UP PROXIMITY (PXS5) ☑	POCKET DOWN PROXIMITY (PXS6) ☑
ARM POSITION 1 PROXIMITY-PS1 (PXS7) ☑	☐
ARM POSITION 2 PROXIMITY-PS2 (PXS58) ☑	ARM POSITION 3 PROXIMITY-PS3 (PXS59) ☑

CHECKED BY:- [signature] VERIFIED BY:- [signature]

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PNUEMATIC ELEMENTS (SPINDLE + ATC)

MAKE; SUCO ☑	TYPE:(24V DC - NO CONTACT) ☑
MAIN AIR PRESSURE SWITCH (PRS 2) ☑	SPINDLE AIR PRESSURE SWITCH (PRS 101) ☑
AIR PURGE PRESSURE SWITCH (PRS 102) ☐	☐
MAKE; FESTO ☑ SMC ☐	TYPE:(24V DC -COIL) ☑
SPINDLE AIR JET SOLENOID (Y1) ☑	SPI AIR PURGE SOLENOID (Y115) ☐
POCKET UP SOLENOID (Y2) ☑	POCKET DOWN SOLENOID (Y3) ☑
TOOL CLAMP SOLENOID (Y51) ☑	TOOL DECLAMP SOLENOID (Y4) ☐
AIR THROUGH NOZZLE SOLENOID (Y99) ☐	☐

HYDRAULIC ELEMENTS (SPINDLE + ATC)

N/A

MAKE; YUKEN ☐ REXROTH ☐	TYPE:(24V DC -COIL) ☐
☐	TOOL DECLAMP SOLENOID (Y4) ☐

UTILITIES(SPINDLE + ATC)

TOOL CHANGE POSITION LIMIT SWITCH- (LS3) :	TEKNIC SG-01-R-610	☑	☐
TOOL DECLAMP PUSH BUTTON :	ILLUMINATED GREEN	☑	☐
ATC / I.TABLE DISPLAY SEL SWITCH :	2 POSITION	☐	☐
ARM/MAG TEST MODE KEY SWITCH (KSW7) :	2 POSITION	☐	☐
POCKET UP/DOWN PB ☐ ARM FWD/REV SEL SW ☐		☐	☐

SPECIAL INFORMATIONS:

AUTO DOOR & CYCLE START DETAILS

MAKE; FESTO ☑	TYPE:(24V DC -COIL) ☑
AUTO DOOR OPEN SOLENOID (Y6) ☑	AUTO DOOR CLOSE SOLENOID (Y7) ☑
AUTO DOOR SAFETYSOLENOID (Y8) ☑	☐
DOOR SWITCHES	TYPE: LS5 - NG1 RS 510 (24V DC -NO CONTACT) ☐
MAKE; TECNIK ☐ FESTO ☑	READ SWITCHES (RSW30& RSW31) ☑
CYCLE START SWITCH	TYPE: LS45 - XCE-154 (24V DC -NO CONTACT) ☐
MAKE; ☐ TELEMECHANIC ☐	☐
DOUBLE HAND CYCLE START BOX	CYCLE START PB (MUSHROOM HEAD GREEN) (02 Nos) ☑
MAKE; TECNIK ☑ ☐	EMERGENCY PB (MUSHROOM HEAD SPRING RETURN) ☑
AUTO DOOR OPERATION	AUTO DOOR OPEN/CLOSE PB (GREEN) ☑
MAKE; TECNIK ☑ ☐	AUTO DOOR INTERLOCK KEY SW (2 POSITION) ☑

SPECIAL INFORMATIONS:

CHECKED BY:-	**VERIFIED BY:-**

COOLANT TANK / CONVEYORS DETAIL

UTILITIES (STD COOLANT)				
COOLANT LEVEL FLOAT SWITCH (PXS4) :	FILPRO(LIM 18PF SC)	☐	D2NS-(LOW) D2NS-(HIGH) LIMIS PR SC	☑
COOLANT GUN SEL SW :	2 POSITION	☐		☐
COOLANT VALVES: PUSH(Y125) /FLUSH(Y126) /ARM FLUSH(Y127) /FLUSH2(Y219) :			MAKE:AVCON / SMC	☐
COOLANT PLUG (HC1) :	16 PIN	☐	32 PIN 24 PIN	☑
LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☑
UTILITIES (HP COOLANT):				
FLOW SWITCH (FLW1) :	IFM (SH0 500)	☑		☐
CLOG SWITCH (FCLG1) :	HIRLEKAR	☑	SALVIN	☐
AIR BLAST SOLENOID (Y5) :	FESTO	☐	ROTEX RELIEF VALVE (Y235)	☑
M/C DOOR CLOSE LIMIT SW (LS5) :	TEKNIC(NG1 RS 510)	☐		☐
UTILITIES (CHIP CONVEYOR):				
CHIP CONVEYOR 1 O/L LIMIT SW (LS12) :	TEKNIC(NG1 RS 510)	☐		☐
CHIP CONVEYOR 2 O/L LIMIT SW (LS13) :	TEKNIC(NG1 RS 510)	☐		☐
CHIP CONVEYOR FWD/REV SEL SW :	3 POSITION(SPRING RETURN)	☐		☐
CHIP CONVEYOR PLUG (HC3) :	10 PIN	☐	24 PIN	☐
LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☐
UTILITIES (COIL/ SCREW CONVEYOR):				
SCREW CONVEYOR 1 O/L PROXIMITY (PXS20) :	M12 X 52MM (CONN. TYPE)	☐	M12 X 72MM (CONN. TYPE)	☐
SCREW CONVEYOR 2 O/L PROXIMITY(PXS104) :	M12 X 52MM (CONN. TYPE)	☐	M12 X 72MM (CONN. TYPE)	☐
SCREW CONVEYOR FWD/REV SEL SW :	3 POSITION(SPRING RETURN)	☐	COMMON WITH CHIP.CNYR	☐

SPECIAL INFORMATIONS:

HYDRAULIC TANK DETAIL

OIL LEVEL FLOAT SWITCH (FL4) :	SHRIDHAN	☑		☐
OIL CLOG SWITCH (CLG2) :	HYDAC	☐	REXROTH	☐
HYDRAULIC THERMAL SWITCH WITH CONTROLLER (TH4) :	SRI ELECTRINICS	☐		☐
HYDRAULIC PLUG (HC2) :	16 PIN	☐	24 PIN	☑
MAKE;INDO ELECTRIC ☑ LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☑

SPECIAL INFORMATIONS:

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LUBRICATION TANK / FRL DETAIL

OIL LEVEL FLOAT SWITCH (FL1) :		CENLUB	☑	GREASE-LUBE	☐
LUB PRESSURE SWITCH (PRS1) :		SUCO	☑	GREASE-LUBE	☐
AUTO DRAIN SOLENOID (Y44) :		ROTEX	☑	FLOAT SW(FL3)	☐

MACHINE LAMPS

MAKE:		PATANKAR	☐	AISHWARYA	☑
MACHINE LAMP1 (ML1) :	11W ☐	9W+9W	☐	36W	☑
MACHINE LAMP2 (ML2) :	11W ☐	9W+9W	☐	36W	☑
MACHINE LAMP3 (ML3) :	11W ☐	9W+9W	☐	36W	☐
MACHINE INDICATION LAMP(PL5) :	1 TIER ☐	3 TIER	☑	3 TIER WITH HOOTER	☐
MAKE:		PATANKAR	☑	LAXMI(VANTECH)	☐

UTILITIES(MAHINE)

EMERGENCY MONITOR (PCB 12)		YSL-EMG-PCB-4-REV3-12/11	☑
HAND HELD MPG:		FANUC ☐	FUTURE(TAIWAN) ☐
	Sl. No.:		
TPM TRACK :		PIONEER CT ☐	☐
	Sl. No.:		
SERVO VOLTAGE STABILIZER:		15KVA ☐	25KVA ☐
INPUT:	OUTPUT:	MAKE: SERVOMAX ☐	☐
	Sl. No.:		
COOLANT / SPINDLE CHILLER UNIT:		MAKE: POINT ☐	☐
POWER:	MCB(F31):10A / 3P	Sl. No.:	
CONNECTOR :	INDO ELECTRIC (10 PIN) ☐	LOCATION: LHS ☐	RHS ☐

FIXTURE DETAIL

MAKE; ☐		PNEUMATIC ☐	HYDRAULIC ☑
STATION 1 CLAMP SOLENOID (Y70)	☑	STATION 1 DECLAMP SOLENOID (Y71)	☑
STATION 2 CLAMP SOLENOID (Y72)	☑	STATION 2 DECLAMP SOLENOID (Y73)	☑
STATION 3 CLAMP SOLENOID (Y84)	☑	STATION 3 DECLAMP SOLENOID (Y85)	☑
STATION 4 CLAMP SOLENOID (Y86)	☑	STATION 4 DECLAMP SOLENOID (Y87)	☑
MAKE; ☐		PNEUMATIC ☐	HYDRAULIC ☑
STATION 1-CLAMP PRESSURE SW (PRS20)	☑	STATION 2 CLAMP PRESSURE SW (PRS22)	☑
STATION 3 CLAMP PRESSURE SW (PRS31)	☑	STATION 4 CLAMP PRESSURE SW (PRS32)	☑

SPECIAL INFORMATIONS: CLAMP PB'S (GREEN) -

DECLAMP PB'S (RED) -

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MACHINE HISTORY - ELECTRICAL

APC (LINEAR TYPE) DETAIL

MANUAL OPERATION PANEL :	YSL-AMS-APC-PCB-REV2-12/11	☐
PALLET SET UP PB :	LUMINOUS GREEN	☐
EMERGENCY PB :	MUSHROOM HEAD RED	☐

PROXIMITY SENSORS:

MAKE; TEKNIC ☐		TYPE: (24V DC - PNP)			☐
PALLET RECEIVED PROXIMITY(PXS11) ☐	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	
ARM 1 IN PROXIMITY (PXS13) ☐	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	
ARM 1 OUT PROXIMITY (PXS14) ☐	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	
ARM 2 IN PROXIMITY (PXS15) ☐	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	
ARM 2 OUT PROXIMITY (PXS16) ☐	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	

PNEUMATIC SOLENOIDS:

MAKE; FESTO ☐	TYPE: 24V DC -COIL ☐
APC DOOR OPEN SOLENOID (Y12) ☐	APC DOOR CLOSE SOLENOID (Y13) ☐
ALLET PAD CLEAN SOLENOID (Y18) ☐	☐

HYDRAULIC SOLENOIDS:

MAKE; YUKEN ☐ REXROTH ☐	TYPE: 24V DC -COIL ☐
PALLET CLAMP SOLENOID (Y10) ☐	PALLET DECLAMP SOLENOID (Y11) ☐
ARM 1 IN SOLENOID (Y14) ☐	ARM 1 OUT SOLENOID (Y15) ☐
ARM 2 IN SOLENOID (Y16) ☐	ARM 2 OUT SOLENOID (Y17) ☐

HYDRAULIC PRESSURE SWITCHES:

MAKE; POLYHYDRON ☐	TYPE: ☐
PALLET CLAMP PRSW (PRS3) ☐	PALLET DECLAMP PRSW (PRS60) ☐

LIMIT SWITCHES:

MAKE; TEKNIC(NG1VS 510) ☐	JAI BALAJI (JLSSPR) ☐
PALLET 1 ON STATION LIMIT SW (LS10) ☐	PALLET 2 ON STATION LIMIT SW (LS11) ☐

READ SWITCHES:

MAKE; FESTO ☐	APC DOOR INTERMEDIATE READ SW (RSW 65) ☐
APC DOOR OPEN READ SW (RSW 58) ☐	APC DOOR CLOSE READ SW (RSW 59) ☐

SPECIAL INFORMATIONS:

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CHECK LIST - ELECTRICAL

M/C NUMBER:- M11-932 DATE:- 14/4/13

I.AXES

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the work order specification for the following	✓	
	a)Servo motor	✓	
	b)Axis stroke	✓	
	c)Rapid rate & Feed rate	✓	
2	Check lub inch & Lubrication timers as required for the machine	✓	
	LM RAIL: **20Min.** HARD.GUIDWAY: **12 Min.** Grease LUB: **6 Hours.**	✓	
3	Check the connection of cable both at motor & drive end.	✓	

II.SPINDLE

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the connection of cable both at motor & drive end.	✓	
2	Specifications of spindle motor /speed /power is as per work order.	✓	
3	Check the spindle motor fan rotation.	✓	
4	Check the following spindle parameters	✓	
	a)Speed locking (P3735,3736)	✓	
	b)Power locking (P4028,4029)	✓	
	c)Rigid tapping (P4016#4,4044,4052,4085,5241,5261)	✓	
5	Check the spindle for following functions		
	a)Maximum & Minimum speed	✓	
	b)Manual Increment & decrement	✓	
	c)Manual stop, clockwise & counter clockwise.	✓	
	d)Star/Delta switching of Contactors for required speed	N/A	
	(switching speed D340 =)		
	e)Spindle coasting	N/A	
	f)Spindle orientation @ maximmum speed.	✓	

III.ATC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the connection of cables.(cabinet, junction box, drive, etc.)	✓	
2	Check ATC for the following functions	✓	
	a)Magzine motor direction	✓	
	b)Arm motor direction	✓	
	c)Tool display for tool no.1,2,4,8,16,32	✓	
	d)Tool declamp in manual mode	✓	
	e)Auto tool change	✓	
	f)Power failure conditions	✓	
3	Check for Test mode,Handheld MPG operation if **servo drive** is used.	N/A	
4	Check for Test mode,parameter locking if **VFD** is used.		

IV. APC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the connection of cables.(cabinet, junction box, etc.)		
2	Check for pallet selection program**(O100).**		
4	Check APC for the following functions		
	a)MOP operations	N/A	
	b)Pallet set up sequence		
	c)Auto pallet change(M80) & related interlocks.		
	d)Power failure conditions		
	e)Fixture interlock		

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CHECK LIST - ELECTRICAL

V.PNUEMATIC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the functions of following elements	✓	
	a)Solenoids	✓	
	b)Pressure switches	✓	
	c)FRL unit & auto drain	✓	
2	Check for solenoid nut,screws & prsw caps.	✓	

VI.HYDRAULIC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for Hydraulic Motor / cooling fan direction.	✓	
2	Check for cables/ hose / plugs termination & routing.	✓	
2	Check for solenoid/pressure sw mounting screws & gasket.	✓	
3	Check the functions of following elements	✓	
	a)Solenoids	✓	
	b)Pressure switches	✓	
	c)Power pack Float , Thermal , clog switch.	✓	

VIII.FIXTURE

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for manual operation through push buttons.	N/A	
2	Check for auto operation through M codes.	✓	
3	Check for timers,bypass bits, seat check,other related interlocks.	✓	

VII.COOLANT

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for direction of all coolant / conveyor motors.	✓	
2	Check for cables/ hose / plugs termination & routing.	✓	
3	Check the functions of following elements	✓	
	a)Manual mode operations.(SKB push buttos, selector sw)	✓	
	b)MDI / Auto mode operations	✓	
	c)Float switch	✓	
	b)Chip/ Screw Conveyor operation(lsw, proximity,,fwd/ rev etc)	✓	
	c)HP coolant functions(clog,flow,air blast).	✓	
	d)Push/Flush sequence through M180 command.	✓	

VIII.ROTARY/ INDEX TABLES

SL.No	PARAMETERS	STATUS	REMARKS
1	Cable/ hoses are routed properly at safe position.	✓	
2	Check for commanded position , table position & shortest path.	✓	
3	Check for tool display & table position in case of index table.	N/A	

IX.CABINET & OPERATING PANEL

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the rating & spec.of following elements as per circuit diagram.	✓	
	a)MPCB's	✓	
	b)Contactors	✓	
	c)MCB's	✓	
	d)Transformer	✓	
	e)PCB's,Relay's & relay boards	✓	
2	Set the MPCB's for 100% of rated current(refer IOM for torque motor)	✓	
3	Check the panel cooler with cab.door interlock.set for 35°C& lock par.	✓	
4	Ensure the rotation of drive heat sink fan & also if any other fans.	✓	

CHECKED BY:- K.V.Sandeep

VERIFIED BY:-

5	Check for the following		
	a)Mounting of elements inside cabinet	✓	
	b)Mounting of elements on operating panel	✓	
	c)All Terminal tightness/ loose connections	✓	
	d)Input/ output short connection	✓	
	e)Ferruling of cable/wire(with adress if necessary)	✓	
	f)Identification stickers	✓	
	g)Moisture free beadings(both in cabinet & operating panel)	✓	

X.GENERAL

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the functions of following		
	a)All emergency push buttons	✓	
	b)Emergency monitor	✓	
	c)Work lamp & m/c indication lamp(3 tier)	✓	
	d)Auto door & m/c door limit switches if provided	✓	
2	Check for - All earthing connections.	✓	
3	Check for -Protective covers.	✓	
4	Check for -PVC/Metal Channel cover fixing.	✓	
5	Check for -Name plate & tags on all elements.	✓	
6	Check for -Routing of cables & hoses.	✓	
7	Check for- Model/Sl.No./ rating in M/C No. plate	✓	

INTERNAL CHECKLIST POINTS

- ~~All~~ Coolant motor tags.
- ~~RC sticker~~.
- TBI-37, 43 marker sheet
- Machine number plate
- ~~Emergency~~ on ARMs to be enabled.
- Fixture to be checked.

XI. PMC DATA

Instruction: Please note all D-bits which is enabled at the time of shipping .

Also give the details if it is used for special function

	7	6	5	4	3	2	1	0		7	6	5	4	3	2	1	0
D0				(4)		(2)		(0)	D16								
D1				4		2		0	D17								
D2				(4)		2		0	D18								
D3				4		2		0	D19								(0)
D4			(5)	4	(3)	2	(1)	0	D20								
D5				4		2		0	D21								
D6	(7)			4		(2)		0	D22								
D7				4	(3)	2		0	D23								
D8				4		(2)		(0)	D24								
D9				4		2		(0)	D25								
D10				4		2		0	D26								
D11				4	(3)	2	(1)	(0)	D27								
D12				4		2		0	D28								
D13				4		2		0	D29								
D14				4		2		0	D30								
D15				4		2		0									

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D110	00000000	1	1	
D000	00000000	3	30	

XII.PROGRAM/ LOGIC VERSION

Instruction: Please ckeck for latest version before making entry.

Logic version No.: TMA01-02-08-M01	Remarks:
Macro version No.: Z-T1-01	Remarks:

Progrm version Nos.:

O9001	A2-2-03	O9010	-	
O9002	B2-2-02	O9011	-	
O9003	-			

XIII.CNC PARAMETERS

REFERENCE POSITION		X	Y	Z
ATC 2nd Ref.	1241	-600.00	0.00	-3.231
APC 3rd Ref.	1242	-	-	-
APC 4th Ref.	1243	-	-	-

RAPID TIME CONSTANT	X	Y	Z
1620	96	112	120
1621	88	96	96

	Check the following with system:		'1'		'0'	
	No.of Ladder steps**	Dgn:1141 # 6	8000	L	5000	**N/A for 0 iMD
	4th axis option**	1279 # 4 OR 1148 # 7	Yes	L	No	**N/A for 0 iMD
	Macro executer option	Dgn:1236 # 1	Yes	L	No	
	MPG option for PMM	Dgn:1292 # 6	Yes		No	L
	AICC1 option**	Dgn:1273 # 0	No		Yes	**N/A for 0 i mateMD
	AICC2 option**	Dgn:1148 # 0	Yes		No	**N/A for 0 i mateMD
	3111#7 = 0 (NPA) - Message display					

XIV.SHIPPING

Instruction: Please perform following activities before shipping .
Also give the details if any special activity is done

SL.No	PARAMETERS	REMARKS
1	Ensure the servo guide back up	
2	Check the various communication like RS232C,PCMCIA	
3	Edit title,& set time, date other original factory set values	
4	Take all programs back up &Lock 8000 & 9000 series programs (P3202)	
5	Take the back up of following :	
	a) CNC parameters	
	b) Pitch error compensation	
	c) PMC parameters	
	d) ID info	
	e) M-info	
	f) Ladder	
	g) Macro	
	h) S-RAM	
6	Make I/O channel '0' for RS232C communication	
7	Keep spare fuse inside the cabinet	
8	Ensure any updation/correction in electrical circuit diagrm	
9	Ensure all the elements are in good condition before shipping	
10	Compare CNC parametrers with standard data & store in backup file	

CHECKED BY:-	VERIFIED BY:-
R.V.Bhat	

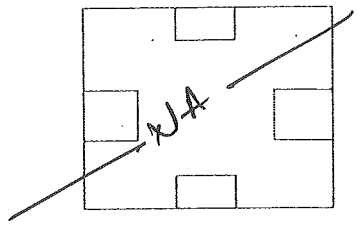
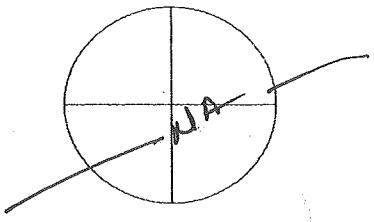
ACE MANUFACTURING SYSTEMS LTD

CHECK LIST - FINAL ASSEMBLY (VMC / CMV) - MECHANICAL

AMS — In Pursuit of Excellence

MACHINE No. : M11-932	CUSTOMER : M/s WABCO INDIA LIMITED

I. Motor Mounting Bracket Face Out and CONCENTRICITY for Direct Drive

	FACE OUT (Recommended ≤ 0.015 mm)	CONCENTRICITY (Recommended ≤ 0.010 mm)
		

S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
1	Spindle Connector Radial Run Out	≤ 0.010 mm	
2	Motor Current at Maximum RPM	β3 < 15%, β6 / α6 & ABOVE <10%	
3	CTS Rod Guide Radial Run Out (DDCTS)	≤ 0.010 mm	
4	Spindle Clamping Force	As per DS-DSN-PROD-01	N/A
5	Spindle Motor Specification	As per Work Order	
II	**INDIRECT DRIVE**		
1	Motor Pulley Radial Run Out	≤ 0.050 mm	0.050
2	Spindle Pulley Radial Run Out	≤ 0.050 mm	0.050
3 a)	Belt Tension	As per DS-DSN-PROD-02	—
b)	Belt Specification	Length : 860 Width : 50	Quantity : 1
4	Motor Current at Maximum RPM	β3 < 20%, β6 / α6 & ABOVE <15%	14 -/.
5	Rotary Joint Radial Run Out (ID CTS)	≤ 0.040 mm	0.030
6	Spindle Clamping Force	As per DS-DSN-PROD-01	8 work gfs
7	Spindle Motor Specification	As per Work Order	β12
8	Pulley Ratio	As per Work Order	1 : 1
III	Axis Dead Stopper Value / Butt Referencing Value (As per Drawing)		
	Axis	Brake ~~MOTOR END~~ End	Motor ~~BRAKE~~ END
1	X	-628.833	18.376
2	Y	16.610	-465.411
3	Z	-519.718	12.902
IV	% Of axis CURRENT Drawn at 100 % Rapid Rate (Recommended as per DS-QA-02)		
	X - Axis	50 -/.	
	Y- Axis	49 -/.	
	Z - Axis Up	68 -/.	
	Z - Axis Down	31 -/.	
V	Accumulator Oil Pressure	As per Circuit Diagram	N/A
	Accumulator Gas Pressure	As per Circuit Diagram	

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ACE MANUFACTURING SYSTEMS LTD			
CHECK LIST - FINAL ASSEMBLY (VMC / CMV) - MECHANICAL			
S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
VI	ATC ALIGNMENT (TURRET TYPE)		
1	Parallelism of gripper Arm face wrto. X - Axis	≤ 0.050 mm	
2	Tool Ejection Stroke	As per DS - DSN-PROD-01	
3	Swing Lever Gap	As per Drawing	
4	Side Cam Alignment	0.020 mm Maximum	
5	Gap between Draw Bar bush & Roller/Screw	1 ± 0.1 mm	
VII	ATC ALIGNMENT (TWIN ARM)		
1	X - Axis	≤ 0.2 mm	0.1
2	Y - Axis	≤ 0.2 mm	0.1
3	Z - Axis	≤ 0.05 mm	0.05
4	Twin Arm bottom face parallelism in Y-Axis	≤ 0.1 mm	0.1
5	Gap between Draw Bar Bush & Roller	1 ± 0.1 mm	1
6	Tool Ejection stroke	As per DS - DSN-PROD-01	0.75
7	Gap between deflection Pads	0.02 TO 0.050 mm	N/A
VIII	APC ALIGNMENT - LINEAR TYPE		
1	Height difference of rest pads	≤ 0.010 mm	
2	Pallet change Time	As per Work Order	
IX	APC ALIGNMENT- ROTARY TYPE		
1	APC Base parallelism w.r.to. X - Axis	≤ 0.1 mm	
	APC Base parallelism w.r.to. Y - Axis	≤ 0.1 mm	
2	Arm Top parallelism w.r.to. X - Axis	≤ 0.05 mm	
	Arm Top parallelism w.r.to. Y - Axis	≤ 0.05 mm	
3	Arm 1 Alignment ' FACE ' w.r.to X- Axis	≤ 0.02 mm	
	Arm 1 Alignment ' Taper ' w.r.to X - Axis	≤ 0.02 mm	
	Arm 2 Alignment ' FACE ' w.r.to X - Axis	≤ 0.02 mm	
	Arm 2 Alignment ' Taper ' w.r.to X - Axis	≤ 0.02 mm	
4	Offset between Arm 1 & Arm 2 - Face	≤ 0.030 mm	
	Offset between Arm 1 & Arm 2 - Taper	≤ 0.030 mm	
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ACE MANUFACTURING SYSTEMS LTD			AMS
CHECK LIST - FINAL ASSEMBLY (VMC / CMV) - MECHANICAL			In Pursuit of Excellence
S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
5	Pallet - 1 Wedge Alignment along X - Axis	≤ 0.020 mm	
	Pallet - 2 Wedge Alignment along X - Axis	≤ 0.020 mm	
6	Gap between Pallet -1 side & Arm along X - Axis	Minimum 0.3 mm	
	Gap between Pallet -1 side & Arm along Z - Axis	0.1 ± 0.02 mm	
	Gap between Pallet -2 side & Arm along X - Axis	Minimum 0.3 mm	
	Gap between Pallet -2 side & Arm along Z - Axis	0.1 ± 0.02 mm	NA
7	Pallet Sagging - Pallet -1 'W.r.to. X & Y - Axis	≤ 1 mm	
	Pallet Sagging - Pallet -2 'W.r.to. X & Y - Axis	≤ 1 mm	
8	Pallet Height Variation	As per TEST CHART	
9	Backlash between Rack & Pinion at distance of 500 mm	≤ 1 mm	
10	Pallet Change Time	As per Work Order	

S.No.	CHECK POINTS	REMARKS
A	AXIS	
1	Ensure geometrical accuracies as per test chart.	OK
2	Ensure plugs are fitted for LM guide ways.	OK
3	Ensure butt reference function.	OK
4	Ensure effective wiping of saddle cover wipers.	OK
5	Ensure lube timer is set 12 min for HG and 20 min for LM Guide ways.	OK
6	Ensure leveling bolt locknuts are tighten after machine leveling.	OK
7	Check the all axes reference arrows are matched & Direction name plate fixed.	
8	Ensure tightness & marking of all screws with Yellow paint marker	
9	Ensure proper setting of Limit switches.	OK
B	ATC	
1	Check oil level is maximum in ATC gear box.	OK
2	Check tool change position limit switch for proper functioning & band width. (Max. 2mm on each side)	OK
3	Ensure smooth function of Tool change without abnormal/unhealthy noise.	OK
4	Ensure ATC bracket & Magazine Dowelling.	OK

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ACE MANUFACTURING SYSTEMS LTD		AMS In Pursuit of Excellence
CHECK LIST - FINAL ASSEMBLY (VMC / CMV) - MECHANICAL		
S.No.	CHECK POINTS	REMARKS
C	SPINDLE	
1	Check the tightness of push stud.	OK
2	Ensure air jet through all the (4 / 8) holes at oriented spindle position	✓
3	Ensure supply of air to spindle Air Purge system (If provided)	N/A
4	Ensure proper setting of proximities.	OK
5	Ensure smooth running of spindle without abnormal/unhealthy noise	OK
6	Ensure dummy plate fixing on CTS slot in Non CTS machine.	N/A
7	Check the Tool not in spindle proximity message.	✓
D	COOLANT	
1	Ensure no coolant leakage around the machine & presence of coolant in lubrication oil drain pipe	✓
2	Ensure proper working of float switch (by shifting the sensor upwards).	✓
3	Ensure all the hoses, fittings tightened and routed properly.	✓
4	Check function of Flood coolant (M7), Chip push (M182), Base flush (M181), CTS (M52) & Coolant gun (through selector switch).	✓
5	Ensure CTS flow sw.setting by opening the bag filter air vent ball valve. (Optional)	✓
6	Ensure suction strainer & Check Valve in CTS tank & pressure gauge working (If provided)	✓
7	Ensure CTS pin locator hole is plugged. (If provided)	✓
8	Ensure that twin arm plunger & CTS pin locator screw do not foul (if provided)	✓
9	Ensure bag filter clog message by closing bag filter outlet ball valve (Optional)	✓
10	Ensure High pressure air blast after CTS switched off (If provided)	✓
11	Ensure drum wash in conveyors (If provided)	✓
12	Check conveyor overload message & Direction (If provided)	✓
13	Check oil in conveyor gear box. (if provided)	N/A
14	Ensure functioning of screw conveyor in forward & reverse directions. (through selector switch)	N/A
15	Ensure splash guard fixed to conveyor.	N/A
E	APC	
1	Ensure proper setting of proximities.	✓
2	Ensure supply of air jet to pallet rest pad cleaning.	✓
3	Ensure pallet change takes place smoothly with specified load (as per work order) without abnormal / unhealthy noise	N/A
4	Ensure APC base Dowelling.	N/A
5	Check pallet initialization in APC Machine	N/A
NAME & SIGN	Rajan B.M / CHECKED BY — APPROVED BY	4 ... 7 Page No

ACE MANUFACTURING SYSTEMS LTD		AMS In Pursuit of Excellence
CHECK LIST - FINAL ASSEMBLY (VMC / CMV) - MECHANICAL		
S.No.	CHECK POINTS	
F	HYDRAULIC, PNEUMATIC & LUBRICATION	
1	Ensure pneumatic main line pressure switch setting (4 bar) & error message are satisfactory	✓
2	Ensure Spindle air jet pressure switch setting (3 bar) & error message are satisfactory	✓
3	Ensure Spindle air purge pressure switch setting (1.2 bar) & error message are satisfactory	- NA -
4	Ensure proper setting of oil flow (1 drop/tool change) in FRL unit. & Oil Level low message	✓
5	Ensure proper routing of FRL drain hose.	✓
6	Ensure identification name plates are fixed on power pack hoses.	= NA —
7	Ensure Ball valve fixed to FRL Unit inlet	✓
8	Ensure pneumatic quick change coupling fixed to 4th axis provision	
9	Ensure no oil leakage in hydraulic valves & fittings	
10	Ensure hydraulic flow control valve grub locking in APC machine.	NA
11	Ensure pneumatic counter balance bracket Dowelling.	
G	GUARDING ASSEMBLY	
1	Ensure all internal guards are assembled.	✓
2	Ensure smooth movement of sliding door without play and unhealthy noise	✓
3	Ensure Nylock nuts are fixed to door rollers.	✓
4	Ensure ATC door LM rail plugs fixed.	NA
5	Ensure Proper tuning of Front auto door / ATC auto door cylinder.	✓
H	ANY OTHER POINTS	
	ATC polycarbon sheet.	
	Circuit, Direction & Arro name plate	
	Bare Guard	
	Auto door cylinder cover.	
NAME & SIGN	Nagaraju B.M CHECKED BY — APPROVED BY	5... 7 Page No.

ACE MANUFACTURING SYSTEMS LTD

GEOMETRICAL TEST CHART(VMC / CMV)

AMS — By Pursuit of Excellence

PERMISSIBLE TOLERANCE AS PER TEST CHART (DS-DSN-PROD-TC-01)

1. MACHINE LEVEL READING:

Along X-Axis	Along Y-Axis
0	0
0	0
0	

2. TABLE TOP READINGS:

Along X-Axis			Along Y-Axis		
0	0	0	-5	0	-6
-4	-3	0	-8	-5	-10
-4	-5	0	0	0	0

3. T SLOT READING:

0	-4	-6

PALLET - II

NA

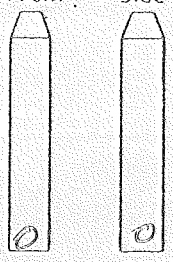
PALLET - II

Along X-Axis	Along Y-Axis
NA	NA

4. COLUMN READINGS :

FRONT	SIDE
+15	-8
0	0

5. MANDREL READINGS:

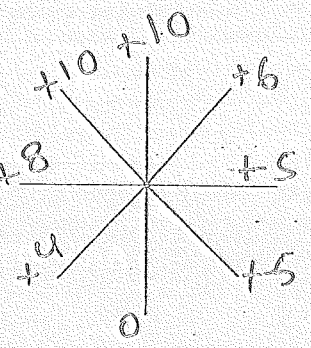


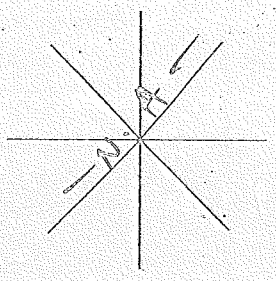
RUNOUT:

a. TOP : 1. 0.003

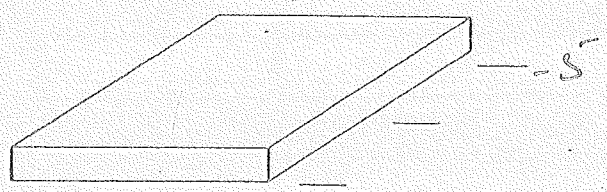
b. BOTTOM : 1. 0.006
2. 0.007
3. 0.005
4. 0.008

6. TURN AROUND READING:





7. SQUARENESS READING (X&Y):



0 +5 0

8. PALLET HEIGHT VARIATION:

Pallet-1:

Pallet-2 : NA

SIGN & DATE	Rajan B M		6.... 7	F-QA-FA-05-R1
	CHECKED BY	APPROVED BY	PAGE NUMBER	DOCUMENT NUMBER

ACE MANUFACTURING SYSTEMS LTD		AMS In Pursuit of Excellence
CHECKLIST - BASE BUILD ASSEMBLY (To be ensured by Final Assembly)		
M/c No. : MF-932	B.B.Team Leader :	Date :

S.No.	CHECK POINTS	STATUS
1	Ensure all leveling Screws & Lock Nuts are fixed to Base.	✓
2	Check for proper tightening of Ball Screw Lock Nut grub screws & marked with Yellow for X, Y & Z - axes Motor end & Brake end.	✓
3	Check for proper tightening of Motor Mounting Bracket, Axis Motor & its coupling screws & Yellow marking (X, Y & Z - Axes)	✓
4	Check for Screw tightening & for Pinning of Ball screw housings & Grub locking with yellow marking (X, Y & Z - Axes)	✓
5	Ensure all Nylon stoppers are being fixed (XYZ - Axes - Motor & Brake end	✓
6	Check for proper Lubrication of guideways & no leakages from fittings/pipe	✓
7	Check for proper tightness of machine lifting bracket & Yellow marking	✓
8	Check for proper machining of "C" frame seating area on base	✓
9	Check for proper machining of seating face of ATC Mounting Bracket & fouling at corners	✓
10	Check for ATC Bracket / Side arm fixing screws tightness & Yellow marking & doweling of side arm	✓
11	Check for Ball Nut Housing tightness & Yellow marking	✓
12	Check for any extra holes are being plugged on base	✓
13	Check for day light area (Refer Work Order)	✓
14	Check for any Dent / Scratch marks on Table top face	✓
15	Check for parallelism of block extension / tightness / Yellow marking	✓
16	Check for proper tightening of Keeper plates screws & Yellow marking	NA
17	Check for locking / tightening of Taper jib screws & Yellow marking	NA
18	Assembly of Telescopic covers of all axes (X, Y & Z axes) Idle running of all axes after fitment (to be carried out by Base Build)	✓
19	Check for geometrical accuracies of Base build	✓
	Remarks if any	
NAME & SIGN	Rajan R.M. CHECKED BY — APPROVED BY	7 ... 7 Page No.

CONTROLLED COPY
Sign: Date: 21/3/12

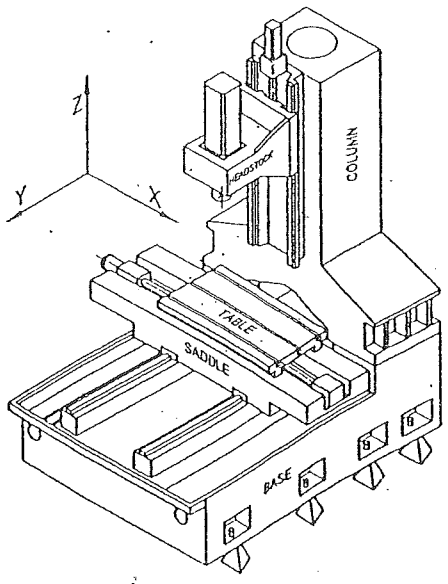
ACE MANUFACTURING SYSTEMS LTD

AMS
In Pursuit of Excellence

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

PRODUCT	:	VERTICAL MACHINING CENTER (FIXED COLUMN)
MODEL NAME	:	MCV - 400
MACHINE NUMBER	:	M11 - 932
DATE OF TESTING	:	15 - 11 - 13



V. G. dd

CHECKED BY
QUALITY ASSURANCE

CHECKED BY
ASSEMBLY - MECHANICAL

CUSTOMER

PMD → Padmanabhan
15/11/13.

WABCO

BSR

SIGN & DATE	BSR	IG	RP	07-02-2012	1 OF 5	DS-DSN-TC-PROD-01-R3
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

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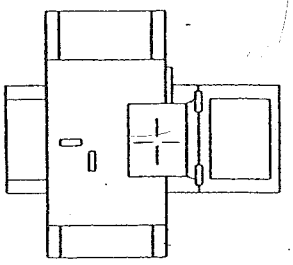
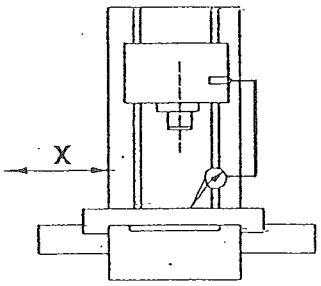
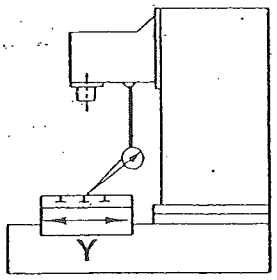
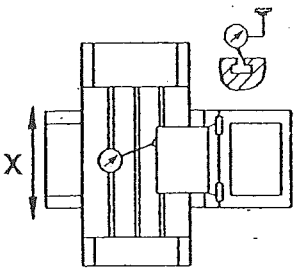
Sign: Date: 21/3/12

ACE MANUFACTURING SYSTEMS LTD

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

All measurements in mm

SL. NO.	Description of the test	Figure	Tolerance	
			Permissible	Actual
G1	LEVELLING OF MACHINE X & Y AXES AT THEIR MIDDLE OF STROKE		0.03 / m ALONG X & Y AXES	LEVEL ALONG X-AXIS : 0 LEVEL ALONG Y-AXIS : 4 micron
G2	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO X-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : .01 mm PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G3	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO Y-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : .012 mm. PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G4	PARALLELISM OF REFERENCE T-SLOT TO X-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : 6 micron PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)

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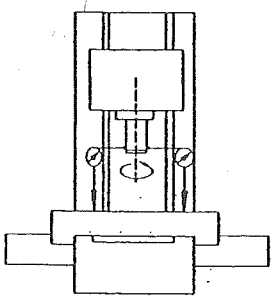
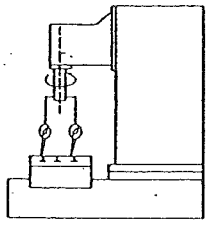
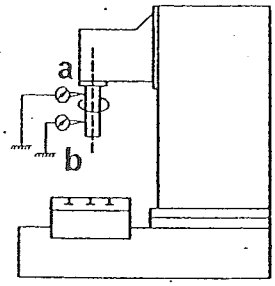
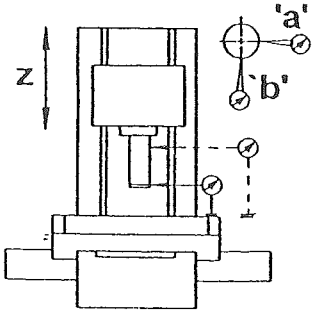
Sign: Date: 21/3/12

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance: Permissible	Actual
G5	SQUARENESS OF SPINDLE AXIS TO TABLE SURFACE IN XZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : .01 / .005/450mm PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G6	SQUARENESS OF SPINDLE AXIS TO TABLE SURFACE IN YZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : .025/450mm PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G7	RUN OUT OF SPINDLE TAPER a) NEAR SPINDLE NOSE b) AT A DISTANCE OF 300 FROM SPINDLE NOSE		a. 0.01 NEAR SPINDLE NOSE b. 0.02 AT A DISTANCE OF 300mm	a. 2 micron b. 4 micron .01 mm
G8	PARALLELISM OF SPINDLE AXIS & Z-AXIS IN a) XZ PLANE b) YZ PLANE TO BE CHECKED AT MEAN POINT OF RUNOUT		a. 0.02 / 300 b. 0.02 / 300	a. 4 micron b. .014 mm

SIGN & DATE	BSR	IG	RP	07-02-2012	3 OF 5	DS-DSN-TC-PROD-01-R3
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

CONTROLLED COPY

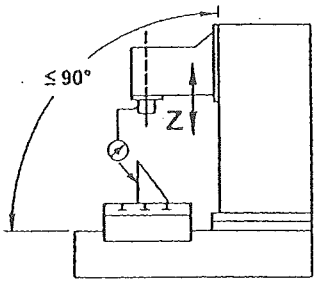
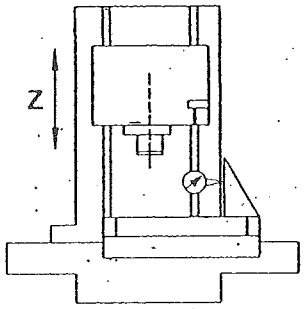
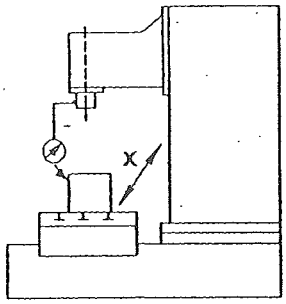
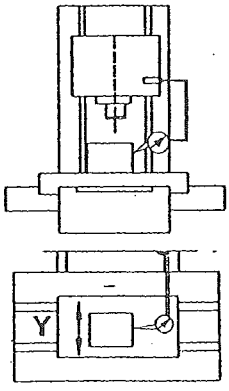
Sign: Date: 21/3/12

ACE MANUFACTURING SYSTEMS LTD

DESIGN VERIFICATION / VALIDATION PLAN CHART

AMS — In Pursuit of Excellence

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance	
			Permissible	Actual
G9	SQUARENESS BETWEEN TABLE SURFACE & Z-AXIS MOVEMENT IN YZ PLANE		0.015 / 300 500	TABLE : + 006 micron. PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G10	SQUARENESS BETWEEN TABLE SURFACE & Z-AXIS MOVEMENT IN XZ PLANE		0.015 / 300 500	TABLE : − 006 mm. PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G11	STRAIGHTNESS OF X-AXIS MOVEMENT IN XY - PLANE		0.010 / 300 500	2micron.
G12	STRAIGHTNESS OF Y-AXIS MOVEMENT IN XY - PLANE		0.010 / 300 500	− 0.01 mm.

ACE MANUFACTURING SYSTEMS LTD.

STANDARD TEST REPORT

1) HEAVY CUT REPORT :

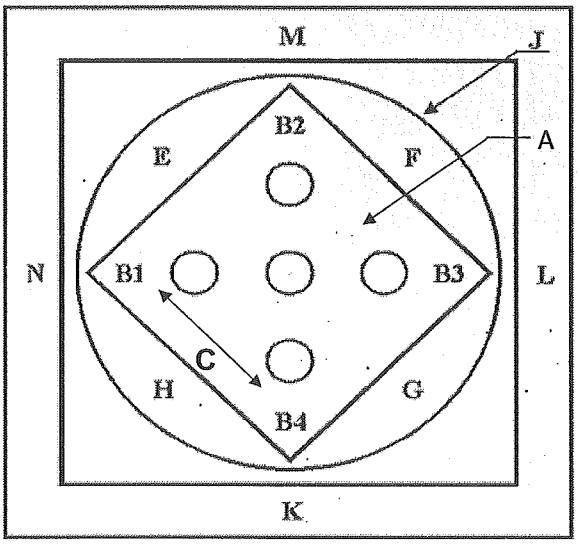
MODEL	: MCV-400	BASE SPEED OF MOTOR	:1100	rpm	MATERIAL	:	C.I.(Gr 25)
M/C NO.	: M11-932	SPINDLE SPEED RANGE	:80-8000	rpm	CUTTER	:	80
TYPE OF MOTOR	: βi 12/8000	MOTOR SPEED RANGE	:80-8000	rpm	WIDTH OF CUT	:	60 mm
PULLEY RATIO	: 1:1	MOTOR CAPACITY	:11/7.5	kW	C.S(V)	:	276.496

Spindle speed in rpm	Feed rate mm/ins.	Feed rate mm/rev.	Feed rate mm/min.	Depth Of Cut in mm	M R R cc/min.	'X' Axis % as per CRT	Spl.load % as per CRT	Remarks
1100	0.15	1.05	1155	2.0	139	21	72	SMOOTH CUTTING
1100	0.15	1.05	1155	4.8	330	39	140	SMOOTH CUTTING

RIGID TAPPING : M8×1.25 RIGID TAPPING TEST CONDUCTED, CHECKED WITH GAUGE & FOUND OK.

2) NAS TEST REPORT :

No.	Description of Test	Test Applied to	Permissible deviation	Actual Readings in mm
1	Flatness of Surface	A	0.015	0.007
2	Parallelism of Surfaces	N-L	0.020	0.008
		M-K	0.020	0.005
		H-F	0.030	0.004
		G-E	0.030	0.007
3	Squareness of side faces	K-L	0.020	0.003
		L-M	0.020	0.004
		M-N	0.020	0.010
		N-K	0.020	0.003
		E-F	0.030	0.014
		F-G	0.030	0.002
		G-H	0.030	0.007
		H-E	0.030	0.005
4	Circularity of Machining	J	0.040	0.007
5	Linear pitch in ' X ' axis	B1-B3 (100)	0.025	-0.006
	Linear pitch in ' Y ' axis	B2-B4 (100)	0.025	-0.001
	Diagonal pitch	C(70.71)	0.035	-0.006
6	Bore size variation		0.020	
	Hole No.	Max.	Min.	Ovality
	B1=	28.984	28.980	0.004
	B2=	28.984	28.981	0.003
	B3=	28.985	28.982	0.003
	B4=	28.985	28.982	0.003
7	Straightness	K	0.010	0.003
		L	0.010	0.002
		M	0.010	0.002
		N	0.010	0.007
		E	0.015	0.009
		F	0.015	0.002
		G	0.015	0.007
		H	0.015	0.004
8	Step difference in axis change over		0.010	



SINGLE AXIS CONTROL K=L=M=N = 240 mm

DOUBLE AXIS CONTROLE=F=G=H = 160 mm

CIRCLE " J " = Ø 236.5 mm

Heavy cut Done By	NAS TEST DONE BY	Department incharge	QC / CMM
MANJUNATH SV	VIJAY		
Date: 5/11/2013	Date:11/11/13	Date:12/11/13	Date: 15/11/13

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-932-X.RTL

Machine:MCV-400.
Date:07:26 Nov 06 2013
Axis:X.

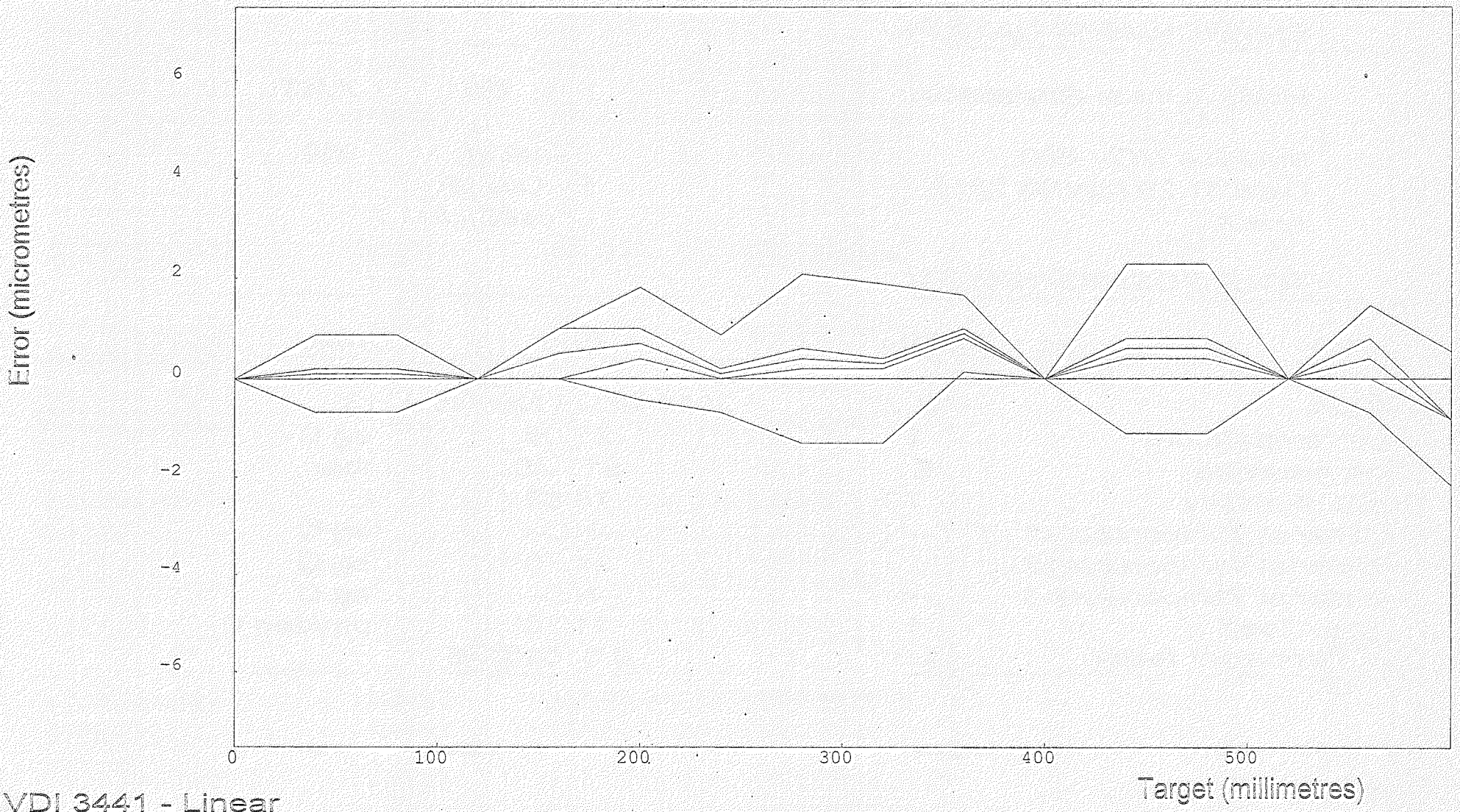
Serial No:M11-932.
By:QAS/RKU.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	07:27 Nov 06 2013	07:36 Nov 06 2013	
Air temperature :	21.08	21.09	deg C
Air pressure :	913.20	913.20	mbar
Rel. humidity :	79.13	78.83	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	24.77	24.70	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31640069	0.31640095	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:X.	U max : 1.000
Serial No:M11-932.	Location:AMS.	P : 4.434
Date:07:26 Nov 06 2013	U mean : 0.288	Ps max : 2.985
By:QAS/RKU.	Ps mean : 1.436	Pa : 1.700

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-932.
Date:07:26 Nov 06 2013 By:QAS/RKU.
Axis:X. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-932-X.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (39.9990 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 2 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

Target 3 (+) (79.9980 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 3 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

Target 4 (+) (119.9970 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 4 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 5 (+) (159.9960 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 5 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 6 (+) (199.9950 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 1.0
Target 6 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 7 (+) (239.9940 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 7 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

Target 8 (+) (279.9930 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 0.0

Target 8 (-)
1) 0.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0

Target 9 (+) (319.9920 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 0.0
Target 9 (-)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 1.0

Target 10 (+) (359.9910 millimetres)
1) 1.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0
Target 10 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 11 (+) (399.9900 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 11 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 12 (+) (439.9890 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 1.0
Target 12 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 13 (+) (479.9880 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 1.0
Target 13 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 14 (+) (519.9870 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 14 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 15 (+) (559.9860 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 15 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 16 (+) (599.9850 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) 0.0
Target 16 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) 0.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-932-Y.RTL

Machine:MCV-400.
Date:08:28 Nov 06 2013
Axis:Y.

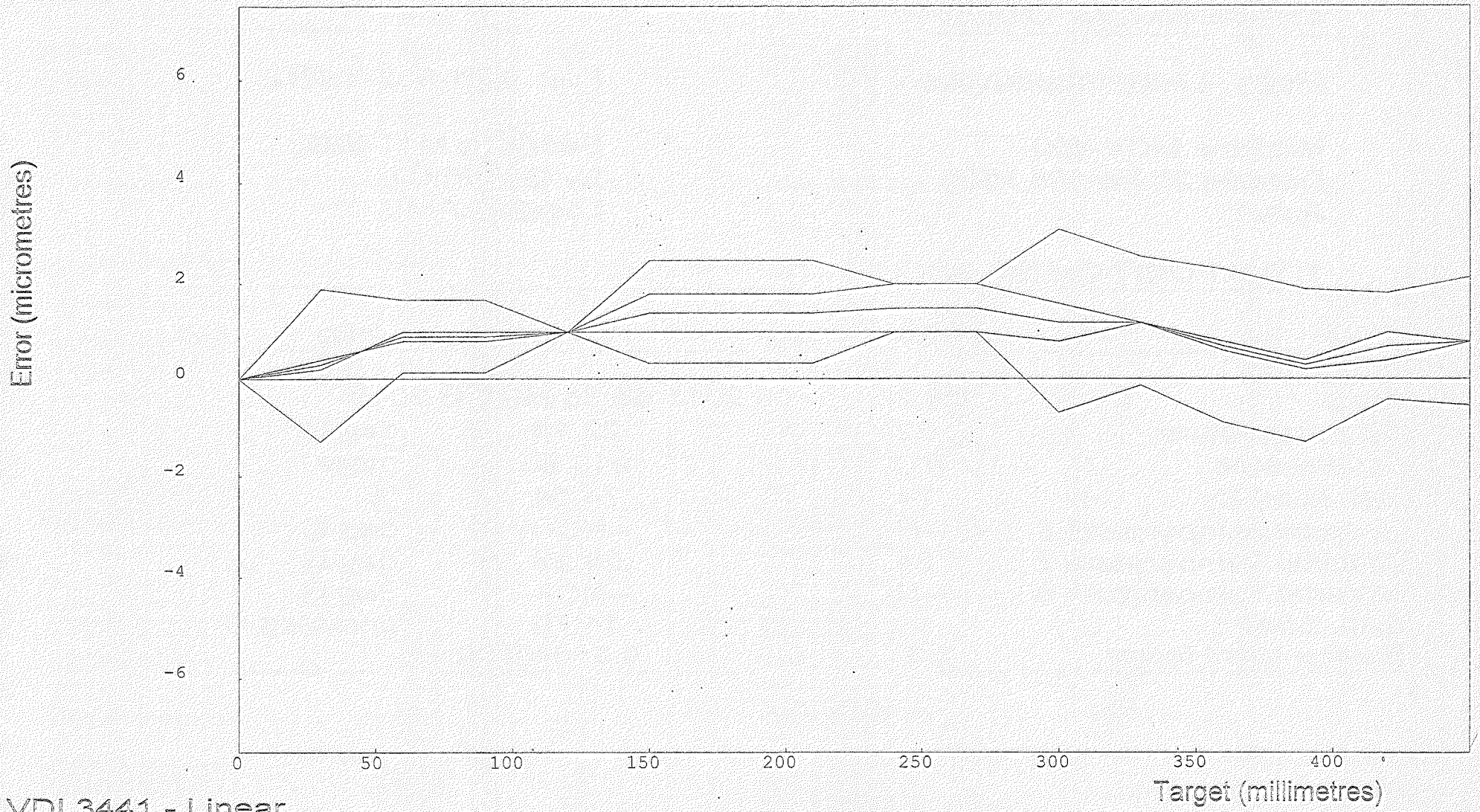
Serial No:M11-932.
By:QAS/RKU.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	08:28 Nov 06 2013	08:38 Nov 06 2013	
Air temperature :	22.03	22.13	deg C
Air pressure :	913.80	913.80	mbar
Rel. humidity :	74.96	74.54	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	24.30	24.25	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31640262	0.31640282	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:Y.	U max : 1.000
Serial No:M11-932.	Location:AMS.	P : 4.385
Date:08:28 Nov 06 2013	U mean : 0.425	Ps max : 2.985
By:QAS/RKU.	Ps mean : 1.604	Pa : 1.500

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-932.
Date:08:28 Nov 06 2013 By:QAS/RKU.
Axis:Y. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-932-Y.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (29.9990 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 0.0
Target 2 (-)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

Target 3 (+) (59.9980 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 3 (-)
1) 1.0 2) 1.0 3) 0.0 4) 1.0
5) 1.0

Target 4 (+) (89.9970 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 4 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 5 (+) (119.9960 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 5 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 6 (+) (149.9950 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 6 (-)
1) 2.0 2) 2.0 3) 2.0 4) 1.0
5) 2.0

Target 7 (+) (179.9940 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 7 (-)
1) 1.0 2) 2.0 3) 2.0 4) 2.0
5) 2.0

Target 8 (+) (209.9930 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 8 (-)
1) 2.0 2) 2.0 3) 2.0 4) 2.0
5) 1.0

Target 9 (+) (239.9920 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 9 (-)
1) 2.0 2) 2.0 3) 2.0 4) 2.0
5) 2.0

Target 10 (+) (269.9910 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 10 (-)
1) 2.0 2) 2.0 3) 2.0 4) 2.0
5) 2.0

Target 11 (+) (299.9900 millimetres)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 11 (-)
1) 1.0 2) 2.0 3) 2.0 4) 2.0
5) 1.0

Target 12 (+) (329.9890 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 2.0
Target 12 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 2.0

Target 13 (+) (359.9880 millimetres)
1) 0.0 2) 1.0 3) 0.0 4) 1.0
5) 1.0
Target 13 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 14 (+) (389.9870 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0
Target 14 (-)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 1.0

Target 15 (+) (419.9860 millimetres)
1) 0.0 2) 0.0 3) 1.0 4) 0.0
5) 1.0
Target 15 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 16 (+) (449.9850 millimetres)
1) 1.0 2) 1.0 3) 0.0 4) 1.0
5) 1.0
Target 16 (-)
1) 1.0 2) 1.0 3) 0.0 4) 1.0
5) 1.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-932-Z.RTL

Machine:MCV-400.
Date:09:53 Nov 06 2013
Axis:Z.

Serial No:M11-932.
By:QAS/RKU.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	09:53 Nov 06 2013	10:02 Nov 06 2013	
Air temperature :	23.45	23.59	deg C
Air pressure :	914.40	914.40	mbar
Rel. humidity :	70.11	70.18	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	24.09	24.09	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31640370	0.31640376	

Target 8 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 9 (+) (266.6640 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 9 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 10 (+) (299.9970 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 10 (-)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 0.0

Target 11 (+) (333.3300 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 11 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 12 (+) (366.6630 millimetres)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 12 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 13 (+) (399.9960 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 13 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

Target 14 (+) (433.3290 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) 0.0
Target 14 (-)
1) 0.0 2) -1.0 3) 0.0 4) 0.0
5) 0.0

Target 15 (+) (466.6620 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 15 (-)
1) 0.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0

Target 16 (+) (499.9950 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 16 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

ACE MANUFACTURING SYSTEM LTD.

AMS
In Pursuit of Excellence

Machine No	:-	M11-932.	Date :- 06:59 Nov 06 2013
Machine Modle	:-	MCV-400.	Cond by:- QAS/RKU.

AXIS CALIBRATION PITCH ERROR REPORT

X				Y				Z			
SL NO 100 (X)	0			SLNO 200 (Y)	0			SL NO 300 (Z)	0		
101	0	141	1	201	1	241	0	301	2	341	1
102	1	142	0	202	1	242	0	302	0	342	0
103	0	143	1	203	1	243	2	303	1	343	1
104	-1	144	0	204	0	244	0	304	0	344	2
105	0	145	1	205	1	245	0	305	2	345	1
106	0	146	1	206	0	246	0	306	1	346	1
107	1	147	0	207	0	247	0	307	0	347	1
108	1	148	0	208	0	248	0	308	1	348	4
109	0	149	1	209	1	249	0	309	0	349	1
110	0	150	0	210	0	250	0	310	1	350	2
111	1	151	0	211	0	251	0	311	1	351	0
112	0	152	0	212	0	252	0	312	1	352	0
113	1	153	-1	213	0	253	0	313	1	353	0
114	1	154	0	214	1	254	0	314	1	354	0
115	0	155	1	215	0	255	0	315	1	355	0
116	0	156	0	216	1	256	0	316	1	356	0
117	0	157	0	217	0	257	0	317	1	357	0
118	1	158	0	218	0	258	0	318	1	358	0
119	0	159	0	219	0	259	0	319	0	359	0
120	1	160	0	220	0	260	0	320	0	360	0
121	0	161	0	221	0	261	0	321	0	361	0
122	0	162	0	222	0	262	0	322	1	362	0
123	0	163	0	223	0	263	0	323	0	363	0
124	0	164	0	224	0	264	0	324	0	364	0
125	1	165	0	225	0	265	0	325	1	365	0
126	0	166	0	226	0	266	0	326	0	366	0
127	0	167	0	227	0	267	0	327	1	367	0
128	-1	168	0	228	0	268	0	328	1	368	0
129	0	169	0	229	-1	269	0	329	1	369	0
130	0	170	0	230	0	270	0	330	0	370	0
131	0	171	0	231	1	271	0	331	1	371	0
132	0	172	0	232	0	272	0	332	1	372	0
133	0	173	0	233	0	273	0	333	1	373	0
134	0	174	0	234	0	274	0	334	0	374	0
135	0	175	0	235	0	275	0	335	1	375	0
136	0	176	0	236	0	276	0	336	0	376	0
137	0	177	0	237	1	277	0	337	1	377	0
138	0	178	0	238	0	278	0	338	1	378	0
139	0	179	0	239	0	279	0	339	2	379	0
140	1	180	0	240	1	280	0	340	1	380	0

SL NO	PARARMETER NO	DISCRIPTION	X	Y	Z
1	3620	P COMP REF NO	160	245	350
2	3621	P COMP MIN NO	100	200	300
3	3622	P COMP MAX NO	165	250	355
4	3623	PCOMP MULTIPLICATION	10	10	10
5	3624	P COMP INTERVAL	10	10	10

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M/c NO :- M11-932.	COND BY:- QAS/RKU.
Machine Modle :- MCV-400.	DATE:- 07-Nov-13

AXIS CALIBRATION BALL BAR PARAMETERS

SERVO PARAMETERS RELATED TO CIRCULARITY FOR FANUC SYSTEM

SL No.	PARAMETER No.	TYPICAL VALUE	DESCRIPTION	X	Y	Z	REMARKS
1	1851	###	BACKLASH IN FEED	25	55	18	
2	1852	###	BACKLASH IN RAPID	15	30	8	
3	1825	8000	SERVO LOOP GAIN	5000	5000	5000	
4	1826	500	IN POSITION WIDTH	500	500	500	
5	2021	256	LOAD INERTIA	192	128	161	******
6	2048	600 (###)	BACK ACCEL COMPENSATION VALUE	20	-50	30	
7	2094	600 (###)	BACK ACCEL COMPENSATION VALUE	20	-50	30	
8	2066	10 TO -20	250 MICRO SEC ACC FEEDBACK	-10	-10	-10	******
9	2068	70 TO700	FEED FWD CO- EFFICIENT	100	100	100	
10	2069	50 TO 200	VELOCITY FEED FWD CO- EFFICIENT FOR RAPID	100	100	100	
11	2071	50 TO 200	TIME DURING WHICH BACK ACCELERATION IS EFFECTIVE	100	100	100	
12	2145	100	VELOCITY FEED FWD CO- EFFICIENT FOR CUTTING	100	100	100	SHOULD NOT BE CHANGED
13	2144	9900	LOOK AHEAD FFW COEFFICIENT FOR CUTTING	9900	9900	9900	
14	2092	9200	LOOK AHEAD FFW COEFFICIENT FOR RAPID	9200	9200	9200	
15	2082	100	STAGE 2 START / END PARAMETER	100	100	100	
16	2039	100	STAGE 2 ACCELERATION AMOUNT	300	300	300	
17	1622	NA	CUT TIME CONSTANT	64	64	64	Same value for 3 axis

NOTE:-
****** DON'T CHANGE THE VALUE
VARIES FOR ALL MACHINES

Ballbar diagnostics table

M11-932-XY-B

Operator: Ramesh.k.ukkali.
Date: 2013-Nov-06 10:57:14

Machine: Quick check
QC20-W: 27K711, Last calibrated: 2010-12-15

Error	Magnitude		Independent circularity		Ranking
Backlash X	▸0.0	◂-0.1µm	0.1µm	(0%)	(13)
Backlash Y	▲0.1	▾0.8µm	0.8µm	(5%)	(7)
Reversal spikes X	▸2.6	◂2.6µm	2.6µm	(18%)	(2)
Reversal spikes Y	▲-3.6	▾-5.0µm	5.0µm	(33%)	(1)
Lateral play X	▸0.7	◂-0.6µm	0.4µm	(2%)	(10)
Lateral play Y	▲0.2	▾-0.1µm	0.1µm	(1%)	(12)
Cyclic error X	↑0.7	↓0.6µm	0.6µm	(4%)	(9)
Cyclic error Y	↑0.9	↓0.9µm	0.8µm	(5%)	(6)
Servo mismatch	0.04ms		0.6µm	(4%)	(8)
Squareness	-1.9µm/m		0.3µm	(2%)	(11)
Straightness X	-1.7µm		0.8µm	(6%)	(5)
Straightness Y	-1.7µm		0.9µm	(6%)	(4)
Scaling mismatch	4.0µm		2.0µm	(13%)	(3)
Scaling error X	30.4ppm				
Scaling error Y	17.1ppm				
Cyclic pitch X	50.8000mm				
Cyclic pitch Y	45.0000mm				
Calculated feedrate	1000.8mm/min				
Centre offset X	-2.2µm				
Centre offset Y	-6.2µm				
Positional tolerance	21.7µm				
Best fit radius	150.0036mm				
Circularity	12.5µm				

Test parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW

ISO 230-4:2005(E)

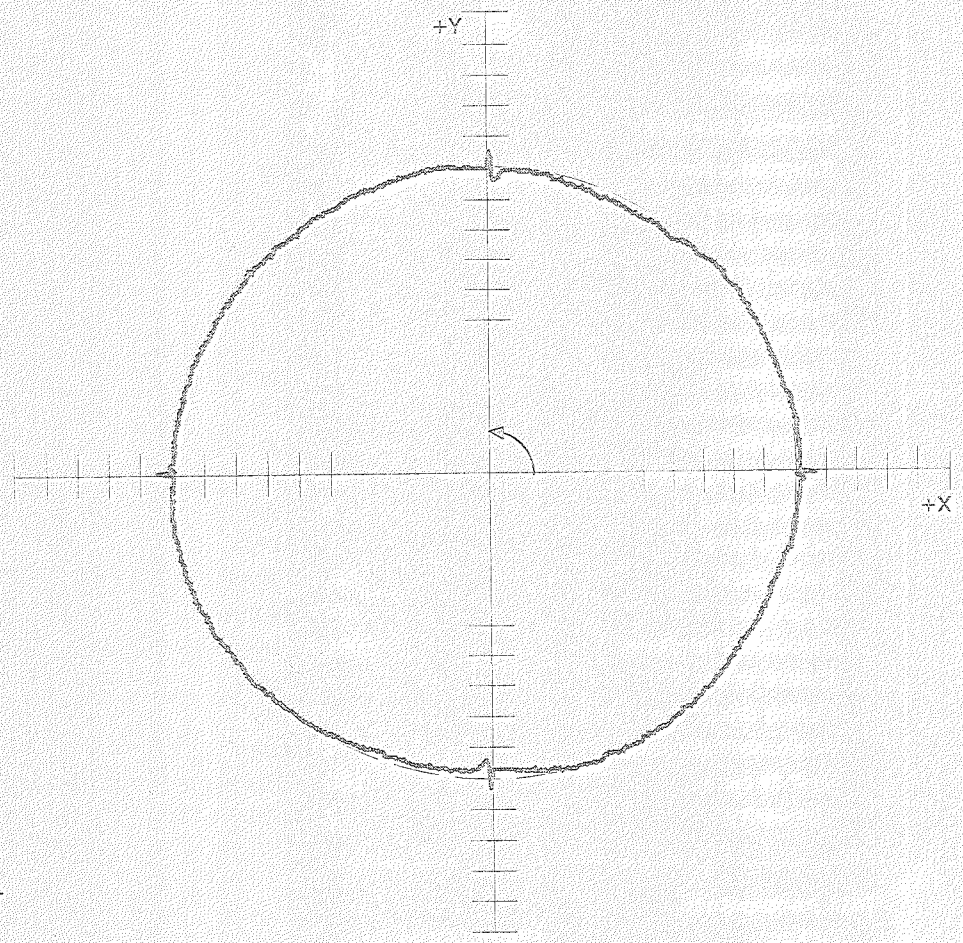
Circular deviation
M11-932-XY-B

Operator: Ramesh.k.ukkali.
Date: 2013-Nov-06 10:57:14

RENISHAW

Machine: Quick check
QC20-W: 27K711, Last calibrated: 2010-12-15

Circular deviation (CW)	
Value	12.5µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	XY
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

10.0µm/div

ISO 230-4:2005(E)

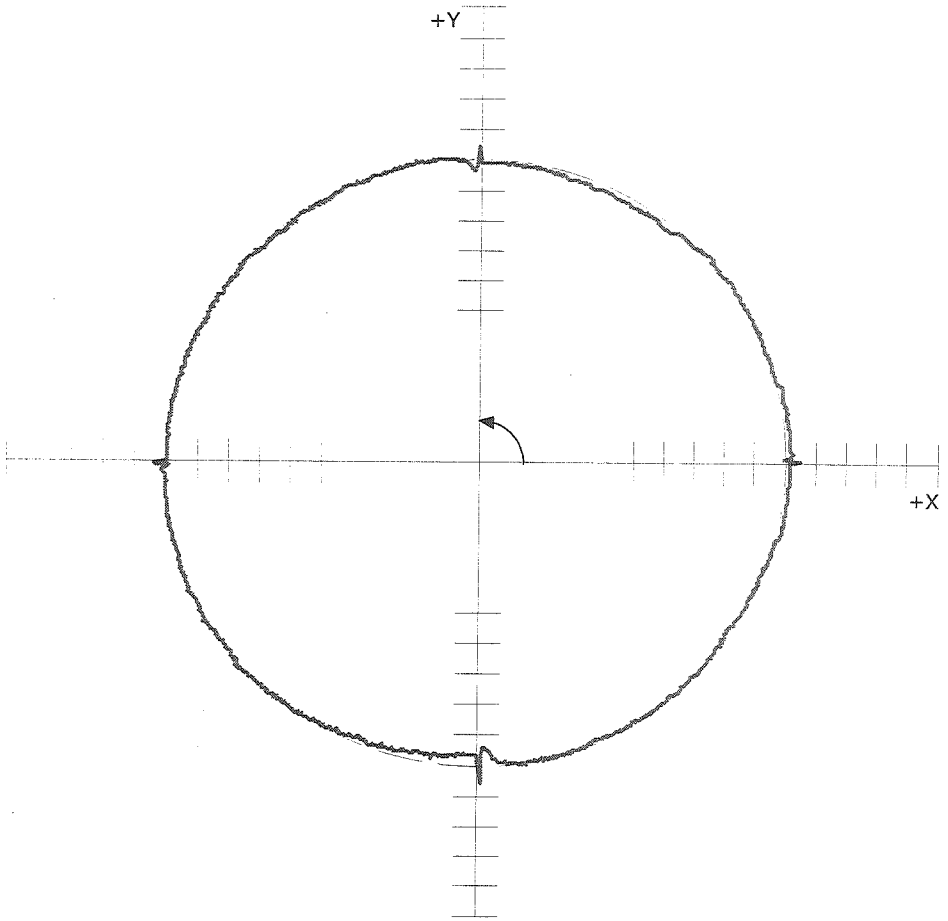
Circular deviation
M11-932-XY-B

Operator: Ramesh.k.ukkali.
Date: 2013-Nov-06 10:57:14

RENISHAW

Machine: Quick check
QC20-W: 27K711, Last calibrated: 2010-12-15

Circular deviation (CCW)	
Value	11.3µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	XY
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

10.0µm/div

Ballbar diagnostics table

M11-932-XZ-B

Operator: Ramesh.k.ukkali.
Date: 2013-Nov-06 11:21:08

RENISHAW

Machine: Quick check
QC20-W: 27K711, Last calibrated: 2010-12-15

Error	Magnitude		Independent circularity		Ranking
Backlash Z	▵0.0	▿0.8µm	0.8µm	(4%)	(8)
Backlash X	‹ 0.0	› 0.0µm	0.0µm	(0%)	(13)
Reversal spikes Z	▵-1.1	▿-1.8µm	1.8µm	(9%)	(5)
Reversal spikes X	‹ 2.8	› 0.5µm	2.8µm	(14%)	(3)
Lateral play Z	▵-0.5	▿0.0µm	0.4µm	(2%)	(11)
Lateral play X	‹ -0.3	› 0.3µm	0.1µm	(1%)	(12)
Cyclic error Z	↑1.3	↓1.2µm	1.2µm	(6%)	(6)
Cyclic error X	↑0.9	↓0.9µm	0.9µm	(5%)	(7)
Servo mismatch	-0.03ms		0.5µm	(2%)	(10)
Squareness	-26.3µm/m		3.9µm	(20%)	(1)
Straightness Z	5.4µm		2.7µm	(14%)	(4)
Straightness X	7.6µm		3.8µm	(19%)	(2)
Scaling mismatch	-1.2µm		0.6µm	(3%)	(9)
Scaling error Z	75.7ppm				
Scaling error X	79.8ppm				
Cyclic pitch Z	45.0000mm				
Cyclic pitch X	22.5000mm				
Calculated feedrate	1000.8mm/min				
Centre offset Z	0.4µm				
Centre offset X	-0.9µm				
Positional tolerance	70.5µm				
Best fit radius	150.0117mm				
Circularity	12.4µm				

Test parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	90°/90°/180°
Run sequence	CW CCW

ISO 230-4:2005(E)

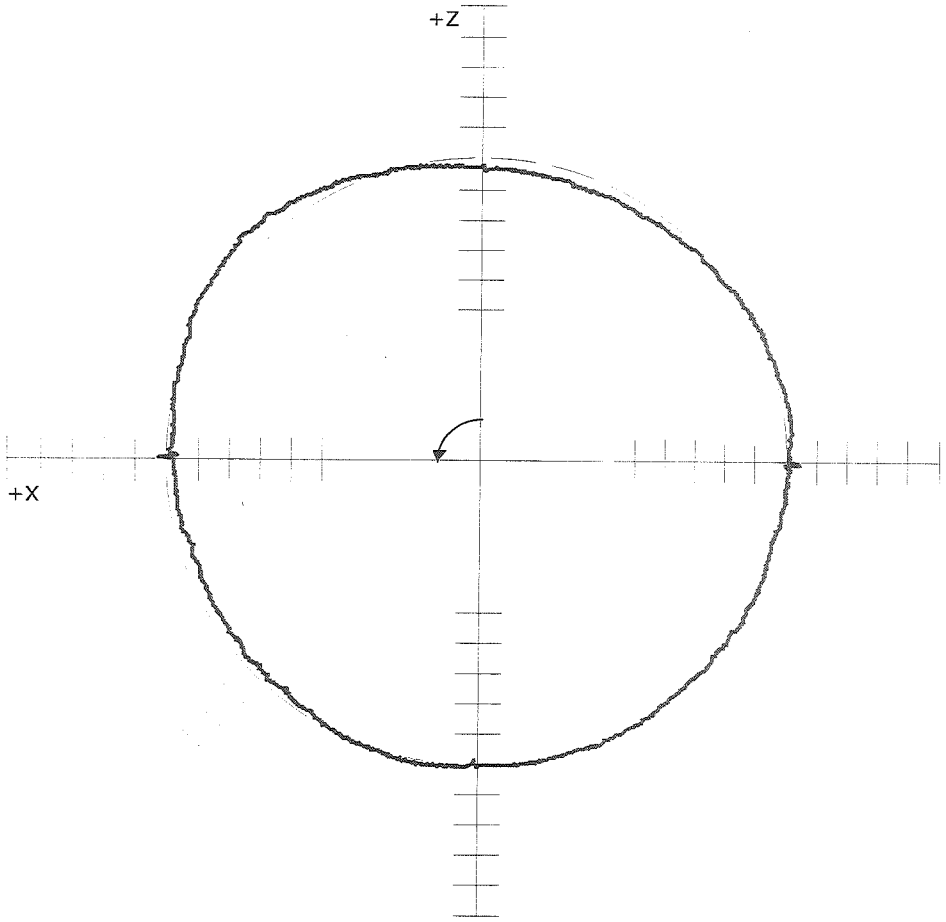
RENISHAW

Circular deviation
M11-932-XZ-B

Operator: Ramesh.k.ukkali.
Date: 2013-Nov-06 11:21:08

Machine: Quick check
QC20-W: 27K711, Last calibrated: 2010-12-15

Circular deviation (CW)	
Value	11.5µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 1

10.0µm/div

ISO 230-4:2005(E)

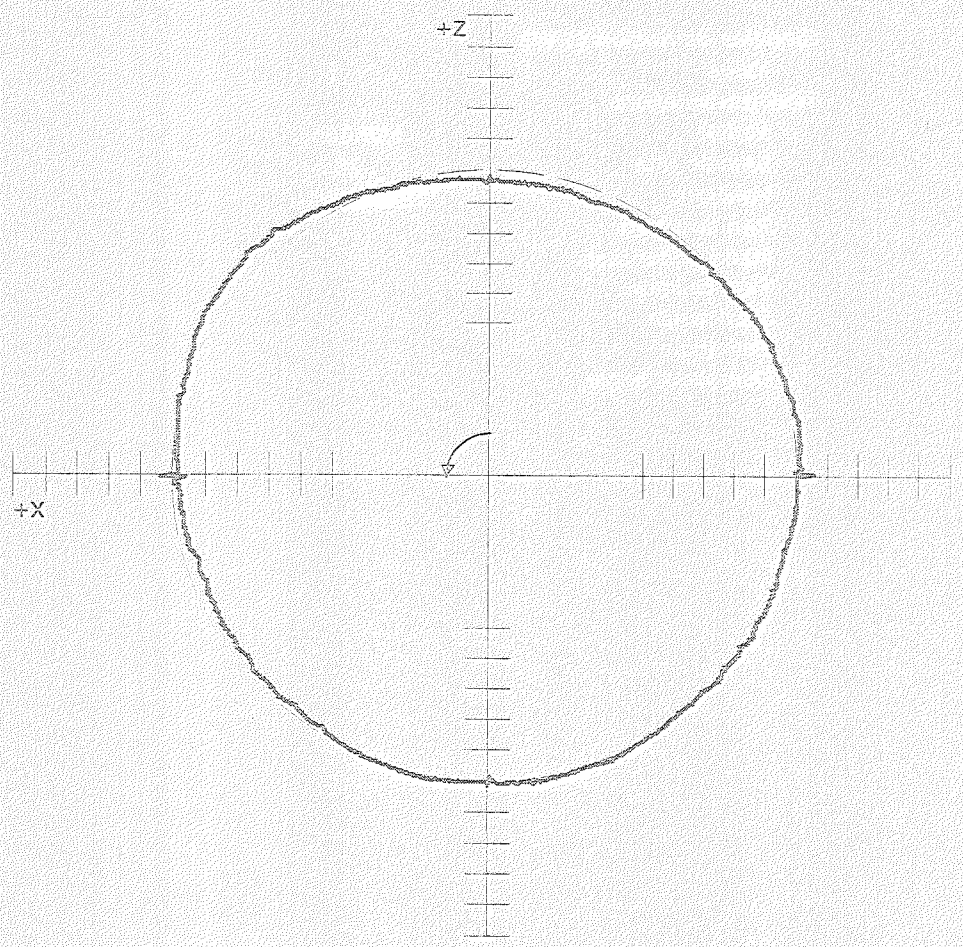
Circular deviation
M11-932-XZ-B

Operator: Ramesh.k.ukkali.
Date: 2013-Nov-06 11:21:08

RENISHAW

Machine: Quick check
QC20-W: 27K711, Last calibrated: 2010-12-15

Circular deviation (CCW)	
Value	11.2µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 2

10.0µm/div

Ballbar diagnostics table

M11-932-YZ-A

Operator: Ramesh.k.ukkali.
Date: 2013-Nov-06 11:38:00

Machine: Quick check
QC20-W: 27K711, Last calibrated: 2010-12-15

Error	Magnitude		Independent circularity		Ranking
Backlash Y	▸0.7	◂0.0µm	0.7µm	(5%)	(8)
Backlash Z	▴-0.3	▾0.6µm	0.6µm	(5%)	(9)
Reversal spikes Y	▸0.5	◂0.8µm	0.8µm	(6%)	(5)
Reversal spikes Z	▴-1.1	▾-1.8µm	1.8µm	(13%)	(2)
Lateral play Y	▸-0.1	◂0.1µm	0.1µm	(0%)	(12)
Lateral play Z	▴1.1	▾0.0µm	0.7µm	(5%)	(7)
Cyclic error Y	↑0.9	↓0.9µm	0.8µm	(6%)	(6)
Cyclic error Z	↑1.0	↓0.8µm	0.9µm	(7%)	(4)
Servo mismatch	0.00ms		0.0µm	(0%)	(13)
Squareness	2.3µm/m		0.4µm	(3%)	(10)
Straightness Y	10.2µm		5.1µm	(38%)	(1)
Straightness Z	-2.5µm		1.2µm	(9%)	(3)
Scaling mismatch	-0.3µm		0.2µm	(1%)	(11)
Scaling error Y	45.9ppm				
Scaling error Z	46.9ppm				
Cyclic pitch Y	45.0000mm				
Cyclic pitch Z	45.0000mm				
Calculated feedrate	1000.8mm/min				
Centre offset Y	0.5µm				
Centre offset Z	-2.1µm				
Positional tolerance	44.8µm				
Best fit radius	150.0070mm				
Circularity	12.7µm				

Test parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW

ISO 230-4:2005(E)

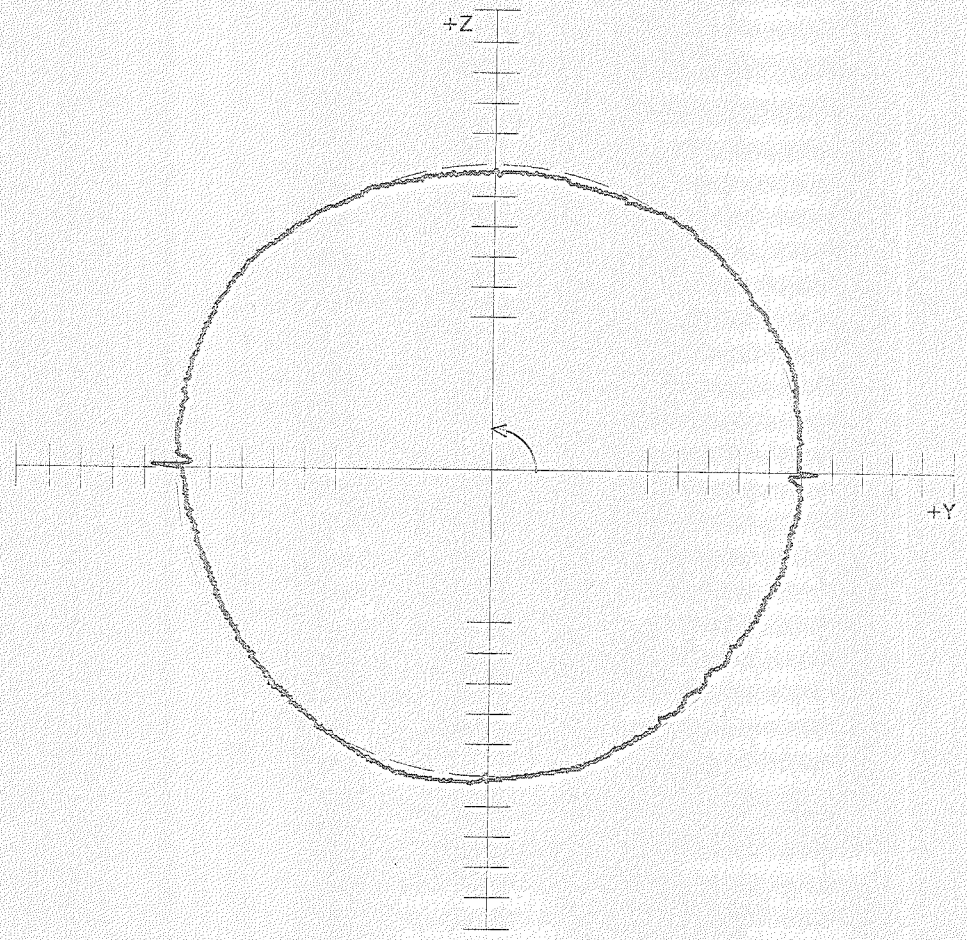
Circular deviation
M11-932-YZ-A

Operator: Ramesh.k.ukkali.
Date: 2013-Nov-06 11:38:00

RENISHAW

Machine: Quick check
QC20-W: 27K711, Last calibrated: 2010-12-15

Circular deviation (CW)	
Value	11.6µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

10.0µm/div

ISO 230-4:2005(E)

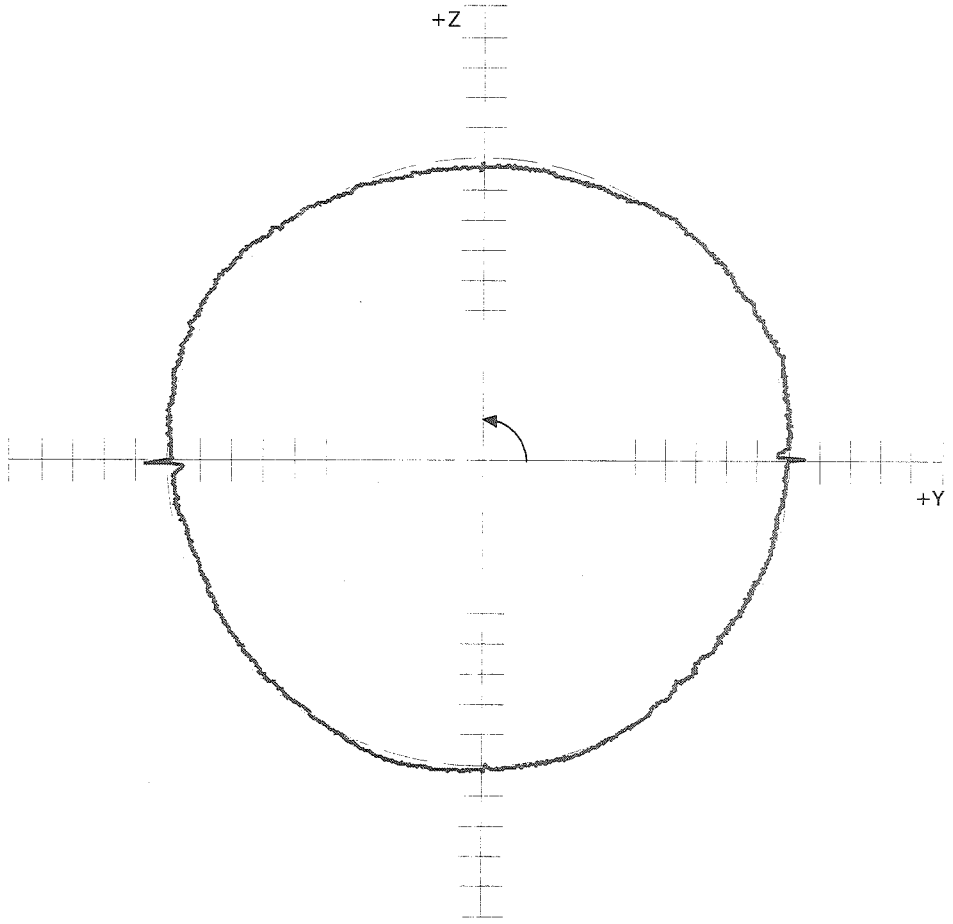
RENISHAW

Circular deviation
M11-932-YZ-A

Operator: Ramesh.k.ukkali.
Date: 2013-Nov-06 11:38:00

Machine: Quick check
QC20-W: 27K711, Last calibrated: 2010-12-15

Circular deviation (CCW)	
Value	11.9µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

10.0µm/div

Nu-932

BZ-1391-13

TEST CERTIFICATE

CENLUB
ENGINEERS OF CENTRALISED LUBRICATION

CENLUB INDUSTRIES LTD.
45-B, 2nd Phase, Peenya Industrial Area,
BANGALORE - 560 058.
Telephone : 23723680, 65584624, 41273544
Telefax : 28374704, E-mail : cenlub@rediffmail.com

Customer Name :- **ACE Manufacturing Systems Pvt Ltd**
DC No. & Date :- **B00925 Date-04.10.2013**
Your P.O. No. Date :- **KA-00-P13-003472 Date-09.08.2012**

With 5ltrs tank

IDENTIFICATION : ALU-03 / ELU / ALU - 05 / ALU - 08
Model of Unit : CIL-B- **2248/13**
Serial No. of Unit : " **Rajmane**' make No. **1411257**
Details of Motor : **0.09 KW / 0.12KW / 0.18KW**

TESTS

Fill unit with Oil (Way lube Oil 68) run for 1/2 hour without load (by putting full flow of pump back to Tank) and observe

e) Abnormal noise : No
f) Abnormal heating of Motor : No
g) Leakage from joints : No
h) Discharge of Pump : **1.12** ltrs./Min

2. Apply load to 25 bar by adjusting pressure relief valve and run the unit of 5 minutes and observe :

f) Abnormal noise : No
g) Leakage from joint : No
h) Stability of pressure : with in 0.5 bar
j) Discharge by pump : **1.10** ltr./Min.
j) Current drawn by motor : **0.42A 0.44A 0.44A**

3. Set Pressure of 15 bar and pressure switch at 8/12 bar, connect pressure switch / timer for intermittent Operation of unit wity cycle of 10 sec. ON, 1 minute OFF: Run for 1 hour and observe:

g) Variation in pressure setting : 0.5 bar
h) Actuation of pressure switch : OK ("Suco"/ "Minihyd" / EURO)
i) Functioning of Pr. reducing valve : OK
j) Functioning of bleed valve : OK
k) Leakage from J Block Joint : No
l) Functioning of oil Pr. Gauge : OK (0-28 Kg/cm2)

4. To check following

f) Leakage of oil from reservoir : No
g) Operation of Float Switch : OK / Vertical 145L / 165L / 175 L / 230 L
h) Functioning of oil level gauge : OK
i) Workmanship : Satisfactory
j) Painting / Plating / Stickers : Satisfactory

Tested by

Approved by

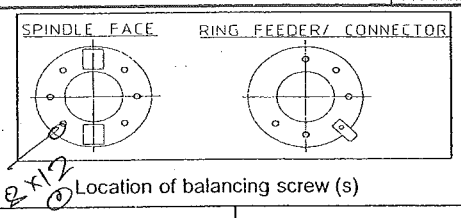
ACE MANUFACTURING SYSTEMS LTD

SPINDLE BALANCING REPORT

AMS In Pursuit of Excellence

Machine Number :	VMC-932
Spindle Serial Number :	13966
Spindle Type :	BD 45
Spindle Speed (Max.) :	8000 rpm

Note : Refer TM-QA-09 for measurement procedure.



Location of balancing screw (s)

Sl No.	Speed in RPM	Velocity of vibration (mm/sec) before Balancing Along X-axis	After Balancing along X-axis <8000 rpm : 0.8 mm/sec max. > 8000 rpm : 1 mm/sec max.	After Balancing along Y-axis <8000 rpm : 0.8 mm/sec max. > 8000 rpm : 1 mm/sec max.
1	500			
2	1000	0.009	0.007	0.005
3	1500	0.005	0.023	
4	2000	0.007	0.004	0.009
5	2500	0.035	0.031	
6	3000	0.054	0.027	0.026
7	3500	0.170	0.130	
8	4000	0.190	0.201	0.019
9	4500	0.334	0.300	
10	5000	0.887	0.710	0.128
11	5500	0.554	0.458	
12	6000	0.360	0.426	0.052
13	6500	0.134	0.280	
14	7000	1.61 – 18°	0.415	0.112
15	7500	1.52 – 309°	0.655	
16	8000	1.20 – 309°	0.658	0.182
17	8500			
18	9000			
19	9500			
20	10000			
21	10500			
22	11000			
23	11500			
24	12000			
25	12500			
26	13000			
27	13500			
28	14000			
29	14500			
30	15000			

Spindle max. vibration mm / sec : **Correction in gms :**

REMARK IF ANY:

nil

Date: 06/11/13 Checked By APPROVED BY

ACE MANUFACTURING SYSTEMS LTD

SPINDLE ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

Machine Number : M11-932	Spindle Unit Number: 13-966
Spindle Drive : ID / ~~ID-CTS~~ / DD / DD-CTS	Spindle Diameter : 70 mm
Pull stud type : DIN / MAS	Max. Spindle Speed : 8000 RPM
Bearing Make & Specification : FAG/HCS7014CTP4SUL	Draw bar Sl. No : 1190
Spindle Number: CH-383	Cartridge Number : 1143

Spindle Nose Taper : ISO 30 / 40 / 50 HSK A50 / A63 / A100

Sl No.	Description	Recommended	Actual
1	Locknut Tightening Torque	Ref.DS-PDN-MEC-SA-08	115Nm
2	Spindle Bearing Preload	Ref.DS-PDN-MEC-SA-07	-0.008mm
3	Spindle Run out on taper	0.002 mm (*)	0.002mm
4	Mandrel Run out near the spindle nose	0.003 mm (*)	0.002mm
5	Mandrel Run out at 4 positions @ 90° at a distance of 300 mm (ISO 40/50, HSK 63/100), 250mm (ISO 30, HSK 50) from spindle nose	10 micron (*)	1. 5 2. 4 3. 4 4. 5
6	Spindle stabilization temperature on Test rig	Up to 15° C above ambient	32°C-27°C=5°C
7	Check for abnormal / unhealthy noise	Abnormal / unhealthy noise not acceptable	OK
8	Pulley ratio (For ID machines)	Refer Work order	1:1 Ratio
9	Spindle pulley Run out (For ID machines)	0.050 mm (*)	0.050mm
10	Spindle connector Run out (For DD machines)	0.010 mm (*)	—
11	Disc spring specification (Single / double stack)	Ref.DS-DSN-PROD-01	Φ140xΦ90.8x3mm
12	Number of disc springs used	Ref.DS-DSN-PROD-01	56Nos
13	Disc spring Compression, Free length & Compressed length	Ref.DS-DSN-PROD-01	031-018=13mm
14	Tool clamping force	Ref.DS-DSN-PROD-01	8kg/cm²
15	ID-CTS Deublin Run out in Assembly condition (For CTS Spindle)	0.030 mm	Deublin Sl. No. : Run out : 0.020mm

(*) To be measured after temperature stabilization of spindle at max. speed.

REMARK IF ANY:

TEST ID:- 0094

Date: 31/10/13	Checked By	Approved By

ACE MANUFACTURING SYSTEMS LTD

HYDRAULIC ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

MACHINE NUMBER : M11-932 WABCO Rexroth 70 Bar Fixture

I. Power Pack Details			II. Pump Details	
1	Make :	Pascal / Hyd Tech / ~~Rexorth~~ / Yuken /Jai Chamundi	Pump Make :	~~Rexroth~~ / yuken /Corotech / vickers
2	Sl No. :	AP - 9280	Pump Sl. No. :	VV 40474
3	Tank capacity :	32 / ~~40~~ / 63 / 70 Liters	Motor Make :	~~Rotomotive~~ / Siemens
4	Cooler make :	Pascal System Technik Pvt Ltd	Motor Sl. No :	N° MO2132333
5	Cooler fan make :	EBM NADI	—	—
6	No.of stations :	1 / 2 / 3 / ~~4~~ / 5 / 6	✓	—

III	CHECK POINTS	Remark
1	Ensure Power pack is assembled as per circuit diagram & same is attached to inspection report.	✓
2	Ensure Accessories like Minnimess coupling, Guages are assembled.	✓
3	Ensure instruction name plates, power pack details, name plate are fitted	✓
4	All ports are tied with identification plate.	✓
5	Ensure there is no visual damage to the elements.	✓
6	Ensure oil level (at level indicator) in the hyd tank is at satisfactory	✓
7	Ensure tightness of all screws & marking neatly with yellow marker.	✓

IV	RUNNING CHECK POINTS	Remark
1	Check Motor direction is as per arrow marking.	✓
2	Ensure direction of rotation of cooler fan is as per arrow marking	✓
3	Ensure hydraulic pressure switch set as per circuit diagram.	✓
4	Ensure Accumulator oil & gas pressure as per circuit diagram (if applicable).	— NA —
5	Ensure no oil leakage from hydraulic system including at the oil level indicator & pipe joints	✓
6	Ensure continuous running of power pack for 4 hrs with solenoid actuation once per minute	✓
7	Power pack oil temp rise is within 30°C above ambient temp or 65 °C maximum.	✓
8	Check for any abnormal / unhealthy noise	✓

REMARK IF ANY: —

22/11/13 Date:	N. Vee Checked by	Approved by	

ACE MANUFACTURING SYSTEMS LTD.

CHECK LIST

SUB ASSEMBLY-HYDRO PNEUMATIC TOOL DECLAMP ASSEMBLY

AMS
In Pursuit of Excellence

MACHINE NUMBER: M11-932 IDCTS

SERIAL NUMBER AND MAKE: 20997 Preac

CYLINDER SPECIFICATION: 9mm stroke

CONTROLLED COPY
Sign. KM Date : 9/10/12

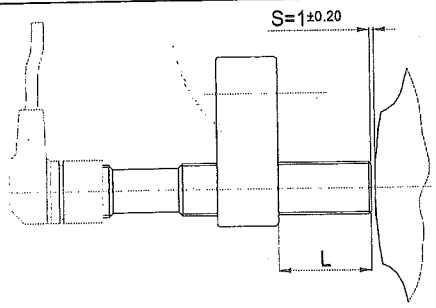
SL.NO	ACTIVITIES TO CHECK	ACTION TAKEN	REMARKS
1.	OIL LEVEL IN THE CONTAINER (FILL TO MAX. OIL LEVEL MARK)	yes	
2.	OIL BLEEDING	yes	
3.	500 CYCLES OF ACTUATION	yes	
4.	OIL LEAKAGE THROUGH BLEED PLUG AFTER 500 CYCLES OF ACTUATION	No	
5.	ALL SILENCER AND QUICK EXHAUST VALVE ARE FIXED	yes	
6.	ANTI ROTATION BRACKET FIXED	yes	
7.	AIR LEAKAGE	NO	
8.	YELLOW MARKING OF ALL SCREWS AND BLEED PLUG	yes	

PRAGATI	AUTOMATIC TOOL CHANGER TEST REPORT	
SL NO: 19458	CUSTOMER I.D:	CUSTOMER: AMS
PRODUCT I.D.: ATC-4030V-053	PRODUCT DESCRIPTION: ATC-4030V-R225 BT40	
DESPATCH DATE: 24-8-13		

TEST RESULTS:

TESTING HOURS	SPECIFIED: 6 Hrs. ✓	ACTUAL: 6 Hours

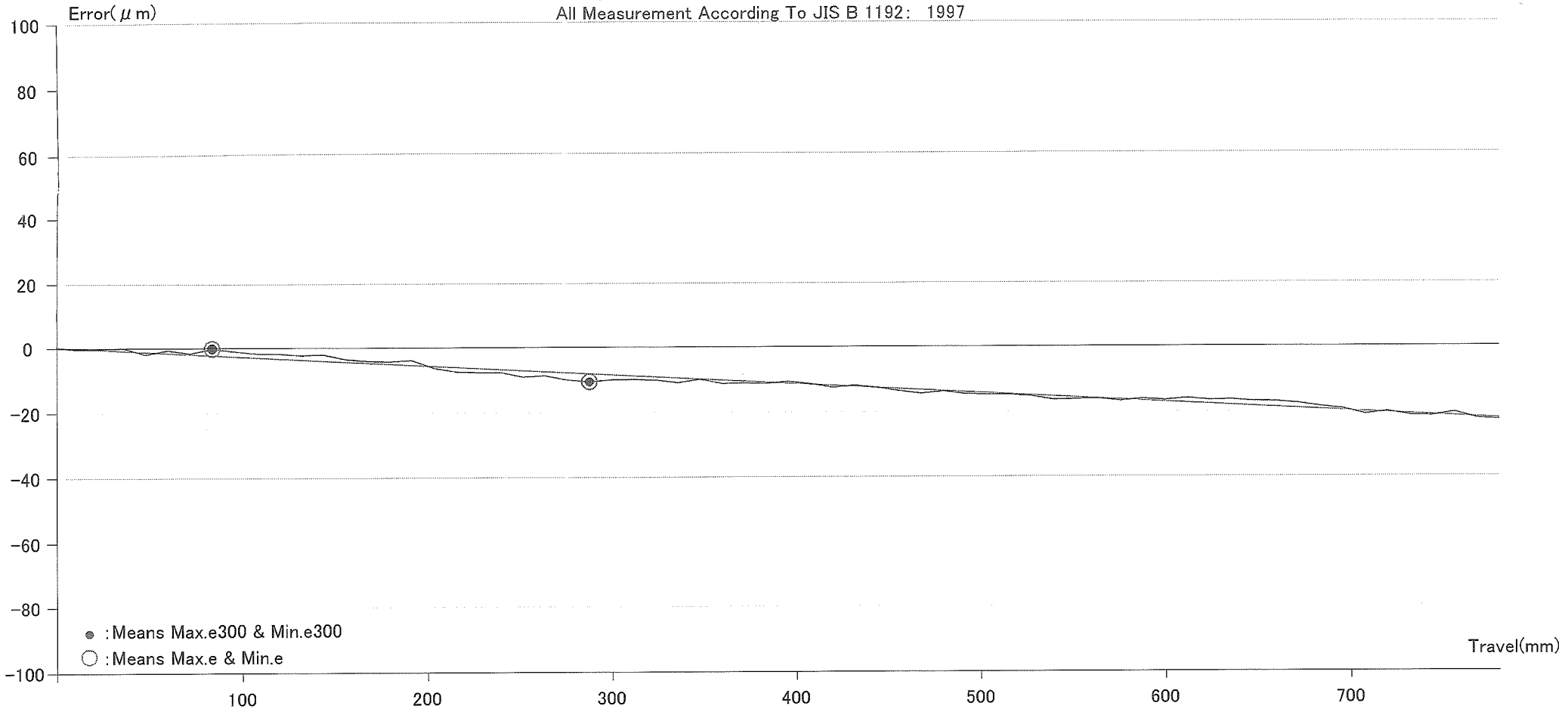
PARAMETERS	SPECIFIED	ACTUAL		
		DURING ASSLY	AFTER 6 Hrs	DURING DESPATCH
ARM STROKE	114-115mm	114.10	114.20	114.20
ARM AXIAL PLAY(IN PARKING)	0.20-1.50mm	0.90	1.00	1.00
ARM RADIAL PLAY @R250 IN PARKING	0.20-1.0mm	0.20	0.30	0.30
ARM RADIAL PLAY @R250 WHILE MOVING OUT	0.20-1.0mm	0.20	0.40	0.40
ARM RADIAL PLAY @R250 WHILE MOVING IN	0.20-1.0mm	0.20	0.40	0.40
ARM PARLLALITY	0-0.20mm	0.05	0.05	0.05
ARM PARK SIGNAL	Visual Check	OK	OK	OK
TOOL DECLAMP SIGNAL	Visual Check	OK	OK	OK
TOOL CLAMP SIGNAL	Visual Check	OK	OK	OK
BACKLASH OF WHEEL @R300	0.05-0.40mm	0.20	0.25	0.25
AXIAL PLAY OF WHEEL	0.01-0.10mm	0.07		

SENSING DISTANCE	PROXIMITY	SENSING GAP(S)	SENSOR LENGTH(L)
	PS1	1.0	23.10
	PS2	1.0	23.20
	PS3	1.0	25.40
	PS4	1.0	30.00
	PS5	1.0	—
	PS6	1.0	—
	PS7	1.0	

DESCRIPTION	NAME	SIGNATURE	DATE
ASSEMBLED BY	Sathish.M.N.		22-8-13
CHECKED BY AFTER 6Hrs	Sathish.M.N.		23-8-13
INSPECTED BY	B.C Boraf	B.C Boraf	23-8-13
APPROVED BY	AKBER		24-8-13

PMI BALLSCREWS — BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	: AMS
·Order No.	: S3-130044661001
·Drawing No.	: BA300510002-0
·Part No.	: *:S12-016-95
·Specification	: R40-12C1-FDWE-846-1110-0.008
·Nominal Diameter (mm)	: 40
·Nominal Pitch (mm)	: 12
·Travel Length (mm)	: 780
·E(μ m)	: ±18
·T(μ m)	: -25

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μ m)	-43～-7	-22.41
·e(μ m)	13	4.56
·e300(μ m)	8	4.56
·Tq(kgf - cm)	4.16～7.28	5.294～6.543
·Accuracy Grade	C3	C1

Manufactured By
PRECISION MOTION IND., INC.
www.pmi-amt.com

Measuring Environment:
·Refernce Temperatrue : 20℃ ± 0.5℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

QC
2013/8/22
PASS

Inspector: ROCKY

M11-932
X'Axis

(001)

ACE MANUFACTURING SYSTEMS LTD

BALL SCREW ASSEMBLY REPORT

AMS — In Pursuit of Excellence

Machine Number : M11-932 | Ball screw Sl No: 931300W4661001 | Axis : X

I. Ball screw Make & Specification : PMI FRM0-1204-F0WE-8N6-1110-0.008

II. Ball Screw Bearing Make & Specification : RAE(BSB-2562-22-SU)

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN - MEC - SA - 02	1.0N/m	1.1N/m
2	Bearing housing Bore diameter	As per drawing	1 to 3μ	2 to 5μ
3	Bearing housing centre height difference	0.01 mm	0.01mm	—
4	Bearing housing bore parallelism with respect to pads	0.01 mm	0.005mm	0.005mm

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	FAG	T3	14.978
4	Bearing width B3	— " —	T4	14.976
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	C24-275-8123	T6	—
7	Housing Bearing depth	C37-103-2123	T7	43.490

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 103/14

T6 = 13.610 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	FAG	T3	14.994
4	Bearing width B3	— " —	T4	14.989
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	C24-360-8123	T6	—
7	Housing Bearing depth	C37-153-2123	T7	43.520

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 153/14

T6 = 13.610 mm

VI. Check points after assembly

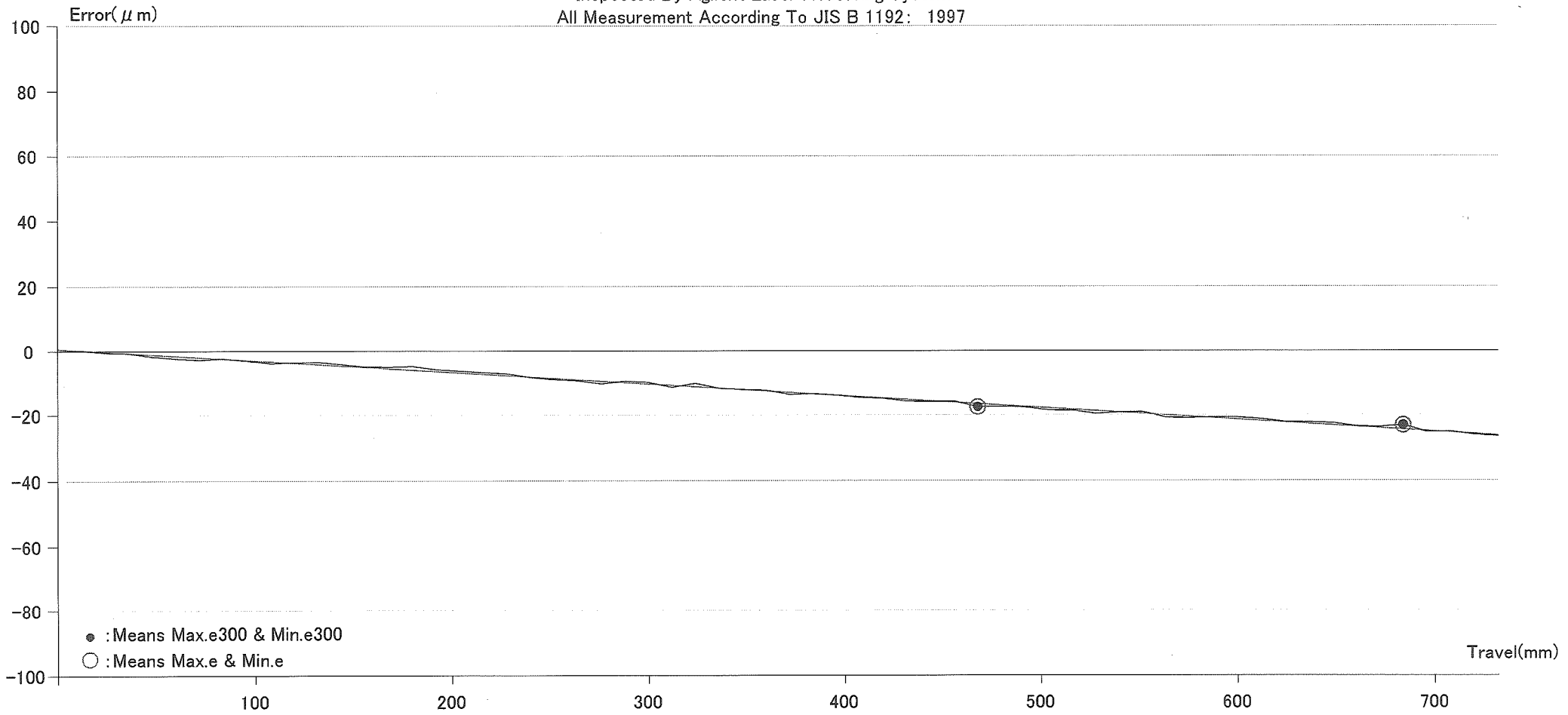
	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	— " —
3	Ensure smooth rotation of ball screw assembly.	— " —
4	Ensure ball screw run out within 0.010 mm	— " —
5	Ensure tightness of all screws & marking with yellow marker	— " —
6	Ensure ball Screw report attached to Inspection Report.	— " —

Date:	Checked by	Approved by

PMI BALLSCREWS

BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	:AMS
·Order No.	:S3-130052882017
·Drawing No.	:BA300510003-0
·Part No.	: *:S12-016-97
·Specification	:R40-12C1-FDWE-796-1060-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:732
·E(μm)	:±18
·T(μm)	:-24

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	-42~-6	-26.81
·e(μm)	13	2.29
·e300(μm)	8	2.29
·Tq(kgf - cm)	4.16~7.28	6.023-7.135
·Accuracy Grade	C3	C0

Manufactured By
PRECISION MOTION IND., INC.
www.pmi-amt.com

Measuring Environment:
·Refernce Temperatrue : 20℃ ± 0.5℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

QC
2013/09/10
PASS

Inspector: JAY R

M11-939

2017

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

BALL SCREW ASSEMBLY REPORT

Machine Number : M11-932	Ball screw Sl No: 331300 5288 2017	Axis : Y

I. Ball screw Make & Specification : PMI RHO- 12C1-TDWC-796-1060-0.008

II. Ball Screw Bearing Make & Specification : FAG BSB-2562-2Z-SU

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN - MEC - SA - 02	0.9N/m	1.0N/m
2	Bearing housing Bore diameter	As per drawing	6+03 µ	8+07 µ
3	Bearing housing centre height difference	0.01 mm	0.01mm	—
4	Bearing housing bore parallelism with respect to pads	0.01 mm	0.005mm	0.005mm

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	FAG	T3	1H 983
4	Bearing width B3	— " —	T4	1H 986
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	C2H-375-8123	T6	—
7	Housing Bearing depth	C37-103-2123	T7	43.510

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 103/907

T6 = 13.610 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	FAG	T3	1H-983
4	Bearing width B3	— " —	T4	1H-981
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	C2H-360-8123	T6	—
7	Housing Bearing depth	C37-153-2123	T7	43.500

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 153/907

T6 = 13.610 mm

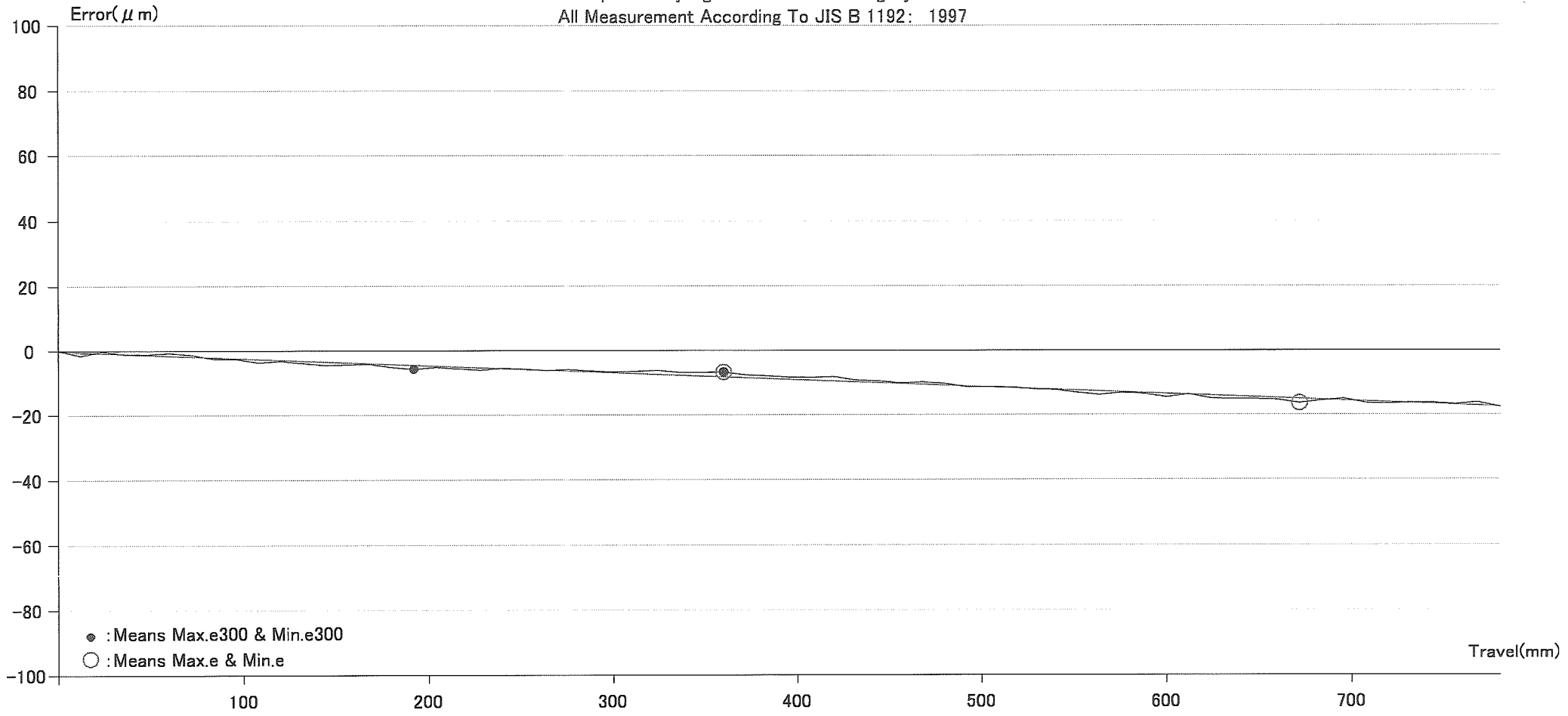
VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	— " —
3	Ensure smooth rotation of ball screw assembly.	— " —
4	Ensure ball screw run out within 0.010 mm	— " —
5	Ensure tightness of all screws & marking with yellow marker	— " —
6	Ensure ball Screw report attached to Inspection Report.	— " —

Date:	Checked by	Approved by

PMI BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	:AMS
·Order No.	:S3-130039032017
·Drawing No.	:BA300510004-0
·Part No.	: *:S12-016-96
·Specification	:R40-12E1-FDWE-841-1105-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:780
·E(μm)	:±18
·T(μm)	:-25

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	-43～-7	-17.33
·e(μm)	13	2.84
·e300(μm)	8	2.80
·Tq(kgf－cm)	5.72～9.88	6.298-8.196
·Accuracy Grade	C3	C1

Manufactured By
PRECISION MOTION IND., INC.
www.pmi-amt.com

Measuring Environment:
·Refernce Temperatrue : 20℃ ± 0.5℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

QC
2013/7/27
PASS

Inspector: ROCKY

2-A++1
111-026

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

BALL SCREW ASSEMBLY REPORT

Machine Number : M11-932 | Ball screw Sl No: S3130039032017 | Axis : Z

I. Ball screw Make & Specification : PMI [RH0-12G1-FDOE-841-1105-0.008]

II. Ball Screw Bearing Make & Specification : FAG [B5B-2562-22-SU]

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN - MEC - SA - 02	1.3Nm/m	1.4Nm/m
2	Bearing housing Bore diameter	As per drawing	3 to 5 µ	3 to 5 µ
3	Bearing housing centre height difference	0.01 mm	-	-
4	Bearing housing bore parallelism with respect to pads	0.01 mm	-	0.005mm

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	C2H-291-8123	T1	13.768
2	Bearing width B1	-	T2	-
3	Bearing width B2	FAG	T3	14.991
4	Bearing width B3	〃	T4	14.992
5	Spacer cylindrical width B4	-	T5	
6	Flange cover (2) spigot depth	C2H-291-8123	T6	
7	Housing Bearing depth	C37-701-2123	T7	56.998

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 701/13 (21)

T6 = 13.320 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	-	T1	-
2	Bearing width B1	-	T2	-
3	Bearing width B2	FAG	T3	14.986
4	Bearing width B3	〃	T4	14.987
5	Spacer cylindrical width B4	-	T5	-
6	Flange cover (2) spigot depth	C2H-280-8123	T6	-
7	Housing Bearing depth	C37-154-2123	T7	43.510

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 154/19

T6 = 13.610 mm

VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	Ok
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	〃
3	Ensure smooth rotation of ball screw assembly.	〃
4	Ensure ball screw run out within 0.010 mm	〃
5	Ensure tightness of all screws & marking with yellow marker	〃
6	Ensure ball Screw report attached to Inspection Report.	〃

Date: | Checked by | Approved by

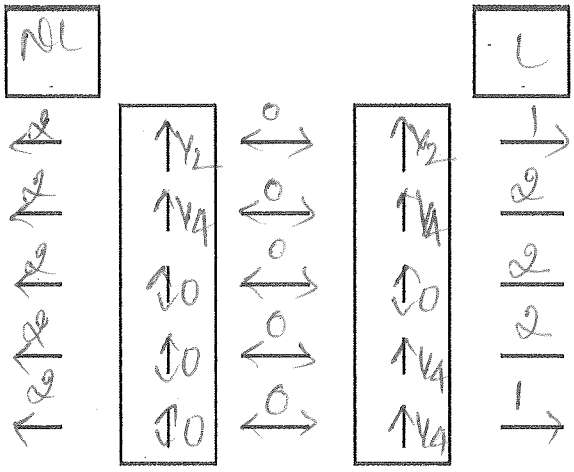
ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS — In Pursuit of Excellence

MACHINE NUMBER: M11-932. DATE: 25/9/13

I. MACHINE BASE LEVELING: Least count of the Instrument: 1 div = 0.020mm / meter



SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	1 div / meter	½ div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	0 div		
3	Leveling - Individual (Transverse)	3 div / meter	2 div		
4	Straightness of locating guide way	± 0.005	0.004		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.005		
6	Ball screw Run out at Motor end	±0.010	0.010		
7	Ball screw Run out at Brake end	±0.010	0.010		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.025 N/m		
			M.E	CEN	B.E
11	Ball screw Torque	0.6 to 1.8 N-m	1.5N/m	1.5N/m	1.5N/m
12	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
13	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
14	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
15	Final slide Torque – box guide way	2.5 to 3.5 N-m			
16	Final slide Torque – L M rail	1.6 to 2.5 N-m	2.5N/m	2.5N/m	2.5N/m
17	Variation in torque for final slide	10%			
18	Remarks if any :				

SIGN & DATE					
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	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER

ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

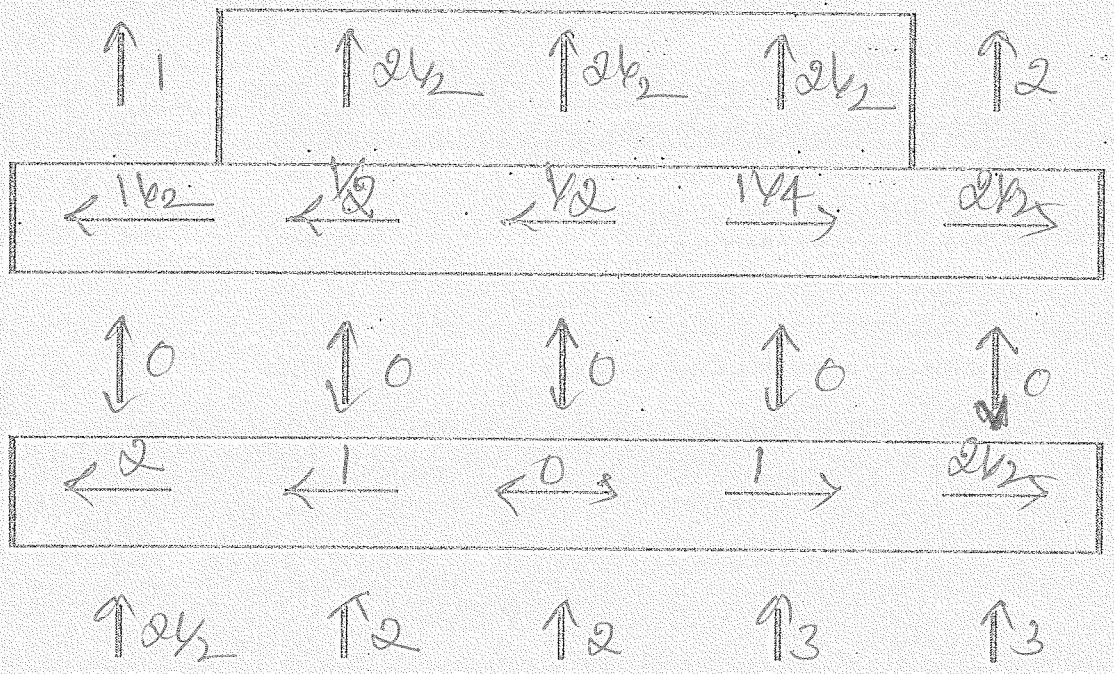
AMS
In Pursuit of Excellence

MACHINE NUMBER: M11-932 DATE: 25/9/13

SL NO	Check Points	Remark
1	Ensure all leveling screws & locknuts are fixed to the base.	✓
2	Ensure guide way butting face blue matching is minimum 70%	✓
3	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	✓
4	Ensure position of LM block ground surface butted to slide butting face.	✓
5	Ensure required height blocks are fitted to the base as per Work Order / Planning chart.	✓
6	Ensure plugs are fitted for L M guide way	✓

DATE: 27/9/13

II. SADDLE LEVEL READING: Least count of the Instrument: 1 div = 0.020mm / meter



Remarks if any :

SIGN & DATE					
	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER
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ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS
In Pursuit of Excellence

MACHINE NUMBER: NM11-932 DATE: 27/9/13

SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	2 div / meter	2 div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	'0' div		
3	Leveling - Individual (Transverse)	3 div / meter	3 div		
4	Straightness of locating guide way	± 0.005	0.002		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.002		
6	Ball screw Run out at Motor end	±0.010	0.010		
7	Ball screw Run out at Brake end	±0.010	0.010		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.02mm/mtr.		
			M.E	CEN	B.E
11	Ball screw Torque	0.6 to 1.8 N-m	1.5N/m	1.5N/m	1.5N/m
12	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
13	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
14	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
15	Final slide Torque – box guide way	2.5 to 3.5 N-m			
16	Final slide Torque – L M rail	1.6 to 2.5 N-m	1.8N/m	1.8N/m	1.8N/m
17	Variation in torque for final slide	10%			

Remarks if any :

SIGN & DATE					
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ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

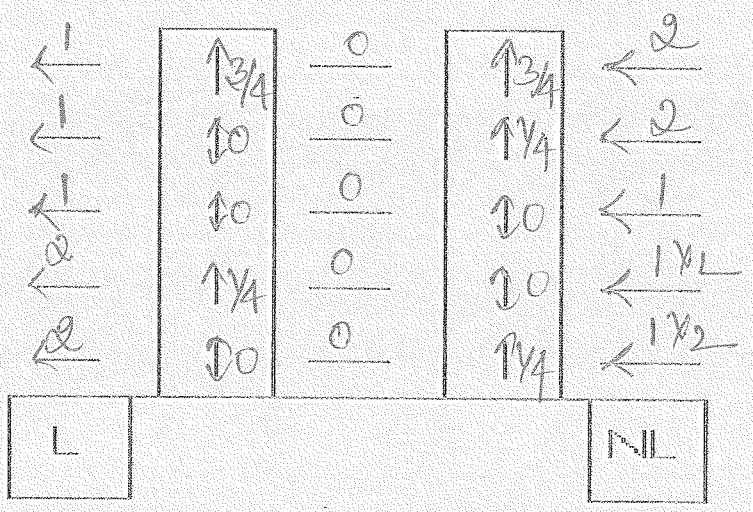
AMS *In Pursuit of Excellence*

MACHINE NUMBER: M911-932 DATE: 27/9/13

SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air & kerosene	✓
2	Ensure guide way butting face & saddle bottom blue matching is minimum 80%	✓
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	NA
4	Ensure proper taper matching of jib. ((H G m/c only)	NA
5	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	✓
6	Ensure position of LM block ground surface butted to slide butting face. (L M m/c only)	✓
7	Ensure Equal gap between ball nut & nut-housing bore.	✓
8	Ensure Nut housing alignment is with in 0.010 mm/meter	✓
9	Ensure White Prussian matching between nut housing & ball nut butting face is80%	✓
10	Ensure bearing housing Dowelling and grub locking.	✓
11	Ensure Ball screw lock nut grub locking with lock tite.	✓
12	Ensure guide way wipers are fixed. (H G m/c only)	NA
13	Ensure Nylon stoppers are fixed as per specification.	✓
14	ensure plugs are fitted for lm guide way & grub screws are fixed to saddle mounting holes	✓

DATE: 25/9/13

III. COLUMN READING: Least count of the Instrument: 1 div = 0.020mm / meter



Remarks if any :

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ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS
In Pursuit of Excellence

MACHINE NUMBER: M11-932 DATE: 25/9/13

SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	1 div / meter	3/4 div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	1 div		
3	Leveling - Individual (Transverse)	3 div / meter	2 div		
4	Straightness of locating guide way	± 0.005	0.002		
.5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.003		
6	Ball screw Run out at Motor end	±0.010	0.010		
7	Ball screw Run out at Brake end	±0.010	0.010		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.025MM/m		
11	Mandrel radial run out	±0.010	0.010		
12	Mandrel reading at Top (parallelism wrt.head stock movement)	±0.010	0.010		
13	Mandrel reading at side (parallelism wrt.head stock movement)	±0.010	0.010		
			M.E	CEN	B.E
14	Ball screw Torque	0.6 to 1.8 N-m	1.5NM	1.6NM	1.6NM
15	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
16	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
17	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
18	Final slide Torque – box guide way	2.5 to 3.5 N-m			
19	Final slide Torque – L M rail	1.6 to 2.5 N-m	1.6NM	1.6NM	1.6NM
20	Variation in torque for final slide	10%			

Remarks if any :

SIGN & DATE	MB 7/8/13			5...10	F-QA-IPR-BBA-02-R0
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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

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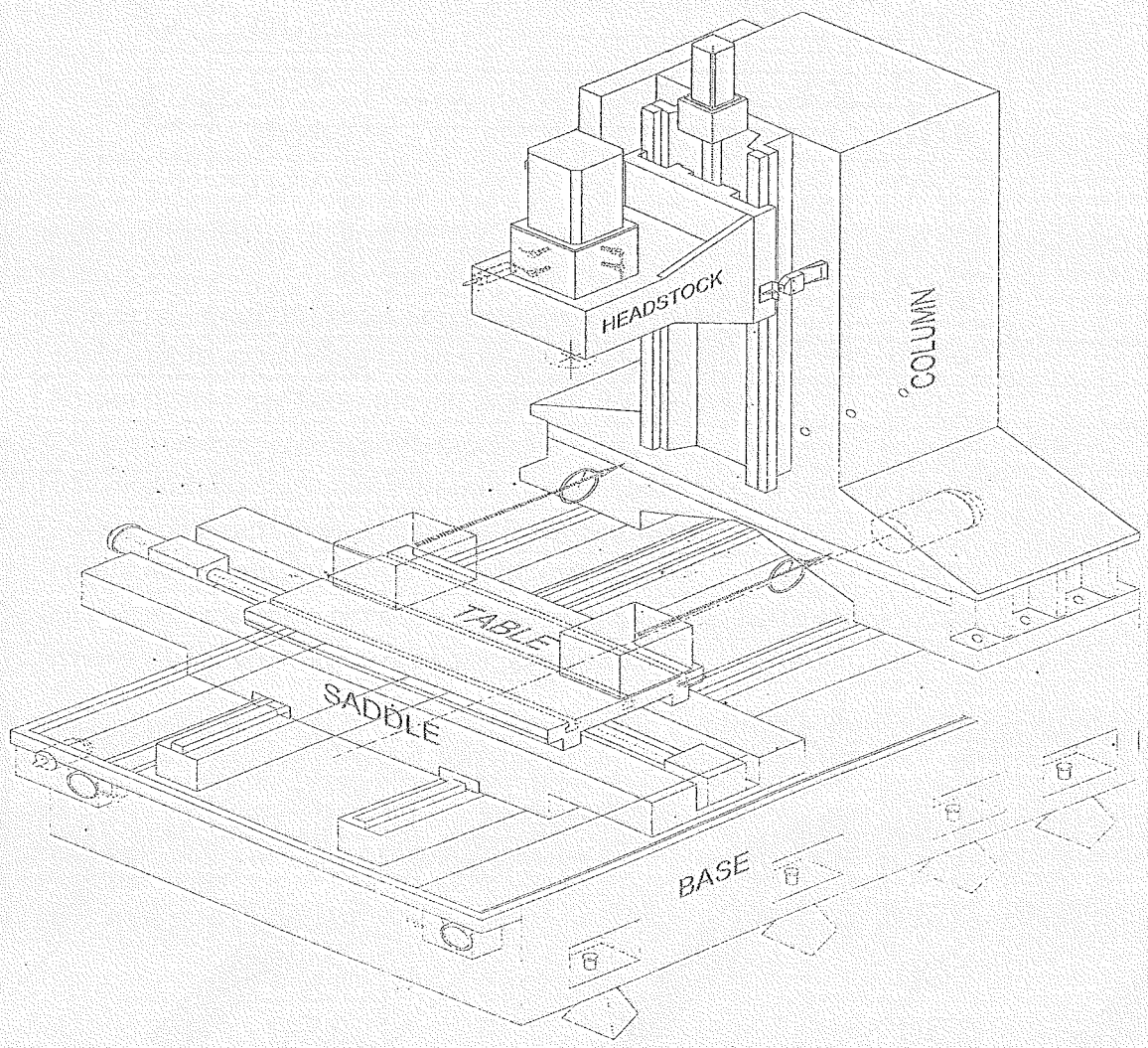
MACHINE NUMBER: NC11-932 DATE: 10/10/13

XI. COLUMN ALINGMENT READING WITH RESPECT TO X-AXIS MOVEMENT:
PERMISSIBLE VALUE = 0.30MM/METRE



0.080 mm.

0	100	200	300	400	500	600					

Remarks if any :

SIGN & DATE					
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ACE MANUFACTURING SYSTEMS LTD.

LUBRICATION ASSEMBLY - INSPECTION REPORT- BBA

AMS *In Pursuit of Excellence*

MACHINE NUMBER : MH-932

I. Lubrication metering cartridge specification (cc of oil / shot) as per DS-PDN-MEC-LUB-01 :

Sl. No.	Axis	Ball screw	Box guideway top / LM block	Box guideway side	Jib	Keeper plate
1	X -Axis	0.25 cc	0.1 cc	—	—	—
2	Y -Axis	0.25 cc	0.1 cc	—	—	—
3	Z-Axis	0.25 cc	0.1 cc	—	—	—

Sl. No.	Check points:	Remark
1	Verify that metering cartridge are fixed as per DS-PDN-MEC-LUB-01.	ok
2	Ensure X,Y & Z guideways and ball screw are lubricating properly	ok
3	Ensure lubrication pressure set to 15 Bar.	ok
4	Ensure there is no leakage of oil from joints, cartridges & manifolds etc.	ok
5	Ensure that the lubrication hoses do not rub or foul with the stationary / moving parts during axis movement.	ok
6	Ensure that the lubrication hoses of appropriate length (as per DS-PDN-MEC-LUB-01) are used and properly clamped and routed.	ok

II Telescopic cover Assembly:

	Sl No	Axis	Extension block Parallelism w.r.t. axis guideway		Extn. Block Length Variation -	DRAWING NUMBER
			Top - Recommended 0.020 mm	Side- Recommended 0.030 mm	Recommended 0.050 mm	(S10 -)
	1	X - Motor End	0.02mm	0.02mm	0.02mm	S10-013-122-06
	2	X - Brake End	0.02mm	0.03mm	0.05mm	S10-013-122-06
	3	Y - Motor End	0.01mm	0.03mm	—	S10-013-269-02
	4	Y - Brake End	0.02mm	0.02mm	0.05mm	S10-013-269-02
	5	Z - Brake End	0.01mm	0.03mm	0.05mm	S10-013-297-00

III. Saddle cover alignment w.r.t. Y axis movement (If applicable) :

	Sl NO	Top - Recommended 0.050 mm	Side - Recommended 0.050 mm	
	1	—	—	

IV. Check Points :

Sl NO	Check points:	Remark
1	Check telescopic cover sliding smoothly	ok
2	Ensure telescopic cover screws are tightened in box close condition	ok.

Date:	PREPARED	CHECKED	APPROVED
24/10/13			

ACE MANUFACTURING SYSTEMS LTD

LUBRICATION ASSEMBLY - INSPECTION REPORT- BBA

AMS *In Pursuit of Excellence*

MACHINE NUMBER : MU- 932.

V. Axis Motor Assembly :

Motor mounting bracket Concentricity & Face out:

Sl No	Axis	Face out -Recommended 0.030 mm	Concentricity Recommended 0.060 mm
1	X- Axis	0.03mm	0.05mm
2	Y- Axis	0.02mm	0.05mm
3	Z- Axis	0.02mm	0.03mm.

Sl NO	Check points:	Remark
1	Ensure motor shaft and coupling bore are free from oil / grease before fitting of coupling	ok
2	Ensure Axis Motor & Motor mounting bracket are fitted as per specification.	ok
3	Ensure that slot on Steel Bellows, Clamping ring & Bush are in line.	ok
4	Ensure tightness of coupling is set to torque value of 17 N-m (M6 screw), 35 N-m (M8 screw).	ok

M/c Axis motor coupling. Mayar Size-2.

VI. Counter balance Assembly (If applicable)

Sl NO	Check points:	Remark
1	Ensure Smooth movement of Z- Axis without fouling of dead weight. (MCV 500 / MCV 600 only)	NA
2	Ensure the parallelism of counter balance cylinder piston is within 0.050 mm in both XZ & YZ planes with respect to Z Axis movement.	—
3	Ensure variation in the vertical axis motor current is within 20% during upward and downward movement of slide.	—
4	Ensure dowelling of counterbalance bracket.	—

REMARK IF ANY:

24/10/13.			
Date:	PREPARED	CHECKED	APPROVED